

E3 Program Report: Performance Analysis of the ATC-ITC Algorithm

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About the Report

This is a note and a report on what I have done during my summer internship at the Adaptive Systems Lab (ASL) in EPFL from 1/7 to 31/8/2025. The work mainly consists of mathematical derivations, with MATLAB simulations supporting the results. The codes to my work can be seen on my [GitHub page](#).

A couple of topics that I have done throughout this internship includes (ordered with respect to the starting time of each project):

- (a) Fish swarm simulation.
- (b) Predator and prey dynamics simulation.
- (c) Football game simulation.
- (d) The stability and performance analysis of the ATC-ITC algorithm on games over a network of multiple teams.

Of course, this report focuses on the last one.

To-Do List:

- (a) Give a more detailed description of the range of step-sizes allowed, and the number of iterations required to reach steady state.
- (b) Read the proofs on variational inequality to ensure the relation between the $\exists!$ of Nash equilibrium and the strong-monotonicity of global gradient.
- (c) Change the local knowledge to zero. Compare the ATC-ITC algorithm with other algorithms.

Nomenclature

Below is a list of all the notations used in this report.

- Defined as: $:=$.
- Equivalent to: $\equiv$.
- Integer sequences: for integers $N \leq M$,

$$[N : M] := \{N, N + 1, \dots, M - 1, M\}.$$

We abbreviate $[1 : M]$ as $[M]$.

- Transpose: $A^\top$. Hermitian conjugate: A^H . Complex conjugate: A^* .
- All vectors are column vectors; row vectors are written with a transpose, e.g., $v^\top$.
- Column array:

$$[a_1; a_2; \dots; a_n] \equiv \text{col}\{a_k\}_{k=1}^n \equiv \text{col}\{a_k\}_{k \in [n]} := \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}.$$

- Row array:

$$[a_1, a_2, \dots, a_n] \equiv \text{row}\{a_k\}_{k=1}^n \equiv \text{row}\{a_k\}_{k \in [n]} := \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}.$$

- Vector of ones: $\mathbf{1}$ or $\mathbf{1}_K$ to specify its length K . Matrix of ones: $\mathbf{1}_{M \times N}$, of size $M \times N$.
- Identity matrix: $\mathbb{1}$ or $\mathbb{1}_K$ to specify its size is $K \times K$.
- Zero matrix: 0 . If its size is to be emphasized, I denote it by $0 \cdot \mathbf{1}_{M \times N}$.
- Stochastic gradient: for a deterministic gradient g , its stochastic counterpart is denoted by $\hat{g}$.
- Vectorization: for a matrix $A = [a_1, \dots, a_N] \in \mathbb{C}^{M \times N}$ with columns $a_i \in \mathbb{C}^{M \times 1}$,

$$\text{vec}(A) \equiv \underline{A} := [a_1; \dots; a_N] \in \mathbb{C}^{MN \times 1}.$$

- Error: for a variable x and its corresponding equilibrium x^* , the error is denoted by $\tilde{x} := x^* - x$.
- Block-diagonal matrix: for square matrices $\{A_k\}_{k \in [N]}$,

$$\text{blkdiag}\{A_k\}_{k \in [N]} = \begin{bmatrix} A_1 & & \\ & \ddots & \\ & & A_N \end{bmatrix}.$$

The sizes of each block need not be the same.

- Matrix element: $[A]_{ij} = a_{ij}$, note that a_{ij} can also be another matrix.
- Matrix symmetrization and skew-symmetrization: for a real square matrix A ,

$$\text{sym}(A) := \frac{A + A^\top}{2}, \text{ and } \text{skew}(A) := \frac{A - A^\top}{2}.$$

- The symbol “...” is added to unfinished equations so that it continues on in the next line: for example, the equations

$$U(x') = \frac{e^{i2\pi uz'}}{i\lambda z'} e^{i\frac{\pi u}{z'}(x'^2+y'^2)} \int_{\mathbb{R}^2} U_0(x, y) e^{-i\frac{2\pi u}{z'}(x'x+y'y)} dx dy.$$

and

$$U(x') = \frac{e^{i2\pi uz'}}{i\lambda z'} e^{i\frac{\pi u}{z'}(x'^2+y'^2)} \int_{\mathbb{R}^2} U_0(x, y) \dots \\ \cdot e^{-i\frac{2\pi u}{z'}(x'x+y'y)} dx dy.$$

are equivalent.

- Matrix component-wise comparison: for matrices A and B of the same size:

$$A \square B \Leftrightarrow [A]_{ij} \square [B]_{ij},$$

where $\square \in \{>, <, \geq, \leq\}$. This notation also extends to vectors.

- Matrix partial-ordering on definiteness: the notations $\succ$, $\prec$, $\succeq$, and $\preceq$ are adopted.
- Kronecker product: denoted by $\otimes$. For $[A]_{ij} = a_{ij}$ and $[B]_{ij} = b_{ij}$,

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots \\ a_{21}B & a_{22}B & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}.$$

Furthermore, for matrices A , B , and C of appropriate sizes,

$$\text{vec}(AXB) = (B^T \otimes A) \cdot \text{vec}(X).$$

- Block-Kronecker product: denoted by $\otimes_b$. For block-partitioned matrices,

$$\begin{bmatrix} A_{11} & A_{12} & \cdots \\ A_{21} & A_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \otimes_b \begin{bmatrix} B_{11} & B_{12} & \cdots \\ B_{21} & B_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \\ = \begin{bmatrix} A_{11} \otimes B_{11} & A_{11} \otimes B_{12} & \cdots & A_{12} \otimes B_{11} & A_{12} \otimes B_{12} & \cdots \\ A_{11} \otimes B_{21} & A_{11} \otimes B_{22} & \cdots & A_{12} \otimes B_{21} & A_{12} \otimes B_{22} & \cdots \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots \\ A_{21} \otimes B_{11} & A_{21} \otimes B_{12} & \cdots & A_{22} \otimes B_{11} & A_{22} \otimes B_{12} & \cdots \\ A_{21} \otimes B_{21} & A_{21} \otimes B_{22} & \cdots & A_{22} \otimes B_{21} & A_{22} \otimes B_{22} & \cdots \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots \end{bmatrix}.$$

The block partition associated depends on the specifications. One should note that block-Kronecker product is related to the usual Kronecker product by a row or column permutation. Note that since if $\mathcal{T}$ denotes a column permutation, then $\mathcal{T}^{-1} = \mathcal{T}^T$ is a row permutation. We have that if both A and B are column-partitioned, there exists a column permutation $\mathcal{T}$ such that

$$A \otimes_b B = (A \otimes B) \mathcal{T};$$

if both A and B are row-partitioned, there exists a row permutation $\mathcal{T}^{-1}$ such that

$$A \otimes_b B = \mathcal{T}^{-1} (A \otimes B);$$

and if both A and B are column- and row-partitioned, there exists column permutations $\mathcal{T}_1$ and $\mathcal{T}_2$ such that

$$A \otimes_b B = \mathcal{T}_2^{-1} (A \otimes B) \mathcal{T}_1.$$

- Block vectorization: denoted by $\text{bvec}(\cdot)$. Same as block-Kronecker product, it is associated to the usual vectorization by a column permutation: for example, if A is row- and column-partitioned, there exists a column permutation $\mathcal{T}$ such that

$$\text{bvec}(A) = \text{bvec} \left(\begin{bmatrix} A_{11} & A_{12} & \cdots \\ A_{21} & A_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \right) = \begin{bmatrix} \text{vec}(A_{11}) \\ \text{vec}(A_{12}) \\ \vdots \\ \text{vec}(A_{21}) \\ \text{vec}(A_{22}) \\ \vdots \end{bmatrix} = \mathcal{T} \text{vec}(A).$$

- Block-1-norm and block- ∞ -norm: for a block-partitioned matrix $[A]_{ij} = A_{ij}$, we define its block-1-norm and block- ∞ -norm respectively as

$$\|A\|_{\text{b},1} := \max_j \sum_i \|A_{ij}\| \quad \text{and} \quad \|A\|_{\text{b},\infty} := \max_i \sum_j \|A_{ij}\|.$$

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1 Introduction

1.1 Games on Networks and the ATC-ITC Algorithm

Learning over networks addresses how each agent can learn a certain information through limited or constrained communication with its neighbors. The study often discusses whether a network configuration allows for agents to reach consensus or reach the ground truth information. A game on networks separates the agents into different teams, requiring each agent to not only estimate and learn its opponents' strategies of the game, but also to optimize for a counter strategy to minimize its loss (or to maximize its gain).

Plenty of algorithms have been proposed to allow for the network as a whole to reach an equilibrium of the game. One analyzes the dynamics of the game: including how fast the game reaches the equilibrium (if it does) and at what point is the equilibrium. Two of these algorithms that are of my utmost interest are the competitive diffusion (CD) algorithm and the adapt-then-combine, infer-then-combine (ATC-ITC) algorithm.

This section gives an introduction to a game over a network of two teams and defines the ATC-ITC algorithm.

1.1.1 Formulation of Games on Network

This first section sets up the formulation to a game on network. The two algorithms: CD and ATC-ITC are also introduced.

Given two teams of agents $\mathcal{N}^{(1)} = [K_1]$ and $\mathcal{N}^{(2)} = [K_2]$ of sizes K_1 and K_2 , respectively. Each agent $k \in \mathcal{N}^{(1)}$ has a strategy $x_k \in \mathbb{R}^{M_1}$, and each agent $\ell \in \mathcal{N}^{(2)}$ has a strategy $y_\ell \in \mathbb{R}^{M_2}$. We associate each agent k in team t (denoted by $\mathcal{N}^{(t)}$) with a loss function $J_k^{(t)}(x, y)$. The task of the agents in team 1 is to minimize the aggregated loss $J^{(1)}$:

$$\min_{x \in \mathbb{R}^{M_1}} J^{(1)}(x, y) = \sum_{k \in \mathcal{N}^{(1)}} p_k^{(1)} J_k^{(1)}(x, y),$$

while the task for those in team 2 is to minimize the aggregated loss $J^{(2)}$:

$$\min_{y \in \mathbb{R}^{M_2}} J^{(2)}(x, y) = \sum_{\ell \in \mathcal{N}^{(2)}} p_\ell^{(2)} J_\ell^{(2)}(x, y).$$

The vectors of coefficients $p^{(1)} \in \mathbb{R}^{K_1 \times 1}$ and $p^{(2)} \in \mathbb{R}^{K_2 \times 1}$ determine the importance of each agent to their respective team. How these coefficients are determined is defined later.

A *Nash equilibrium* of the game above is a strategy $z^* = [x^*; y^*]$ such that

$$J^{(1)}(x^*, y^*) \leq J^{(1)}(x, y^*) \text{ and } J^{(2)}(x^*, y^*) \leq J^{(2)}(x^*, y) \quad (\forall x, y), \quad (1.1)$$

i.e., no single agent would like to move away from the current strategy as it would increase the aggregated losses.

I assume the existence and uniqueness of the Nash equilibrium throughout this discussion. The question we care about is whether or not each x_k ($k \in \mathcal{N}^{(1)}$) and each y_ℓ ($\ell \in \mathcal{N}^{(2)}$) can converge to x^* and y^* , respectively?

1.1.2 ATC-ITC Algorithm

The adapt-then-combine, inference-then-combine (ATC-ITC) algorithm is proposed [ZCS25] to solve the problem above. The idea of the ATC-ITC algorithm is as below: at each round, each player will update their strategy based on stochastic gradient descent, it further shares its updated strategy with

its neighbors to obtain an aggregated strategy. However, in order to compute the gradient and to find a good counter-strategy in face of the opponent's action, one must estimate what the action of its opponent is. Therefore, besides having one's own strategy, say x_k for $k \in \mathcal{N}^{(1)}$, one also has an estimate y_k of the strategy of the opponents. An estimate can also be updated by inference or by sharing information with one's neighbors.

The ATC-ITC algorithm is summarized as below:

- (a) At each iteration i , an agent, say $k \in \mathcal{N}^{(1)}$ updates its strategy $x_{k,i}$ by a gradient descent with step size $\mu > 0$ using its local gradient:

$$\psi_{k,i} = x_{k,i-1} - \mu J_k^{(1)}(x_{k,i-1}, y_{k,i-1}). \quad (1.2)$$

The variables $x_{k,i-1}$ and $y_{k,i-1}$ are, respectively, agent k 's strategy and estimate of opposing team strategy at the $(i-1)$ th iteration.

- (b) Next, it aggregates the strategies from its neighbors that are of the same team by a linear combination:

$$x_{k,i} = \sum_{j \in \mathcal{N}^{(1)}} a_{jk}^{(1)} \psi_{j,i} = \sum_{j \in \mathcal{N}^{(1)}} a_{jk}^{(1)} \left(x_{j,i-1} - \mu \nabla_x J_k^{(1)}(x_{j,i-1}, y_{j,i-1}) \right). \quad (1.3)$$

The weights $a_{jk}^{(1)} \geq 0$ forms a left-stochastic matrix $A^{(1)} = [a_{jk}^{(1)}]_{jk}$ representing the influence of agent j on agent k . Left-stochasticity means that $\mathbf{1}_{K_1}^\top A^{(1)} = \mathbf{1}_{K_1}^\top$, where $\mathbf{1}_{K_1}$ is the column vector of ones of length K_1 .

- (c) Similarly, an agent $\ell \in \mathcal{N}^{(2)}$ updates via

$$y_{\ell,i} = \sum_{j \in \mathcal{N}^{(2)}} a_{j\ell}^{(2)} \left(y_{j,i-1} - \mu \nabla_y J^{(2)}(x_{j,i-1}, y_{j,i-1}) \right). \quad (1.4)$$

The weights $a_{j\ell}^{(2)} \geq 0$ forms a left-stochastic matrix $A^{(2)} = [a_{j\ell}^{(2)}]_{j\ell}$.

- (d) Next, each agent (say $k \in \mathcal{N}^{(1)}$) updates its estimate on the strategy of the opposing team by a descent using its local knowledge $g_{k,i-1}$:

$$\phi_{k,i} = y_{k,i-1} - \mu g_{k,i-1}. \quad (1.5)$$

The form of the local knowledge $g_{k,i-1}$ determines which learning algorithm we are adopting. The exact form to the local knowledge vector will be explained later.

- (e) Each agent then aggregate the estimates from its neighbors and opponents to obtain the updated estimate as

$$\begin{aligned} y_{k,i} &= \sum_{j \in \mathcal{N}^{(1)}} c_{jk}^{(11)} \phi_{j,i} + \sum_{\ell \in \mathcal{N}^{(2)}} c_{\ell k}^{(21)} y_{\ell,i} \\ &= \sum_{j \in \mathcal{N}^{(1)}} c_{jk}^{(11)} (y_{j,i-1} - \mu g_{j,i-1}) + \sum_{\ell \in \mathcal{N}^{(2)}} c_{\ell k}^{(21)} y_{\ell,i}. \end{aligned} \quad (1.6)$$

The coefficient matrix $C^{(11)} = [c_{jk}^{(11)}]_{jk}$ and $C^{(21)} = [c_{\ell k}^{(21)}]_{\ell k}$ forms a left-stochastic matrix $[C^{(11)}; C^{(21)}]$ by stacking them in a column.

- (f) Similarly, an agent $\ell \in \mathcal{N}^{(2)}$ updates its estimate on team 1 via

$$x_{\ell,i} = \sum_{k \in \mathcal{N}^{(1)}} c_{k\ell}^{(12)} x_{k,i} + \sum_{j \in \mathcal{N}^{(2)}} c_{j\ell}^{(22)} (x_{j,i-1} - \mu g_{j,i-1}). \quad (1.7)$$

The matrix $[C^{(12)}; C^{(22)}]$ is again left-stochastic where $C^{(12)} = [c_{k\ell}^{(12)}]_{k\ell}$ and $C^{(22)} = [c_{j\ell}^{(22)}]_{j\ell}$.

For the ATC-ITC algorithm, we choose that the local knowledge vector is

$$g_{k,i-1} = \nabla_y J_{\ell_k}^{(2)}(x_{k,i-1}, y_{k,i-1}) \quad (1.8)$$

for agents $k \in \mathcal{N}^{(1)}$ and a correspondence $\ell_k \in \mathcal{N}^{(2)}$; the local knowledge vector is

$$g_{\ell,i-1} = \nabla_x J_{k_\ell}^{(1)}(x_{\ell,i-1}, y_{\ell,i-1}) \quad (1.9)$$

for agents $\ell \in \mathcal{N}^{(2)}$ and a correspondence $k_\ell \in \mathcal{N}^{(1)}$.

1.2 Goals

The chapters to this report are organized as follows:

- In [Chapter 1](#), I gave an introduction to all the topics, this includes a game on a two-team network.
- In [Chapter 2](#), the scheme of a game on two-team network is generalized to a T -team network. Furthermore, theoretical analyses on an algorithm (ATC-ITC) that reaches the Nash equilibrium of the game is given, these includes stability, first-order MSD performance, and the rate of convergence to steady state.
- In [Chapter 3](#), the step-sizes is now team-dependent. This change adds nothing new, simply making the discussion more general. The derivation provided has the goal to be much cleaner than its previous chapter. Further, I also considered the case of diminishing step-sizes, showing its convergence to the Nash equilibrium.
- Lastly, in [Chapter 4](#), the performances (this includes the MSD and the rate of convergence) to the ATC-ITC algorithm is compared with other algorithms.

2 Games on Network of Multiple Teams

Looking at the derivation and results from the last chapter, many begging for a generalization. This chapters serves for this exact purpose.

Furthermore, the results from the network-of-two-team scheme seems to also be applicable in the case of multiple teams, as an agent in a team can simply regard its opponent as all other agents from all other teams, thus reducing to a two-team case.

There are multiple ways to generalize the ATC-ITC algorithm when there are multiple teams involved, their differences depending on how one updates its estimates on the strategy of opposing teams.

- In the two-team ATC-ITC algorithm: At iteration i , one updates its estimate using the *updated* strategy of the opponent from iteration i .
- In the two-team CD algorithm: At iteration i , one updates its estimate using the *not-yet-updated* strategy of the opponent from iteration $i - 1$.

When there are multiple teams involved, to estimate the strategy of team 1, not only can we utilize the partial information on the strategy of team 1, but we can also obtain information from other teams' estimates to the strategy of team 1. The ITC step has many possible generalizations, to name a few: for an agent $k \in \mathcal{N}^{(t)}$, it wants to update its estimate of the strategy of team τ by:

- (a) Aggregating the updated strategy in team τ :

$$x_{k,i}^\tau = \sum_{\ell \in \mathcal{N}^{(t)}} c_{\ell k}^{(tt)} \left(x_{\ell,i-1}^\tau - \mu g_{\ell,i-1}^\tau \right) + \sum_{\ell \in \mathcal{N}^{(\tau)}} c_{\ell k}^{(\tau t)} x_{\ell,i}^\tau.$$

- (b) Aggregating the not-yet-updated strategy in team τ :

$$x_{k,i}^\tau = \sum_{\ell \in \mathcal{N}^{(t)}} c_{\ell k}^{(tt)} \left(x_{\ell,i-1}^\tau - \mu g_{\ell,i-1}^\tau \right) + \sum_{\ell \in \mathcal{N}^{(\tau)}} c_{\ell k}^{(\tau t)} x_{\ell,i-1}^\tau.$$

- (c) Aggregate every agent's not-yet-updated estimate on team τ :

$$x_{k,i}^\tau = \sum_{\ell \in \mathcal{N}^{(t)}} c_{\ell k}^{(tt)} \left(x_{\ell,i-1}^\tau - \mu g_{\ell,i-1}^\tau \right) + \sum_{s \in [T] \setminus \{t\}} \sum_{\ell \in \mathcal{N}^{(s)}} c_{\tau, \ell k}^{(st)} x_{\ell,i-1}^\tau.$$

There can be much more such as changing the index of team τ from $i - 1$ to i , or that make $c_{\ell k}^{(st)}$ to be τ -independent. I shall adopt the first one in the following discussions as it bears the closest resemblance to the two-team case.

2.1 Formulation of a Game between Multiple Teams

Consider playing a spontaneous game with $T \geq 2$ teams. Each team $t \in [T]$ has a strategy $x^t \in \mathbb{R}^{M_t}$ ($M_t \geq 1$) to play, and its goal is to minimize the loss function $J^{(t)}(x^1, x^2, \dots, x^T) = J^{(t)}(x) = J^{(t)}(x^t, x^{-t})$, where $x := \{x^\tau\}_{\tau \in [T]}$ is the strategy of all teams and $x^{-t} := \{x^\tau\}_{\tau \in [T] \setminus \{t\}}$ is the strategies of all other teams. How can we reach a Nash equilibrium of this game?

Furthermore, consider each team t consisting of $K_t \geq 1$ agents, we denote all agents within team t by the set $\mathcal{N}^{(t)}$. Each agent $k \in \mathcal{N}^{(t)}$ can, too, play the strategy $x_k^t \in \mathbb{R}^{M_t}$. We associate each agent $k \in \mathcal{N}^{(t)}$ with a local loss function $J_k^{(t)}(x^1, \dots, x^T)$. The task of team t is now to minimize the aggregated global loss function

$$\min_{x^t \in \mathbb{R}^{M_t}} J^{(t)}(x^t, x^{-t}) = \sum_{k \in \mathcal{N}^{(t)}} p_k^{(t)} J_k^{(t)}(x^t, x^{-t}), \quad (2.1)$$

where $p_k^{(t)}$ is some weights that add up to 1 and x^t is an aggregation of the individual strategies x_k^t from agents within the team. Besides asking whether $x = \{x^t\}_{t \in [T]}$ reaches the Nash equilibrium or not, we also want to know if agents of the same team can reach a consensus or not, viz., whether $x_k^t = x_\ell^t$ for all $k, \ell \in \mathcal{N}^{(t)}$?

The number of total agents is $K := K_1 + \dots + K_T$, and the size of all strategies is $M := M_1 + \dots + M_T$. The set of all agents is denoted by $\mathcal{N} = \mathcal{N}^{(1)} \sqcup \dots \sqcup \mathcal{N}^{(T)}$ as all teams are disjointed.

The *Nash equilibrium* of the game above is a strategy $x^* = \text{col}\{x^{t,*}\}_{t \in [T]}$ such that

$$J^{(t)}(x^*) \leq J^{(t)}(x^t, x^{-t,*}) \text{ for all } t \in [T] \text{ and } x^t \in \mathbb{R}^{M_t}. \quad (2.2)$$

2.1.1 Multi-Team ATC-ITC Algorithm

Note from above that the strategies x_k^t is only defined if $k \in \mathcal{N}^{(t)}$. Besides bearing a strategy, however, each agent must also bear estimates of the strategies of opposing teams in order to select a good response. Therefore, for agent $k \in \mathcal{N}^{(t)}$, we define its *estimate* on the strategy of team $\tau \neq t$ as the vector $x_k^\tau \in \mathbb{R}^{M_\tau}$.

At each iteration i , the agent $k \in \mathcal{N}^{(t)}$ bears its strategy $x_{k,i}^t$ and its estimates $x_{k,i}^\tau$ of the strategy of team $\tau \neq t$. By combining the strategy and estimates born by a single agent k into a single vector $x_{k,i} = \text{col}\{x_{k,i}^\tau\}_{\tau \in [T]}$, we again ask the questions whether $x_{k,i}$ converges to the Nash equilibrium as $i \rightarrow \infty$, and whether $x_{k,i}$ and $x_{\ell,i}$ reaches consensus for $k, \ell \in \mathcal{N}$ as $i \rightarrow \infty$?

The ATC-ITC algorithm proposed is a gradient descent method for reaching the Nash equilibrium and consensus. A simple adaptation of the ATC-ITC algorithm for a game over multiple teams is described as below: at each iteration i , the agent $k \in \mathcal{N}^{(t)}$ updates both its strategy x_k^t and estimates x_k^τ ($\tau \neq t$) by the steps below:

- (a) Strategy update (ATC): for $k \in \mathcal{N}^{(t)}$,

$$x_{k,i}^t = \sum_{\ell \in \mathcal{N}^{(t)}} a_{\ell k}^{(t)} \left(x_{\ell,i-1}^t - \mu \nabla_t J_\ell^{(t)}(x_{\ell,i-1}) \right) \quad (2.3)$$

for some constant step-size μ . The gradient ∇_t is applied with respect to the strategy x^t . The combination coefficients $a_{\ell k}^{(t)}$ sum up to 1 with respect to ℓ .

- (b) Estimate update (ITC): for $k \in \mathcal{N}^{(t)}$ and all $\tau \neq t$,

$$x_{k,i}^\tau = \sum_{\ell \in \mathcal{N}^{(t)}} c_{\ell k}^{(tt)} \left(x_{\ell,i-1}^\tau - \mu g_{\ell,i-1}^\tau \right) + \sum_{\ell \in \mathcal{N}^{(\tau)}} c_{\ell k}^{(\tau t)} x_{\ell,i}^\tau. \quad (2.4)$$

The vector $g_{\ell,i-1}^\tau$ is some local knowledge of agent $\ell \in \mathcal{N}^{(t)}$ has on team τ . The estimate is updated by aggregating *only* the information from neighbors of the same team and agents in team τ . Again, for a fix k and τ , the combination coefficients $c_{\ell k}^{(tt)}$ and $c_{\ell k}^{(\tau t)}$ should sum up to 1.

An assumption was made explicitly for the ATC-ITC algorithm:

Assumption 2.1: Local Knowledge of Opposing Team

All agents have the local knowledge of the gradient of a local loss in every opposing team, i.e., for agent $k \in \mathcal{N}^{(t)}$ and $\tau \neq t$,

$$g_{k,i-1}^\tau := \nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}) \quad (2.5)$$

for some correspondence $\ell_k \in \mathcal{N}^{(\tau)}$.

As a side note, the case where the step-sizes vary across teams is discussed in [Section 3.1](#). In this section, I will give a more lengthy but detailed description of the proofs and variables. For a more concise version, one can refer to [Chapter 3](#).

A further side note, the exact form to the local knowledge is not really important in the following discussion as its effects are of high order, and we only consider the *first-order* effects. But the effect induced by different choices of the local knowledge is elaborated in [Chapter 4](#).

2.1.2 Assumptions on Network Structure

Assumption 2.2: Network Structure

Consider the non-negative combination matrices $[A^{(t)}]_{\ell k} = a_{\ell k}^{(t)}$ and $[C^{(st)}]_{\ell k} = c_{\ell k}^{(st)}$:

- (a) For all $t \in [T]$, the matrix $A^{(t)}$ is left-stochastic and primitive.
- (b) For all $t \in [T]$, the matrix $C^{(tt)}$ is irreducible.
- (c) For all $t, \tau \in [T]$ and $t \neq \tau$, the matrix $[C^{(tt)}; C^{(\tau t)}]$ is left stochastic.
- (d) For all $t, \tau \in [T]$, the matrix $C^{(\tau t)}$ is non-zero.

Intuitively speaking, each of the assumptions on the network structure attributes to:

- (a) The left-stochasticity is required so that the weights used to update one's strategy from each contribution adds up to 1. The primitivity not only ensures irreducibility, but it also ensures only the eigenvalue 1 is on its spectral circle, facilitating the existence of a limit to the power of the matrix.
- (b) Irreducibility of $C^{(tt)}$ implies that the information of estimates born by any agent in team t goes around every other agents in the team eventually as the iteration goes on.
- (c) The left-stochasticity is required so that the weights used to update one's estimate from each contribution adds up to 1.
- (d) The information on the strategy of team τ can flow to any other team t .

Pragmatically speaking, the above assumptions lead to the following result on the spectra of the combination matrices

Theorem 2.1

The matrix

$$P^{(\tau)} = \begin{bmatrix} C^{(11)} & & & & & \\ & \ddots & & & & \\ & & C^{(\tau-1, \tau-1)} & & & \\ \hline \bar{A}^{(\tau)} \bar{C}^{(\tau 1)} & \dots & \bar{A}^{(\tau)} \bar{C}^{(\tau, \tau-1)} & \bar{A}^{(\tau)} & \bar{A}^{(\tau)} \bar{C}^{(\tau, \tau+1)} & \dots & \bar{A}^{(\tau)} \bar{C}^{(\tau T)} \\ \hline & & & C^{(\tau+1, \tau+1)} & & & \\ & & & & \ddots & & \\ & & & & & & C^{(TT)} \end{bmatrix} \quad (2.6)$$

is left-stochastic and has a single eigenvalue on the spectral circle for all $\tau \in [T]$.

Proof. The left-stochasticity is trivial, and it is then immediately obvious that the spectral radius to $P^{(\tau)}$ is 1. We will focus our attention on characterizing the spectrum, showing that only one eigenvalue exists on the spectral circle, having a value of exactly 1.

By a permutation of the rows and columns, we know that $P^{(\tau)}$ is similar to

$$\begin{bmatrix} A^{(\tau)} & A^{(\tau)} C^{(\tau 1)} & \dots & A^{(\tau)} C^{(\tau T)} \\ \hline & C^{(11)} & & \\ & & \ddots & \\ & & & C^{(TT)} \end{bmatrix}, \quad (2.7)$$

meaning that the spectrum of $P^{(\tau)}$ is equal to the union of the spectra of $A^{(\tau)}$ and $C^{(tt)}$'s (with $t \neq \tau$).

Since $A^{(\tau)}$ is primitive, it has a simple (single) eigenvalue 1 on its spectral circle. Further, since $C^{(\tau t)}$ is non-zero for all $t \neq \tau$, the columns to $C^{(tt)}$ adds up to a value less than or equal to 1, and at least the sum of a column has a strict inequality. It can hence be shown that $\rho(C^{(tt)}) < 1$. We will prove it by contradiction: Consider an eigenvalue and eigenvector pair (λ, x) to $C^{(tt)}$ where $|\lambda| = 1$. Define the component-wise absolute value to x as $y := |x|$. Then

$$C^{(tt)} y = \left| C^{(tt)} |x| \right| \geq \left| C^{(tt)} x \right| = |\lambda x| = |\lambda| |x| = y$$

by the triangle inequality. Summing up all the elements, we have

$$\mathbf{1}^\top y \leq \mathbf{1}^\top C^{(tt)} y < \mathbf{1}^\top y,$$

since there exists at least a column to $C^{(tt)}$ with a column sum strictly less than 1. A contradiction is met.

All eigenvalues to $C^{(tt)}$ have magnitude strictly less than 1. Thus, we have proven the statement. ■

2.1.3 Assumptions on Loss Functions

Some assumptions on the loss functions are required to make our problem be analyzable.

Assumption 2.3: Local Loss Functions

For all $t \in [T]$ and $k \in \mathcal{N}^{(t)}$,

- (a) The local risk functions $J_k^{(t)}$ are twice differentiable with respect to all arguments.
 (b) Smoothness: for all $x_1, x_2 \in \mathbb{R}^{M \times 1}$,

$$\left\| \nabla_t J_k^{(t)}(x_1) - \nabla_t J_k^{(t)}(x_2) \right\| \leq \delta^{(t)} \|x_1 - x_2\| \quad (2.8)$$

holds for some $\delta^{(t)} > 0$.

- (c) Hessian is locally Lipschitz continuous about the Nash equilibrium: for any $\tau \neq t$,

$$\left\| \nabla_t^2 J_k^{(t)}(x^* + \Delta x) - \nabla_t^2 J_k^{(t)}(x^*) \right\| \leq \kappa \|\Delta x\|, \quad (2.9)$$

$$\left\| \nabla_{\tau t}^2 J_k^{(t)}(x^* + \Delta x) - \nabla_{\tau t}^2 J_k^{(t)}(x^*) \right\| \leq \kappa \|\Delta x\| \quad (2.10)$$

hold for small $\|\Delta x\| \leq \varepsilon_H$ and $\kappa > 0$.

If stated explicitly, the matrix norms are taken to be the operator norm, and the vector norms are taken to be the Euclidean 2-norm.

Theorem 2.2: Global Lipschitz Smoothness of Hessians

Under Assumption 2.3, the Hessians are actually globally Lipschitz: for any $t, \tau \in [T]$,

$$\left\| \nabla_{\tau t}^2 J_k^{(t)}(x) - \nabla_{\tau t}^2 J_k^{(t)}(x^*) \right\| \leq \kappa' \|x - x^*\| \quad (2.11)$$

with some constant $\kappa' > 0$.

Proof. From the first two conditions, we know that the Hessians are bounded: for all $t, \tau \in [T]$,

$$\left\| \nabla_{\tau t}^2 J_k^{(t)}(x) \right\| \leq \delta^{(t)}.$$

Furthermore, for any $u \in \mathbb{R}^{M_t}$ and any $v \in \mathbb{R}^{M_\tau}$,

$$u^\top \left(\nabla_{\tau t}^2 J_k^{(t)}(x) - \nabla_{\tau t}^2 J_k^{(t)}(x^*) \right) v \leq \delta^{(t)} \|u\| \|v\| - (-\delta^{(t)}) \|u\| \|v\| = 2\delta^{(t)} \|u\| \|v\|.$$

Henceforth,

$$\left\| \nabla_{\tau t}^2 J_k^{(t)}(x) - \nabla_{\tau t}^2 J_k^{(t)}(x^*) \right\| \leq \begin{cases} \kappa \|x - x^*\|, & \|x - x^*\| \leq \varepsilon_H; \\ 2\delta^{(t)} \leq \frac{2\delta^{(t)}}{\varepsilon_H} \|x - x^*\|, & \|x - x^*\| > \varepsilon_H. \end{cases}$$

We can thus choose

$$\kappa' = \max \left\{ \kappa, \frac{2\delta^{(t)}}{\varepsilon_H} \right\}. \quad (2.12)$$

■

Assumption 2.4: Aggregated Second Derivatives

Assume for all $t \in [T]$ and for all $\{x_k\}_{k \in \mathcal{N}^{(t)}}$, there exists $\varepsilon_J > 0$ such that

$$\sum_{k \in \mathcal{N}^{(t)}} p_k^{(t)} \nabla_t^2 J_k^{(t)}(x_k) \succeq \varepsilon_J \mathbb{1}_{M_t}. \quad (2.13)$$

Furthermore, for all $t, \tau \in [T]$, $t \neq \tau$, assume there exists constants $\nu^{(t)} > 0$, $\delta^{(t\tau)} > 0$ such that

$$\nu^{(t)} := \sup_{\{x_k\}_{k \in \mathcal{N}^{(t)}}} \left\| \sum_{k \in \mathcal{N}^{(t)}} p_k^{(t)} \nabla_t^2 J_k^{(t)}(x_k) \right\|, \quad (2.14)$$

$$\delta^{(t\tau)} := \sup_{\{x_k\}_{k \in \mathcal{N}^{(t)}}} \left\| \sum_{k \in \mathcal{N}^{(t)}} p_k^{(t)} \nabla_{\tau t}^2 J_k^{(t)}(x_k) \right\|. \quad (2.15)$$

We denote $\nu_J := \min_t \nu^{(t)}$ and $\delta_J := \max_{t, \tau} \delta^{(t\tau)}$. It can be assumed that the following relation holds:

$$\varepsilon_J > (T - 1)\delta_J. \quad (2.16)$$

It might seem like Equation (2.16) appeared out of nowhere. However, the motivation for introducing this inequality can be seen in the proof to Theorem 2.3 below. Similarly, in that proof, we will see how Equation (2.16) is a sufficient condition to the stability of a matrix, but it is not necessary.

Theorem 2.3: Strong-Monotonicity of Global Gradients

Under Assumption 2.4, it is implied that the global gradient

$$F(x) := \text{col} \left\{ \nabla_t J^{(t)}(x) \right\}_{t \in [T]} \quad (2.17)$$

is ν_R -strongly monotonic with respect to $x := \text{col}\{x^t\}_{t \in [T]}$. I.e.,

$$\langle F(x_1) - F(x_2), x_1 - x_2 \rangle \geq \nu_R \|x_1 - x_2\|^2 \quad (2.18)$$

for any x_1, x_2 , and a constant $\nu_R > 0$. It can be found that

$$\nu_R = \lambda_{\min} \left(\text{sym} \left(\begin{bmatrix} \nabla_1^2 J^{(1)}(x) & \nabla_{21}^2 J^{(1)}(x) & \cdots & \nabla_{T1}^2 J^{(1)}(x) \\ \nabla_{12}^2 J^{(2)}(x) & \nabla_2^2 J^{(2)}(x) & \cdots & \nabla_{T2}^2 J^{(2)}(x) \\ \vdots & \vdots & \ddots & \vdots \\ \nabla_{1T}^2 J^{(T)}(x) & \nabla_{2T}^2 J^{(T)}(x) & \cdots & \nabla_T^2 J^{(T)}(x) \end{bmatrix} \right) \right), \quad (2.19)$$

which, under Equation (2.16), can be lower bounded by

$$\nu_R \geq \varepsilon_J - (T - 1)\delta_J. \quad (2.20)$$

Proof. By the smoothness assumed on the local loss functions, the strong monotonicity of F is equivalently established if we consider the infinitesimal version, showing the positivity of the quadratic form generated by $\nabla_x F(x)$. The quadratic form generated by $\nabla_x F(x)$ is its symmetrized version

$$S(x) := \text{sym}(\nabla_x F(x)) = \frac{1}{2} \left[\nabla_x F(x) + (\nabla_x F(x))^T \right]$$

for all x . The matrix form of S is partitioned into blocks where the (tt') block is of size $M_t \times M_{t'}$ (with $t, t' \in [T]$). The diagonal block at the t th row is of the form

$$[S]_{tt} = \nabla_t^2 J^{(t)}(x) \succeq \varepsilon_J \mathbb{1}_{M_t}.$$

The off-diagonal block at position (tt') is of the form

$$[S]_{tt'} = \frac{1}{2} \left(\nabla_{t't}^2 J^{(t)}(x) + \nabla_{t't}^2 J^{(t')}(x) \right),$$

and can be bounded in spectral norm by

$$\|[S]_{tt'}\| = \left\| \frac{1}{2} \left(\nabla_{t't}^2 J^{(t)}(x) + \nabla_{t't}^2 J^{(t')}(x) \right) \right\| \leq \delta_J.$$

For any vector $v = [v_1; \dots; v_T] \in \mathbb{R}^M$, where $v_t \in \mathbb{R}^{M_t}$, a lower bound to the eigenvalue of S can be established by

$$\begin{aligned} \nu_R := \lambda_{\min}(S) &= \min_{\|v\|=1} v^\top S v = \sum_{t,t' \in [T]} v_t^\top [S]_{tt'} v_{t'} \\ &= \sum_{t \in [T]} v_t^\top [S]_{tt} v_t + \sum_{t \neq t' \in [T]} v_t^\top [S]_{tt'} v_{t'} \\ &\geq \sum_{t \in [T]} \varepsilon_J \|v_t\|^2 - \sum_{t \neq t' \in [T]} \delta_J \|v_t\| \|v_{t'}\| \\ &= \varepsilon_J - \frac{1}{2} \delta_J \sum_{t \neq t' \in [T]} 2 \|v_t\| \|v_{t'}\| \\ &\geq \varepsilon_J - \frac{1}{2} \delta_J \sum_{t \neq t' \in [T]} (\|v_t\|^2 + \|v_{t'}\|^2) \\ &= \varepsilon_J - (T-1) \delta_J. \end{aligned}$$

Thus, $\nu_R \geq \varepsilon_J - (T-1) \delta_J > 0$, with its positivity guaranteed by [Equation \(2.16\)](#).

Since $v^\top S(x) v > \nu_R$, we also have $v^\top \nabla_x F(x) v > \nu_R$. ■

With the same notation as in the proof to [Theorem 2.3](#), although $S(x)$ is symmetric and has real eigenvalues, $\nabla_x F(x)$ is not. Although since it is a real matrix, the complex eigenvalues come in conjugate pairs, and the imaginary parts cancel each other out when computing the quadratic form.

However, we can state more about its eigenvalue:

Theorem 2.4: Spectrum of $\nabla_x F(x)$

Under [Assumption 2.4](#) and [Theorem 2.3](#), the eigenvalues to $\nabla_x F(x)$ is within the region:

$$\{ \lambda \in \mathbb{C} \mid \nu_R \leq \operatorname{Re}\{\lambda\} \leq \nu_S, |\operatorname{Im}\{\lambda\}| \leq \nu_I \}, \quad (2.21)$$

where ν_S and ν_I are constants bounded by

$$\nu_S \leq \nu_J + (T-1) \delta_J, \quad (2.22)$$

$$\nu_I \leq (T-1) \delta_J. \quad (2.23)$$

Proof. First, the bound on the real parts. Let an eigenvalue to $\nabla_x F(x)$ be $\lambda \in \mathbb{C}$ with an associated eigenvector $v \in \mathbb{C}^M$ satisfying $\|v\| = 1$,

$$\operatorname{Re}\{\lambda\} = \operatorname{Re} \left\{ v^H \nabla_x F(x) v \right\} = \frac{v^H \left(\nabla_x F(x) + (\nabla_x F(x))^T \right) v}{2} = v^H S(x) v \geq \nu_R.$$

Thus, a lower bound on the real parts of the eigenvalues to $\nabla_x F(x)$ is found. Further,

$$\nu_S := \max \operatorname{Re}\{\lambda\} = \max_{\|v\|=1} v^H S(x) v \leq \nu_J + (T-1)\delta_J.$$

Next, the imaginary parts. The imaginary parts to the eigenvalues of $\nabla_x F(x)$ originated from its skew-symmetric part

$$A(x) := \operatorname{skew}(\nabla_x F(x)) = \frac{1}{2} [\nabla_x F(x) - (\nabla_x F(x))^T]$$

for all x . For any eigenpair $\nabla_x F(x)v = \lambda v$ with $\|v\| = 1$,

$$\operatorname{Im}\{\lambda\} = v^H A(x) v \Rightarrow |\operatorname{Im}\{\lambda\}| \leq \|A(x)\|.$$

Note that for $t \neq t'$, i.e., the off-diagonal blocks, we have

$$\|[A]_{tt'}\| = \left\| \frac{1}{2} \left(\nabla_{t't}^2 J^{(t)}(x) - \nabla_{tt'}^2 J^{(t')}(x) \right) \right\| \leq \delta_J.$$

We can finally bound the spectral norm to A by its *block-1-norm* and *block- ∞ -norm*:

$$\nu_I := \max |\operatorname{Im}\{\lambda\}| \leq \|A\| \leq (T-1)\delta_J.$$

Thus it is proven. ■

Lemma 2.5: Existence and Uniqueness of Nash Equilibrium

Under [Assumption 2.4](#), the global gradient is strongly monotonic by [Theorem 2.3](#), then there exists a unique equilibrium for the game in [Equation \(2.1\)](#).

Proof. In [\[Thm 2.3.3\]](#) of [\[Fra03\]](#). ■

2.1.4 Assumptions on Gradient Noise

Let us define the gradient noise of agent $k \in \mathcal{N}^{(t)}$ on the strategy x^τ as

$$s_{k,i-1}^\tau = \widehat{g}_{k,i-1}^\tau - g_{k,i-1}^\tau = \widehat{\nabla}_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}) - \nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}). \quad (2.24)$$

Some assumptions are made on the gradient noise:

Assumption 2.5: Gradient Noise Moment

Define $\mathcal{F}_{i-1} = \{x_{k,j} \mid k \in \mathcal{N}, 1 \leq j \leq i-1\}$ as the filtration of all the past strategies and estimates made before the i th iteration. Then for all $k, \ell \in \mathcal{N}$, for all $t, \tau \in [T]$, and all i ,

$$\mathbb{E} \left[s_{k,i}^t \mid \mathcal{F}_{i-1} \right] = 0, \quad (2.25)$$

$$\mathbb{E} \left[s_{k,i}^t s_{\ell,w,i}^\tau \mid \mathcal{F}_{i-1} \right] = 0 \quad \forall k \neq \ell \text{ or } t \neq \tau, \quad (2.26)$$

$$\mathbb{E} \left[\|s_{k,i}^t\|^4 \mid \mathcal{F}_{i-1} \right] \leq \left(\overline{\beta}_k^t \right)^4 \|x_{k,i-1}\|^4 + \left(\overline{\sigma}_k^t \right)^4. \quad (2.27)$$

It should be noted that $s_{k,i}^t$ is a function on $x_{k,i-1}$, hence the dependency of the inequality on it.

Theorem 2.6: Bounded Second Moment of Gradient Noise

Under [Assumption 2.5](#), the following holds for all $t \in [T]$ and all $k \in \mathcal{N}$:

$$\mathbb{E} \left[\left\| s_{k,i}^t \right\|^2 \middle| \mathcal{F}_{i-1} \right] \leq \left(\bar{\beta}_k^t \right)^2 \|x_{k,i-1}\|^2 + \left(\bar{\sigma}_k^t \right)^2. \quad (2.28)$$

Proof. Apply Jensen's inequality on the function $\|\cdot\|^2$ and further note that for any positive a and b , we have $a^2 + b^2 \leq (a + b)^2$. ■

Theorem 2.7: Bounded Moments Centered about Nash Equilibrium

Under [Assumption 2.5](#), the following equations also hold:

$$\mathbb{E} \left[\left\| s_{k,i}^t \right\|^4 \middle| \mathcal{F}_{i-1} \right] \leq \left(\beta_k^t \right)^4 \|x_{k,i-1} - x^*\|^4 + \left(\sigma_k^t \right)^4, \quad (2.29)$$

$$\mathbb{E} \left[\left\| s_{k,i}^t \right\|^2 \middle| \mathcal{F}_{i-1} \right] \leq \left(\beta_k^t \right)^2 \|x_{k,i-1} - x^*\|^2 + \left(\sigma_k^t \right)^2. \quad (2.30)$$

for some constants β_k^t and σ_k^t .

Proof. We will only proof the first equation, the second can be derived similarly. Since for any vectors a and b , we can apply Jensen's inequality on $\|\cdot\|^4$ to obtain

$$\|a + b\|^4 \leq 8\|a\|^4 + 8\|b\|^4,$$

we have

$$\begin{aligned} \left(\bar{\beta}_k^t \right)^4 \|x_{k,i-1}\|^4 + \left(\bar{\sigma}_k^t \right)^4 &= \left(\bar{\beta}_k^t \right)^4 \|x_{k,i-1} - x^* - x^*\|^4 + \left(\bar{\sigma}_k^t \right)^4 \\ &\leq \underbrace{8 \left(\bar{\beta}_k^t \right)^4 \|x_{k,i-1} - x^*\|^4}_{=:(\beta_k^t)^4} + \underbrace{8 \left(\bar{\beta}_k^t \right)^4 \|x^*\|^4 + \left(\bar{\sigma}_k^t \right)^4}_{=:(\sigma_k^t)^4}. \end{aligned}$$

Thus it is proven. ■

Assumption 2.6: Local Smoothness of Gradient Noise Covariance

Define the covariance at the Nash equilibrium x^* associated with the gradient noise $s_{k,i}^t$ to be

$$G_k^t := \lim_{i \rightarrow \infty} \mathbb{E} \left[s_{k,i}^t(x^*) s_{k,i}^t(x^*)^\top \middle| \mathcal{F}_{i-1} \right]. \quad (2.31)$$

I assume the gradient noises satisfy

$$\left\| \lim_{i \rightarrow \infty} \mathbb{E} \left[s_{k,i}^t(x^* + \Delta x) s_{k,i}^t(x^* + \Delta x)^\top \middle| \mathcal{F}_{i-1} \right] - G_k^t \right\| \leq L \|\Delta x\|^\gamma \quad (2.32)$$

for some $\gamma \in (0, 4]$, positive constant $L > 0$, and small perturbations $\|\Delta x\| \leq \varepsilon_R$.

Theorem 2.8: Global Smoothness of Gradient Noise Covariance

Under [Assumption 2.5](#) and [Assumption 2.6](#), the gradient noise covariances globally satisfy the smoothness relation:

$$\left\| \lim_{i \rightarrow \infty} \mathbb{E} \left[s_{k,i}^t(x) s_{k,i}^t(x)^\top \mid \mathcal{F}_{i-1} \right] - G_k^t \right\| \leq \max \left\{ L \|x - x^*\|^\gamma, L' \|x - x^*\|^2 \right\}, \quad (2.33)$$

where $L' > 0$ is some constant.

Proof. Consider any $x \in \mathcal{F}_{i-1}$ such that $\|x - x^*\| > \varepsilon_R$. Then

$$\left\| \mathbb{E} \left[s_{k,i}^t(x) s_{k,i}^t(x)^\top \mid \mathcal{F}_{i-1} \right] - G_k^t \right\| \leq \left\| \mathbb{E} \left[s_{k,i}^t(x) s_{k,i}^t(x)^\top \mid \mathcal{F}_{i-1} \right] \right\| + \|G_k^t\|.$$

Note that since $s_{k,i}^t(x) s_{k,i}^t(x)^\top$ is positive semi-definite,

$$\begin{aligned} \left\| \mathbb{E} \left[s_{k,i}^t(x) s_{k,i}^t(x)^\top \mid \mathcal{F}_{i-1} \right] \right\| &\leq \text{Tr} \left\{ \mathbb{E} \left[s_{k,i}^t(x) s_{k,i}^t(x)^\top \mid \mathcal{F}_{i-1} \right] \right\} \\ &= \mathbb{E} \left[\left\| s_{k,i}^t(x) \right\|^2 \mid \mathcal{F}_{i-1} \right] \\ &\leq \left(\beta_k^t \right)^2 \|x - x^*\|^2 + \left(\sigma_k^t \right)^2. \end{aligned}$$

Hence,

$$\begin{aligned} \left\| \mathbb{E} \left[s_{k,i}^t(x) s_{k,i}^t(x)^\top \mid \mathcal{F}_{i-1} \right] - G_k^t \right\| &\leq \left(\beta_k^t \right)^2 \|x - x^*\|^2 + 2 \left(\sigma_k^t \right)^2 \\ &\leq \left(\beta_k^t \right)^2 \|x - x^*\|^2 + \frac{2 \left(\sigma_k^t \right)^2}{\varepsilon_R^2} \|x - x^*\|^2 \\ &= \underbrace{\left(\left(\beta_k^t \right)^2 + \frac{2 \left(\sigma_k^t \right)^2}{\varepsilon_R^2} \right)}_{=: L'_{k,t}} \|x - x^*\|^2. \end{aligned}$$

Define $L' = \max_{k,t} L'_{k,t}$, we thus obtain the statement. ■

Definition 2.1

Define

$$R_k^t := \mathbb{E} \left[G_k^t \right] = \mathbb{E} \left[s_{k,i}^t(x^*) s_{k,i}^t(x^*)^\top \right]. \quad (2.34)$$

2.2 Second Moment Stability of ATC-ITC

Let us analyze the second moment stability of the ATC-ITC algorithm. Let us rewrite the ATC-ITC algorithm in a form that will make our derivations clean.

Collect all the strategies and all the estimates by all the agents in a single column vector and denote it by $\underline{X}$ in the follow ordering

$$\underline{X} := \text{col} \left\{ \text{col} \left\{ \text{col} \left\{ x_k^\tau \right\}_{k \in \mathcal{N}^{(t)}} \right\}_{t \in [T]} \right\}_{\tau \in [T]}. \quad (2.35)$$

We shall call this vector the state vector. In index notation, it can also be represented as

$$[\underline{X}]_{\tau t} = \text{col} \left\{ x_k^\tau \right\}_{k \in \mathcal{N}^{(t)}} \quad (\forall \tau, t \in [T]). \quad (2.36)$$

For example, consider $T = 2$ and $K_1 = K_2 = 2$. The agents with index 1 and 2 are in team 1, while those with index 3 and 4 are in team 2, then

$$\underline{X} = \left[\underbrace{x_1^1; x_2^1; x_3^1; x_4^1}_{KM_1}; \underbrace{x_1^2; x_2^2; x_3^2; x_4^2}_{KM_2} \right] \in \mathbb{R}^{KM}.$$

Definition 2.2

The evolution of state vector $\underline{X}_i$ follows the recursion relation

$$\underline{X}_i = \mathcal{CA} \left(\underline{X}_{i-1} - \mu \widehat{\underline{G}}_i \right) = \mathcal{CA} \left(\underline{X}_{i-1} - \mu \underline{G}_i \right) - \mu \mathcal{CA} \underline{S}_i. \quad (2.37)$$

The terms in the recursion above are defined as: for $\tau, \tau', t, t' \in [T]$,

$$[\mathcal{A}]_{\tau\tau'tt'} := \begin{cases} A^{(\tau)\top} \otimes \mathbb{1}_{M_\tau}, & \tau = \tau' = t = t'; \\ \mathbb{1}_{K_t} \otimes \mathbb{1}_{M_\tau}, & \tau = \tau' \neq t = t'; \\ 0 \cdot \mathbf{1}_{K_t \times K_{t'}} \otimes \mathbf{1}_{M_\tau \times M_{\tau'}}, & \text{otherwise,} \end{cases} \quad (2.38)$$

$$[\mathcal{C}]_{\tau\tau'tt'} := \begin{cases} \mathbb{1}_{K_\tau} \otimes \mathbb{1}_{M_\tau}, & \tau = \tau' = t = t'; \\ C^{(tt)\top} \otimes \mathbb{1}_{M_\tau}, & \tau = \tau' \neq t = t'; \\ C^{(t't)\top} \otimes \mathbb{1}_{M_\tau}, & \tau = \tau' = t' \neq t; \\ 0 \cdot \mathbf{1}_{K_t \times K_{t'}} \otimes \mathbf{1}_{M_\tau \times M_{\tau'}}, & \text{otherwise,} \end{cases} \quad (2.39)$$

$$[\underline{G}_i]_{\tau t} := \text{col} \left\{ g_{k,i-1}^\tau \right\}_{k \in \mathcal{N}^{(t)}}, \quad (2.40)$$

$$[\widehat{\underline{G}}_i]_{\tau t} := \text{col} \left\{ \widehat{g}_{k,i-1}^\tau \right\}_{k \in \mathcal{N}^{(t)}}, \quad (2.41)$$

$$\underline{S}_i := \widehat{\underline{G}}_i - \underline{G}_i. \quad (2.42)$$

We allow the abuse of notation

$$g_{k,i-1}^\tau = \nabla_\tau J_k^{(\tau)}(x_{k,i-1}) \quad (2.43)$$

if $k \in \mathcal{N}^{(t)}$, else it follows [Assumption 2.1](#).

This tensor-indexing notation will be proven to be easier for us to manipulate around in some of the proofs. Note that the terms of the form $0 \cdot \mathbf{1}$ can be abbreviated to just 0. But for the sake of clarity throughout the following proofs, the size of the zero matrices are written out explicitly.

The definitions to $\mathcal{A}$ and $\mathcal{C}$ seems a bit complicated, they are separated into $T \times T$ blocks of different sizes, with the (τ, τ') th block being a matrix of size $K M_\tau \times K M_{\tau'}$. Here is an example to clear things up: consider $T = 3$, then

$$\mathcal{CA} = \begin{bmatrix} \mathcal{A}^{(1)} & & & & & \\ \mathcal{C}^{(12)} \mathcal{A}^{(1)} & \mathcal{C}_1^{(22)} & & & & \\ \mathcal{C}^{(13)} \mathcal{A}^{(1)} & & \mathcal{C}_1^{(33)} & & & \\ \hline & & & \mathcal{C}_2^{(11)} & \mathcal{C}^{(21)} \mathcal{A}^{(2)} & \\ & & & & \mathcal{A}^{(2)} & \\ & & & & \mathcal{C}^{(23)} \mathcal{A}^{(2)} & \mathcal{C}_2^{(33)} \\ \hline & & & & & \mathcal{C}_3^{(11)} & \mathcal{C}^{(21)} \mathcal{A}^{(3)} \\ & & & & & & \mathcal{C}_3^{(22)} & \mathcal{C}^{(32)} \mathcal{A}^{(3)} \\ & & & & & & & \mathcal{A}^{(3)} \end{bmatrix},$$

where (only here) I abbreviate $A^{(\tau)\top} \otimes \mathbb{1}_{M_\tau} =: \mathcal{A}^{(\tau)}$, $C^{(t't)\top} \otimes \mathbb{1}_{M_{t'}} =: \mathcal{C}^{(t't)}$, and $C^{(tt)\top} \otimes \mathbb{1}_{M_t} =: \mathcal{C}_t^{(tt)}$. The teal indices represent the $(\tau\tau'tt')$ indices to each block matrix as in Definition 2.2.

Theorem 2.9: Eigenvectors of $\mathcal{CA}$

Under Assumption 2.2, consider the unique Perron vector of $A^{(\tau')}$ as $p^{(\tau')}$. For all $\tau' \in [T]$, the matrices $\mathbf{p}^{(\tau')}$ and $\mathbf{q}^{(\tau')}$ of size $\mathbb{R}^{KM \times M_{\tau'}}$ defined as

$$[\mathbf{p}^{(\tau')}]_{\tau t} := \begin{cases} p^{(\tau')} \otimes \mathbb{1}_{M_{\tau'}}, & \tau = t = \tau'; \\ 0 \cdot \mathbf{1}_{K_t} \otimes \mathbf{1}_{M_\tau \times M_{\tau'}}, & \text{otherwise,} \end{cases} \quad (2.44)$$

$$[\mathbf{q}^{(\tau')}]_{\tau t} := \begin{cases} \mathbf{1}_{K_t} \otimes \mathbb{1}_{M_{\tau'}}, & \tau = \tau'; \\ 0 \cdot \mathbf{1}_{K_t} \otimes \mathbf{1}_{M_\tau \times M_{\tau'}}, & \text{otherwise} \end{cases} \quad (2.45)$$

satisfies

$$\mathbf{p}^{(\tau')\top} \mathcal{CA} = \mathbf{p}^{(\tau')\top}, \quad \mathcal{CA} \mathbf{q}^{(\tau')} = \mathbf{q}^{(\tau')}. \quad (2.46)$$

Proof. The proof is by brute force calculation and careful index manipulations:

$$\begin{aligned} [\mathbf{p}^{(\tau')\top} \mathcal{CA}]_{\tau t} &= \sum_{\tau'', t'' \in [T]} [\mathbf{p}^{(\tau')}]_{t'' \tau''}^\top [\mathcal{CA}]_{\tau'' \tau' t'' t} \\ &= \sum_{\tau'', t'' \in [T]} \left\{ p^{(\tau')} \otimes \mathbb{1}_{M_{\tau'}}, \quad t'' = \tau'' = \tau'; \right. \\ &\quad \left. 0 \cdot \mathbf{1}_{K_{t''}} \otimes \mathbf{1}_{M_{\tau'} \times M_{\tau''}}, \quad \text{otherwise} \right\} [\mathcal{CA}]_{\tau'' \tau' t'' t} \\ &= (p^{(\tau')\top} \otimes \mathbb{1}_{M_{\tau'}}) [\mathcal{CA}]_{\tau' \tau' t} = (p^{(\tau')\top} \otimes \mathbb{1}_{M_{\tau'}}) \sum_{\tau'', t'' \in [T]} [\mathcal{C}]_{\tau' \tau'' \tau' t''} [\mathcal{A}]_{\tau'' \tau' t'' t} \\ &= \sum_{\tau'', t'' \in [T]} (p^{(\tau')\top} \otimes \mathbb{1}_{M_{\tau'}}) \left\{ \mathbb{1}_{K_{\tau'}} \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau' = \tau'' = t''; \right. \\ &\quad \left. 0 \cdot \mathbf{1}_{K_{\tau'} \times K_{t''}} \otimes \mathbf{1}_{M_{\tau'} \times M_{\tau''}}, \quad \text{otherwise} \right\} [\mathcal{A}]_{\tau'' \tau' t'' t} \end{aligned}$$

$$\begin{aligned}
&= \sum_{\tau'', t'' \in [T]} \left\{ p^{(\tau')}^\top \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau' = \tau'' = t''; \right. \\
&\quad \left. 0 \cdot \mathbf{1}_{K_{t''}}^\top \otimes \mathbf{1}_{M_{\tau'} \times M_{\tau''}}, \quad \text{otherwise} \right\} \cdots \\
&\quad \cdot \left\{ A^{(\tau)}^\top \otimes \mathbb{1}_{M_\tau}, \quad \tau'' = \tau = t'' = t; \right. \\
&\quad \left. \mathbb{1}_{K_{t''}} \otimes \mathbb{1}_{M_{\tau''}}, \quad \tau'' = \tau \neq t'' = t; \right. \\
&\quad \left. 0 \cdot \mathbf{1}_{K_{t''} \times K_t} \otimes \mathbf{1}_{M_{\tau''} \times M_\tau}, \quad \text{otherwise} \right\} \\
&= \left\{ p^{(\tau')}^\top A^{(\tau')}^\top \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau = t = \tau'; \right. \\
&\quad \left. 0 \cdot \mathbf{1}_{K_t}^\top \otimes \mathbf{1}_{M_{\tau'} \times M_\tau}, \quad \text{otherwise} \right\} = \mathbf{p}^{(\tau')^\top},
\end{aligned}$$

where the last equality is by the fact that $A^{(\tau')} p^{(\tau')} = p^{(\tau')}$. Similarly,

$$\begin{aligned}
[\mathcal{CA}q^{(\tau')}]_{\tau t} &= \sum_{\bar{\tau}, \bar{t} \in [T]} [\mathcal{CA}]_{\tau \bar{\tau} \bar{t} \bar{t}} [q^{(\tau')}]_{\bar{\tau} \bar{t}} \\
&= \sum_{\bar{\tau}, \bar{t} \in [T]} [\mathcal{CA}]_{\tau \bar{\tau} \bar{t} \bar{t}} \left\{ \mathbf{1}_{K_{\bar{t}}} \otimes \mathbb{1}_{M_{\tau'}}, \quad \bar{\tau} = \tau'; \right. \\
&\quad \left. 0 \cdot \mathbf{1}_{K_{\bar{t}}} \otimes \mathbf{1}_{M_{\bar{\tau}} \times M_{\tau'}}, \quad \text{otherwise} \right\} \\
&= \sum_{\bar{t} \in [T]} [\mathcal{CA}]_{\tau \tau' \bar{t} \bar{t}} (\mathbf{1}_{K_{\bar{t}}} \otimes \mathbb{1}_{M_{\tau'}}) = \sum_{\bar{t}, \tau'', t'' \in [T]} [\mathcal{C}]_{\tau \tau'' t'' \bar{t}} [\mathcal{A}]_{\tau'' \tau' t'' \bar{t}} (\mathbf{1}_{K_{\bar{t}}} \otimes \mathbb{1}_{M_{\tau'}}) \\
&= \sum_{\bar{t}, \tau'', t'' \in [T]} [\mathcal{C}]_{\tau \tau'' t'' \bar{t}} \left\{ A^{(\tau')}^\top \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau'' = \tau' = t'' = \bar{t}; \right. \\
&\quad \left. \mathbb{1}_{K_{t''}} \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau'' = \tau' \neq t'' = \bar{t}; \right. \\
&\quad \left. 0 \cdot \mathbf{1}_{K_{t''} \times K_{\bar{t}}} \otimes \mathbf{1}_{M_{\tau''} \times M_{\tau'}}, \quad \text{otherwise} \right\} (\mathbf{1}_{K_{\bar{t}}} \otimes \mathbb{1}_{M_{\tau'}}) \\
&= \sum_{\tau'', t'' \in [T]} \left\{ \mathbb{1}_{K_\tau} \otimes \mathbb{1}_{M_\tau}, \quad \tau = \tau'' = t = t''; \right. \\
&\quad \left. C^{(tt)}^\top \otimes \mathbb{1}_{M_\tau}, \quad \tau = \tau'' \neq t = t''; \right. \\
&\quad \left. C^{(t''t)}^\top \otimes \mathbb{1}_{M_\tau}, \quad \tau = \tau'' = t'' \neq t; \right. \\
&\quad \left. 0 \cdot \mathbf{1}_{K_t \times K_{t''}} \otimes \mathbf{1}_{M_\tau \times M_{\tau''}}, \quad \text{otherwise}, \right\} \cdots \\
&\quad \cdot \left\{ \mathbf{1}_{K_{t''}} \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau'' = \tau'; \right. \\
&\quad \left. 0 \cdot \mathbf{1}_{K_{t''}} \otimes \mathbf{1}_{M_{\tau''} \times M_{\tau'}}, \quad \text{otherwise} \right\} \\
&= \sum_{\tau'', t'' \in [T]} \left\{ \mathbf{1}_{K_{\tau'}} \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau = \tau'' = t = t'' = \tau'; \right. \\
&\quad \left. C^{(tt)}^\top \mathbf{1}_{K_t} \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau = \tau'' = \tau' \neq t = t''; \right. \\
&\quad \left. C^{(t''t)}^\top \mathbf{1}_{K_{t''}} \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau = \tau'' = t'' = \tau' \neq t; \right. \\
&\quad \left. 0 \cdot \mathbf{1}_{K_t} \otimes \mathbf{1}_{M_\tau \times M_{\tau'}}, \quad \text{otherwise}, \right\} \\
&= \left\{ \mathbf{1}_{K_{\tau'}} \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau = t = \tau'; \right. \\
&\quad \left. \left(C^{(tt)}^\top \mathbf{1}_{K_t} + C^{(\tau't)}^\top \mathbf{1}_{K_{\tau'}} \right) \otimes \mathbb{1}_{M_{\tau'}}, \quad \tau = \tau' \neq t; \right. \\
&\quad \left. 0 \cdot \mathbf{1}_{K_t} \otimes \mathbf{1}_{M_\tau \times M_{\tau'}}, \quad \text{otherwise}, \right\} = [q^{(\tau')}]_{\tau t}.
\end{aligned}$$

The sixth and the last equality utilize the fact that $A^{(\tau')}$ and $[C^{(\tau't)}; C^{(tt)}]$ are left-stochastic. ■

2.2.1 Representing Gradients by Hessians

Next, we re-express the gradients by the state vector multiplied with mixed Hessians. This is a nice trick that allows us to translate the analysis over Hessians to the analysis of the state multiplied by a

bounded matrix. Note that all the possible gradients are of the form

$$g_{k,i-1}^\tau = \nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}),$$

denoting the gradient used to update x^t (either be strategy or estimation) by agent $k \in \mathcal{N}^{(\tau)}$ using information at iteration $i-1$, where $\ell_k \in \mathcal{N}^{(t)}$ is $\ell_k = k$ if $t = \tau$ or else it follows the correspondence established in [Assumption 2.1](#).

Theorem 2.10

All the gradients can be written in the form of

$$\nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}) = \nabla_\tau J_{\ell_k}^{(\tau)}(x^*) - \sum_{t \in [T]} H_{k,\tau t,i-1} \tilde{x}_{k,i-1}^t, \quad (2.47)$$

where $\tilde{x}_{k,i-1}^t := x^{t,*} - x_{k,i-1}^t$ is the error vector between the Nash equilibrium and agent k 's estimate of the strategy of team t , and $H_{k,\tau t,i-1}$ is an integral of $\nabla_{t\tau}^2 J_{\ell_k}^{(\tau)}$.

Proof. The proof is direct by applying the fundamental theorem of calculus on $x_{k,i-1}^1$ to $x_{k,i-1}^T$. The order in which you apply the fundamental theorem of calculus will change the value to the matrices $H_{k,\tau t,i-1}$. However, this does not affect the following derivations and results. ■

Here, a simple example to the case of $T = 3$ is demonstrated, one can generalize this to arbitrary T 's:

$$\begin{aligned} & \nabla_\tau J_{\ell_k}^{(\tau)}(x^{1,*}, x^{2,*}, x^{3,*}) - \nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}) \dots \\ &= \left(\nabla_\tau J_{\ell_k}^{(\tau)}(x^{1,*}, x^{2,*}, x^{3,*}) - \nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}^1, x^{2,*}, x^{3,*}) \right) \dots \\ & \quad + \left(\nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}^1, x^{2,*}, x^{3,*}) - \nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}^1, x_{k,i-1}^2, x^{3,*}) \right) \dots \\ & \quad + \left(\nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}^1, x_{k,i-1}^2, x^{3,*}) - \nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}^1, x_{k,i-1}^2, x_{k,i-1}^3) \right) \\ &= \underbrace{\int_0^1 \nabla_{1,\tau}^2 J_{\ell_k}^{(\tau)}(x^{1,*} - t\tilde{x}_{k,i-1}^1, x^{2,*}, x^{3,*}) dt \tilde{x}_{k,i-1}^1}_{H_{k,\tau 1,i-1}} \dots \\ & \quad + \underbrace{\int_0^1 \nabla_{2\tau}^2 J_{\ell_k}^{(\tau)}(x_{k,i-1}^1, x^{2,*} - t\tilde{x}_{k,i-1}^2, x^{3,*}) dt \tilde{x}_{k,i-1}^2}_{H_{k,\tau 2,i-1}} \dots \\ & \quad + \underbrace{\int_0^1 \nabla_{3\tau}^2 J_{\ell_k}^{(\tau)}(x_{k,i-1}^1, x_{k,i-1}^2, x^{3,*} - t\tilde{x}_{k,i-1}^3) dt \tilde{x}_{k,i-1}^3}_{H_{k,\tau 3,i-1}}. \end{aligned}$$

Another possible choice of applying the fundamental theorem of calculus is

$$\nabla_\tau J_{\ell_k}^{(\tau)}(x^*) - \nabla_\tau J_{\ell_k}^{(\tau)}(x_{k,i-1}) = \underbrace{\int_0^1 \nabla \nabla_\tau J_{\ell_k}^{(\tau)}(x^* - t\tilde{x}_{k,i-1}) dt}_{\begin{bmatrix} H_{k,\tau 1,i-1} & H_{k,\tau 2,i-1} & H_{k,\tau 3,i-1} \end{bmatrix}} \underbrace{\tilde{x}_{k,i-1}}_{\text{col}\{\tilde{x}_{k,i-1}^t\}_{t \in [3]}}.$$

For the simplicity of our discussion, one can assume that I used the scheme in the first example, as it makes the discussion of [Theorem 2.16](#) appearing later simpler.

With these mixed Hessians $H_{k,\tau t,i-1}$ that translates $\tilde{x}_{k,i-1}^t$ (the error vector between Nash equilibrium and the estimate of agent k on the strategy of team t) to $\nabla_{\tau} J_{\ell_k}^{(\tau)}(x_{k,i-1})$ (the gradient vector of agent k on the estimate of team t), we can re-write the iteration again into its final form that will be of most use.

Theorem 2.11: Error State Recursion

Under [Assumption 2.4](#), define the unique Nash equilibrium vector $\underline{X}^*$ as

$$[\underline{X}^*]_{\tau t} := \text{col} \{x^{\tau,*}\}_{k \in \mathcal{N}(t)} \quad (2.48)$$

Then the error vector $\tilde{\underline{X}}_i := \underline{X}^* - \underline{X}_i$ evolves according to the following recursion relation:

$$\tilde{\underline{X}}_i = \mathcal{CA} \left(\tilde{\underline{X}}_{i-1} + \mu \underline{G}_i \right) + \mu \mathcal{CAS}_i \quad (2.49)$$

$$= \mathcal{CA} (\mathbb{I}_{KM} - \mu \mathcal{H}_{i-1}) \tilde{\underline{X}}_{i-1} + \mu \mathcal{CAG}^* + \mu \mathcal{CAS}_i, \quad (2.50)$$

where

$$[\underline{G}^*]_{\tau t} := \text{col} \{g_k^{\tau,*}\}_{k \in \mathcal{N}(t)} = \text{col} \left\{ \nabla_{\tau} J_{\ell_k}^{(\tau)}(x^*) \right\}_{k \in \mathcal{N}(t)}, \quad (2.51)$$

$$[\mathcal{H}_{i-1}]_{\tau\tau'tt'} := \begin{cases} \mathcal{H}_{\tau\tau',i-1}^{(t)}, & t = t'; \\ 0 \cdot \mathbf{1}_{K_t \times K_{t'}} \otimes \mathbf{1}_{M_{\tau} \times M_{\tau'}}, & \text{otherwise,} \end{cases} \quad (2.52)$$

$$\mathcal{H}_{\tau\tau',i-1}^{(t)} := \text{blkdiag} \{H_{k,\tau\tau',i-1}\}_{k \in \mathcal{N}(t)}. \quad (2.53)$$

Proof. First, let us show that [Equation \(2.49\)](#) holds. Note that the Nash equilibrium can be written as

$$[\underline{X}^*]_{\tau t} = \text{col} \{x^{\tau,*}\}_{k \in \mathcal{N}(t)} = \mathbf{1}_{K_t} \otimes x^{\tau,*} = [\mathbf{q}^{(\tau)}]_{\tau t} \cdot x^{\tau,*} = \sum_{\tau' \in [T]} [\mathbf{q}^{(\tau')}]_{\tau t} \cdot x^{\tau',*}.$$

Therefore, we have:

$$\begin{aligned} [\mathcal{CA}\underline{X}^*]_{\tau t} &= \sum_{\tau',t' \in [T]} [\mathcal{CA}]_{\tau\tau'tt'} [\underline{X}^*]_{\tau't'} = \sum_{\tau',t' \in [T]} [\mathcal{CA}]_{\tau\tau'tt'} \sum_{\bar{\tau} \in [T]} [\mathbf{q}^{(\bar{\tau})}]_{\tau't'} \cdot x^{\bar{\tau},*} \\ &= \sum_{\bar{\tau} \in [T]} \left(\sum_{\tau',t' \in [T]} [\mathcal{CA}]_{\tau\tau'tt'} [\mathbf{q}^{(\bar{\tau})}]_{\tau't'} \right) \cdot x^{\bar{\tau},*} = \sum_{\bar{\tau} \in [T]} [\mathbf{q}^{(\bar{\tau})}]_{\tau t} \cdot x^{\bar{\tau},*} = [\underline{X}^*]_{\tau t}. \end{aligned}$$

The second to last equality is by [Theorem 2.9](#). With $\mathcal{CA}\underline{X}^* = \underline{X}^*$, [Equation \(2.49\)](#) is easily proven. Next, let us show that [Equation \(2.50\)](#) holds. Since

$$\begin{aligned} [\underline{G}_i]_{\tau t} &= \text{col} \left\{ \nabla_{\tau} J_{\ell_k}^{(\tau)}(x_{k,i-1}) \right\}_{k \in \mathcal{N}(t)} \\ &= \text{col} \left\{ \nabla_{\tau} J_{\ell_k}^{(\tau)}(x^*) \right\}_{k \in \mathcal{N}(t)} - \text{col} \left\{ \sum_{\tau' \in [T]} H_{k,\tau\tau',i-1} \tilde{x}_{k,i-1}^{\tau'} \right\}_{k \in \mathcal{N}(t)} \\ &= [\underline{G}^*]_{\tau t} - \sum_{\tau' \in [T]} \text{col} \left\{ H_{k,\tau\tau',i-1} \tilde{x}_{k,i-1}^{\tau'} \right\}_{k \in \mathcal{N}(t)} \\ &= [\underline{G}^*]_{\tau t} - \sum_{\tau' \in [T]} \text{blkdiag} \{H_{k,\tau\tau',i-1}\}_{k \in \mathcal{N}(t)} \text{col} \left\{ \tilde{x}_{k,i-1}^{\tau'} \right\}_{k \in \mathcal{N}(t)} \end{aligned}$$

$$= [\underline{G}^*]_{\tau t} - \sum_{\tau' \in [T]} [\mathcal{H}_{i-1}]_{\tau \tau' t t'} [\tilde{X}_{i-1}]_{\tau' t} = [\underline{G}^*]_{\tau t} - \sum_{\tau', t' \in [T]} [\mathcal{H}_{i-1}]_{\tau \tau' t t'} [\tilde{X}_{i-1}]_{\tau' t'}$$

Thus it is shown that $\underline{G}_i = \underline{G}^* - \mathcal{H} \tilde{X}_{i-1}$. ■

For comparison, an example of $\mathcal{H}$ is given: for $T = 3$, we have

$$\mathcal{H}_{i-1} = \begin{bmatrix} \mathcal{H}_{11,i-1}^{(1)} & 0 & 0 & \mathcal{H}_{12,i-1}^{(1)} & 0 & 0 & \mathcal{H}_{13,i-1}^{(1)} & 0 & 0 \\ 0 & \mathcal{H}_{11,i-1}^{(2)} & 0 & 0 & \mathcal{H}_{12,i-1}^{(2)} & 0 & 0 & \mathcal{H}_{13,i-1}^{(2)} & 0 \\ 0 & 0 & \mathcal{H}_{11,i-1}^{(3)} & 0 & 0 & \mathcal{H}_{12,i-1}^{(3)} & 0 & 0 & \mathcal{H}_{13,i-1}^{(3)} \\ \hline \mathcal{H}_{21,i-1}^{(1)} & 0 & 0 & \mathcal{H}_{22,i-1}^{(1)} & 0 & 0 & \mathcal{H}_{23,i-1}^{(1)} & 0 & 0 \\ 0 & \mathcal{H}_{21,i-1}^{(2)} & 0 & 0 & \mathcal{H}_{22,i-1}^{(2)} & 0 & 0 & \mathcal{H}_{23,i-1}^{(2)} & 0 \\ 0 & 0 & \mathcal{H}_{21,i-1}^{(3)} & 0 & 0 & \mathcal{H}_{22,i-1}^{(3)} & 0 & 0 & \mathcal{H}_{23,i-1}^{(3)} \\ \hline \mathcal{H}_{31,i-1}^{(1)} & 0 & 0 & \mathcal{H}_{32,i-1}^{(1)} & 0 & 0 & \mathcal{H}_{33,i-1}^{(1)} & 0 & 0 \\ 0 & \mathcal{H}_{31,i-1}^{(2)} & 0 & 0 & \mathcal{H}_{32,i-1}^{(2)} & 0 & 0 & \mathcal{H}_{33,i-1}^{(2)} & 0 \\ 0 & 0 & \mathcal{H}_{31,i-1}^{(3)} & 0 & 0 & \mathcal{H}_{32,i-1}^{(3)} & 0 & 0 & \mathcal{H}_{33,i-1}^{(3)} \end{bmatrix}.$$

2.2.2 Change of Coordinates

To establish the second moment stability of the state vector under Equation (2.50), it is required that we bound the size of the matrix $\mathcal{B}_{i-1} = \mathcal{CA}(\mathbb{I} - \mu \mathcal{H}_{i-1})$.

Consider the following Jordan decomposition to each $(\tau \tau')$ -block of $\mathcal{CA}$:

$$\begin{aligned} \therefore [\mathcal{CA}]_{\tau \tau' t t'} &= \begin{cases} A^{(\tau)} \mathbb{T} \otimes \mathbb{I}_{M_\tau}, & \tau = \tau' = t = t'; \\ C^{(tt)} \mathbb{T} \otimes \mathbb{I}_{M_\tau}, & \tau = \tau' \neq t = t'; \\ C^{(\tau t)} \mathbb{T} \otimes \mathbb{I}_{M_\tau}, & \tau = \tau' = t' \neq t; \\ 0 \cdot \mathbf{1}_{K_t \times K_{t'}} \otimes \mathbf{1}_{M_\tau \times M_{\tau'}}, & \text{otherwise,} \end{cases} \\ \therefore [\mathcal{CA}]_{\tau \tau} &= \underbrace{\begin{bmatrix} C^{(11)} \mathbb{T} & & & (A^{(\tau)} C^{(\tau 1)}) \mathbb{T} \\ & \ddots & & \vdots \\ & & C^{(\tau-1, \tau-1)} \mathbb{T} & (A^{(\tau)} C^{(\tau, \tau-1)}) \mathbb{T} \\ \hline & & & A^{(\tau)} \mathbb{T} \\ \hline & & & (A^{(\tau)} C^{(\tau, \tau+1)}) \mathbb{T} & C^{(\tau+1, \tau+1)} \mathbb{T} \\ & & & \vdots & \ddots \\ & & & (A^{(\tau)} C^{(\tau T)}) \mathbb{T} & & C^{(TT)} \mathbb{T} \end{bmatrix}}_{=: P^{(\tau)} \mathbb{T}} \otimes \mathbb{I}_{M_\tau}. \end{aligned}$$

The $K \times K$ matrix $P^{(\tau)}$ represents how the information on the strategy of team τ is diffused across the whole network. Note that the matrices $[\mathbf{p}^{(\tau')}]_\tau$ and $[\mathbf{q}^{(\tau')}]_\tau$ are both of the form $v \otimes \mathbb{I}_{M_\tau}$ where v is a vector of length K . The vectors v from $[\mathbf{p}^{(\tau')}]_\tau$ and $[\mathbf{q}^{(\tau')}]_\tau$ are in fact, respectively, the left- and right-eigenvectors of $P^{(\tau)} \mathbb{T}$, they can be used to perform the Jordan decomposition on $P^{(\tau)} \mathbb{T}$.

Let us define

$$\begin{aligned}
P^{(\tau)\top} &= \begin{bmatrix} | & & \\ \mathbf{1}_K & & V_{L,\tau} \\ | & & \end{bmatrix} \begin{bmatrix} 1 & & & \\ & \lambda_2 & \varepsilon_\tau & \\ & & \ddots & \\ & & & \lambda_2 & \ddots \\ & & & & \ddots \end{bmatrix} \begin{bmatrix} 0 \cdot \mathbf{1}_{(K_1+\dots+K_{T-1})}^\top & p^{(\tau)\top} & 0 \cdot \mathbf{1}_{(K_{T+1}+\dots+K_T)}^\top \\ & V_{R,\tau}^\mathbf{H} & \end{bmatrix} \\
&= \begin{bmatrix} \mathbf{1}_K & V_{L,\tau} \end{bmatrix} \begin{bmatrix} 1 & \\ & J_{\varepsilon_\tau} \end{bmatrix} \begin{bmatrix} \bar{p}^{(\tau)\top} \\ V_{R,\tau}^\mathbf{H} \end{bmatrix} \\
&= V_\tau^{-\mathbf{H}} J_\tau V_\tau^\mathbf{H},
\end{aligned} \tag{2.54}$$

where the extended Perron vector $\bar{p}^{(\tau)} \in \mathbb{R}^{K \times 1}$ is introduced, and the matrices $V_{R,\tau}$ and $V_{L,\tau}$ satisfy

$$V_{R,\tau}^\mathbf{H} \mathbf{1}_K = 0, \quad V_{L,\tau}^\mathbf{H} \bar{p}^{(\tau)} = 0, \quad V_{L,\tau}^\mathbf{H} V_{R,\tau} = \mathbb{1}_{K-1}. \tag{2.55}$$

Note that under [Assumption 2.2](#) and the derived [Theorem 2.1](#), the diagonal elements to J_{ε_τ} 's have magnitudes strictly less than 1 (with the largest being λ_2). The first upper-diagonal element ε_τ is a free parameter that will be determined later in [Equation \(2.82\)](#).

With the above decomposition, we can decompose the whole $\mathcal{CA}$ by

$$\mathcal{CA} = \mathcal{V}^{-\mathbf{H}} \mathcal{J} \mathcal{V}^\mathbf{H}, \tag{2.56}$$

$$\mathcal{J} = \text{blkdiag} \left\{ \text{blkdiag} \{1, J_{\varepsilon_\tau}\} \otimes \mathbb{1}_{M_\tau} \right\}_{\tau \in [T]}, \tag{2.57}$$

$$\mathcal{V} = \text{blkdiag} \left\{ [\bar{p}^{(\tau)}, V_{R,\tau}] \otimes \mathbb{1}_{M_\tau} \right\}_{\tau \in [T]}, \tag{2.58}$$

$$\mathcal{V}^{-\mathbf{H}} = \text{blkdiag} \left\{ [\mathbf{1}_K, V_{L,\tau}] \otimes \mathbb{1}_{M_\tau} \right\}_{\tau \in [T]}. \tag{2.59}$$

Finally, let us perform a permutation on the rows and columns such that the components associated with the extended Perron vectors are grouped together: let us define a column-permutation by setting $\underline{e}_k \in \mathbb{R}^{2T \times 1}$ ($k \in [2T]$) as the k th basis vector of $\mathbb{R}^{2T}$, then

$$\mathcal{T} := \text{row} \left\{ \text{row} \left\{ \underline{e}_{2\tau-1} \otimes \mathbb{1}_{M_\tau} \right\}_{\tau \in [T]}, \text{row} \left\{ \underline{e}_{2\tau} \otimes \mathbb{1}_{(K-1)M_\tau} \right\}_{\tau \in [T]} \right\}. \tag{2.60}$$

Since $\mathcal{T}^{-1} = \mathcal{T}^\top$ for any permutation $\mathcal{T}$, we have

$$\mathcal{CA} = \tilde{\mathcal{V}}^{-\mathbf{H}} \tilde{\mathcal{J}} \tilde{\mathcal{V}}^\mathbf{H} = \left(\mathcal{V}^{-\mathbf{H}} \mathcal{T} \right) \left(\mathcal{T}^{-1} \mathcal{J} \mathcal{T} \right) (\mathcal{V} \mathcal{T})^\mathbf{H} \tag{2.61}$$

$$\tilde{\mathcal{J}} = \mathcal{T}^{-1} \mathcal{J} \mathcal{T} = \text{blkdiag} \left\{ \text{blkdiag} \{ \mathbb{1}_{M_\tau} \}_{\tau \in [T]}, \text{blkdiag} \{ J_{\varepsilon_\tau} \otimes \mathbb{1}_{M_\tau} \}_{\tau \in [T]} \right\}, \tag{2.62}$$

$$[\tilde{\mathcal{V}}]_{\tau m} = \begin{cases} \bar{p}^{(m)} \otimes \mathbb{1}_{M_m}, & m \in [1 : T], \tau = m; \\ 0 \cdot \mathbf{1}_K \otimes \mathbf{1}_{M_\tau \times M_m}, & m \in [1 : T], \tau \neq m; \\ V_{R,m-T} \otimes \mathbb{1}_{M_{m-T}}, & m \in [T+1 : 2T], \tau = m-T; \\ 0 \cdot \mathbf{1}_{K \times (K-1)} \otimes \mathbf{1}_{M_\tau \times M_{m-T}}, & m \in [T+1 : 2T], \tau \neq m-T, \end{cases} \tag{2.63}$$

$$[\tilde{\mathcal{V}}^{-\mathbf{H}}]_{\tau m} = \begin{cases} \mathbf{1}_K \otimes \mathbb{1}_{M_m}, & m \in [1 : T], \tau = m; \\ 0 \cdot \mathbf{1}_K \otimes \mathbf{1}_{M_\tau \times M_m}, & m \in [1 : T], \tau \neq m; \\ V_{L,m-T} \otimes \mathbb{1}_{M_{m-T}}, & m \in [T+1 : 2T], \tau = m-T; \\ 0 \cdot \mathbf{1}_{K \times (K-1)} \otimes \mathbf{1}_{M_\tau \times M_{m-T}}, & m \in [T+1 : 2T], \tau \neq m-T. \end{cases} \tag{2.64}$$

Remark that so far we have used 3 types of indexing: $\tau, \tau' \in [T]$ for the indexing of larger blocks, each of size $K M_\tau \times K M_{\tau'}$; $t, t' \in [T]$ for the indexing in each $(\tau \tau')$ -block, each of size $K_t M_\tau \times K_{t'} M_{\tau'}$;

and now the new indices $m, n \in [2T]$ which replaces the indices $(\tau\tau')$, representing the blocks in a new set of coordinates. The block matrices in this new set of coordinates have sizes following those illustrated in Equation (2.63), Equation (2.64), and Equation (2.66).

Next, consider applying the same decomposition on $\mathcal{H}_{i-1}$, we have

$$\mathcal{H}_{i-1} = \tilde{\mathcal{V}}^{-H} \mathcal{D}_{i-1} \tilde{\mathcal{V}}^H. \quad (2.65)$$

The structure to $\mathcal{D}_{i-1}$ is of our top interest: $\mathcal{D}_{i-1}$ is separated into $2T \times 2T$ blocks, with the (mn) -block being denoted as $D_{mn,i-1}$, it is of size

$$D_{mn,i-1} \in \begin{cases} \mathbb{R}^{M_m \times M_n}, & m, n \in [1 : T]; \\ \mathbb{C}^{M_m \times (K-1)M_{n-T}}, & m \in [1 : T], n \in [T+1 : 2T]; \\ \mathbb{C}^{(K-1)M_{m-T} \times M_n}, & m \in [T+1 : 2T], n \in [1 : T]; \\ \mathbb{C}^{(K-1)M_{m-T} \times (K-1)M_{n-T}}, & m, n \in [T+1 : 2T]. \end{cases} \quad (2.66)$$

The size of each block actually separates $\mathcal{D}_{i-1}$ into four quadrants, revealing similar structure for $D_{mn,i-1}$ within each block:

Theorem 2.12: Block Structure of $\mathcal{D}_{i-1}$

The blocks of $\mathcal{D}_{i-1}$ can be explicitly written as

- For $m, n \in [1 : T]$:

$$D_{mn,i-1} = \sum_{k \in \mathcal{N}^{(m)}} p_k^{(m)} H_{k,mn,i-1}.$$

- For $m \in [1 : T], n \in [T+1 : 2T]$:

$$D_{mn,i-1} = \left(\bar{p}^{(m)} \otimes \mathbb{1}_{M_m} \right)^T \text{blkdiag} \left\{ \mathcal{H}_{m(n-T),i-1}^{(t)} \right\}_{t \in [T]} \left(V_{L,n-T} \otimes \mathbb{1}_{M_{n-T}} \right). \quad (2.67)$$

- For $m \in [T+1 : 2T], n \in [1 : T]$:

$$D_{mn,i-1} = \left(V_{R,m-T} \otimes \mathbb{1}_{M_{m-T}} \right)^H \text{blkdiag} \left\{ \mathcal{H}_{(m-T)n,i-1}^{(t)} \right\}_{t \in [T]} (\mathbf{1}_K \otimes \mathbb{1}_{M_n}). \quad (2.68)$$

- For $k, \ell \in [T+1 : 2T]$:

$$D_{mn,i-1} = \left(V_{R,m-T} \otimes \mathbb{1}_{M_{m-T}} \right)^H \text{blkdiag} \left\{ \mathcal{H}_{(m-T)(n-T),i-1}^{(t)} \right\}_{t \in [T]} \left(V_{L,n-T} \otimes \mathbb{1}_{M_{n-T}} \right). \quad (2.69)$$

Proof. I shall only show the proof for the quadrant $m, n \in [1 : T]$ as it is the only one where we actually care about the explicit form. Since

$$[\mathcal{D}_{i-1}]_{mn} = [\tilde{\mathcal{V}}^H \mathcal{H}_{i-1} \tilde{\mathcal{V}}^{-H}]_{mn} = \sum_{\tau, \tau' \in [T]} [\tilde{\mathcal{V}}]_{\tau m}^H [\mathcal{H}_{i-1}]_{\tau \tau'} [\tilde{\mathcal{V}}^{-H}]_{\tau' n},$$

we have

$$D_{mn,i-1} = \sum_{\tau, \tau' \in [T]} \left\{ \begin{array}{ll} \bar{p}^{(m)} \otimes \mathbb{1}_{M_m}, & \tau = m; \\ 0 \cdot \mathbf{1}_{KM_\tau \times M_m}, & \text{otherwise} \end{array} \right\}^H \cdot \text{blkdiag} \left\{ \mathcal{H}_{\tau \tau', i-1}^{(t)} \right\}_{t \in [T]} \cdots$$

$$\begin{aligned}
& \cdot \begin{cases} \mathbf{1}_K \otimes \mathbb{1}_{M_n}, & \tau' = n; \\ 0 \cdot \mathbf{1}_{KM_{\tau'} \times M_n}, & \text{otherwise} \end{cases} \\
&= (\bar{p}^{(m)\mathbf{H}} \otimes \mathbb{1}_{M_m}) \cdot \text{blkdiag}\{\mathcal{H}_{mn,i-1}^{(t)}\}_{t \in [T]} \cdot (\mathbf{1}_K \otimes \mathbb{1}_{M_n}) \\
&= (p^{(m)\mathbf{H}} \otimes \mathbb{1}_{M_m}) \cdot \mathcal{H}_{mn,i-1}^{(m)} \cdot (\mathbf{1}_{K_m} \otimes \mathbb{1}_{M_n}) \\
&= \sum_{k \in \mathcal{N}^{(m)}} p_k^{(m)} H_{k,mn,i-1}.
\end{aligned}$$

The last equality is by identifying $\mathcal{H}_{mn,i-1}^{(t)} = \text{blkdiag}\{H_{k,mn,i-1}\}_{k \in \mathcal{N}^{(t)}}$ and each block $H_{k,mn,i-1}$ is of size $M_m \times M_n$. Thus it is proven. $\blacksquare$

2.2.3 Second Moment Iteration

This subsection establishes the following result:

Theorem 2.13: Second Moment Stability

Under the ATC-ITC recursion as stated in [Theorem 2.11](#), the error vector to Nash equilibrium is stable in the sense of its second moment for small enough step-sizes:

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \tilde{\underline{X}}_i \right\|^2 \right] = O(\mu). \quad (2.70)$$

Let us return to [Equation \(2.50\)](#) and rewrite the equation in the transformed coordinates:

$$\tilde{\underline{W}}_i := \tilde{\mathcal{V}}^{\mathbf{H}} \tilde{\underline{X}}_i \quad (2.71)$$

$$= \left(\tilde{\mathcal{J}} - \mu \tilde{\mathcal{D}}_{i-1} \right) \tilde{\underline{W}}_{i-1} + \mu \underline{\mathcal{G}}^* + \mu \underline{\mathcal{S}}_i, \quad (2.72)$$

where

$$\tilde{\mathcal{D}}_{i-1} := \tilde{\mathcal{J}} \mathcal{D}_{i-1}, \quad \underline{\mathcal{G}}^* := \tilde{\mathcal{J}} \tilde{\mathcal{V}}^{\mathbf{H}} \underline{\mathcal{G}}^*, \quad \underline{\mathcal{S}}_i := \tilde{\mathcal{J}} \tilde{\mathcal{V}}^{\mathbf{H}} \underline{\mathcal{S}}_i. \quad (2.73)$$

The entries to $\tilde{\mathcal{D}}_{i-1}$ are

$$\tilde{D}_{mn,i-1} = [\tilde{\mathcal{D}}_{i-1}]_{mn} = [\tilde{\mathcal{J}}]_{mm} [\mathcal{D}_{i-1}]_{mn} = \begin{cases} D_{mn,i-1}, & m \in [1 : T]; \\ (J_{\varepsilon_m} \otimes \mathbb{1}_{M_{m-T}}) D_{mn,i-1}, & m \in [T+1 : 2T]. \end{cases} \quad (2.74)$$

Just like how $\tilde{\mathcal{J}}$ and $\tilde{\mathcal{D}}_{i-1}$ are partitioned into four quadrants, $\tilde{\underline{W}}_i$ can also be partitioned into two parts: the top half $\tilde{\underline{W}}_i^{(1)}$ of length M representing index $m \in [1 : T]$ and the bottom half $\tilde{\underline{W}}_i^{(2)}$ of length $(K-1)M$ representing index $m \in [T+1 : 2T]$. We thus have the following recursion:

Theorem 2.14: Error Recursion in a New Coordinate

Partition the transformed error vector $\tilde{\underline{W}}_i = \tilde{\mathcal{V}}^{\mathbf{H}} \tilde{\underline{X}}_i$ into top and bottom half of length M and $(K-1)M$, respectively. Then the recursion [Equation \(2.72\)](#) can be rewritten as

$$\begin{bmatrix} \tilde{\underline{W}}_i^{(1)} \\ \tilde{\underline{W}}_i^{(2)} \end{bmatrix} = \left(\begin{bmatrix} \tilde{\mathcal{J}}^{(11)} & 0 \\ 0 & \tilde{\mathcal{J}}^{(22)} \end{bmatrix} - \mu \begin{bmatrix} \tilde{\mathcal{D}}_{i-1}^{(11)} & \tilde{\mathcal{D}}_{i-1}^{(12)} \\ \tilde{\mathcal{D}}_{i-1}^{(21)} & \tilde{\mathcal{D}}_{i-1}^{(22)} \end{bmatrix} \right) \begin{bmatrix} \tilde{\underline{W}}_{i-1}^{(1)} \\ \tilde{\underline{W}}_{i-1}^{(2)} \end{bmatrix} + \mu \begin{bmatrix} 0 \\ \underline{\mathcal{G}}^{*,(2)} \end{bmatrix} + \mu \begin{bmatrix} \underline{\mathcal{S}}_i^{(1)} \\ \underline{\mathcal{S}}_i^{(2)} \end{bmatrix}, \quad (2.75)$$

where

$$\begin{aligned}
\widetilde{W}_i^{(1)} &= [\widetilde{W}_i]_{m \in [1:T]}, \\
\widetilde{W}_i^{(2)} &= [\widetilde{W}_i]_{m \in [T+1:2T]}, \\
\widetilde{\mathcal{J}}^{(11)} &= \mathbb{1}_M, \\
\widetilde{\mathcal{J}}^{(22)} &= \text{blkdiag} \{ J_{\varepsilon_\tau} \otimes \mathbb{1}_{M_\tau} \}_{\tau \in [T]} \\
\widetilde{\mathcal{D}}_{i-1}^{(11)} &= [\widetilde{\mathcal{D}}_{i-1}]_{m,n \in [1:T]}, \\
\widetilde{\mathcal{D}}_{i-1}^{(12)} &= [\widetilde{\mathcal{D}}_{i-1}]_{m \in [1:T], n \in [T+1:2T]}, \\
\widetilde{\mathcal{D}}_{i-1}^{(21)} &= [\widetilde{\mathcal{D}}_{i-1}]_{m \in [T+1:2T], n \in [1:T]}, \\
\widetilde{\mathcal{D}}_{i-1}^{(22)} &= [\widetilde{\mathcal{D}}_{i-1}]_{m,n \in [T+1:2T]}, \\
\underline{\mathcal{G}}^{*,(2)} &= [\widetilde{\mathcal{J}} \widetilde{\mathcal{V}}^H \underline{G}^*]_{m \in [T+1:2T]}, \\
\underline{\mathcal{S}}_i^{(1)} &= [\widetilde{\mathcal{J}} \widetilde{\mathcal{V}}^H \underline{G}_i]_{m \in [1:T]}, \\
\underline{\mathcal{S}}_i^{(2)} &= [\widetilde{\mathcal{J}} \widetilde{\mathcal{V}}^H \underline{G}_i]_{m \in [T+1:2T]}.
\end{aligned}$$

Proof. The partitioning is trivial, and the only thing that requires further examination is why the gradient $\underline{\mathcal{G}}^{*,(1)} = 0$. This is shown below: for any $m \in [1:T]$,

$$\begin{aligned}
[\underline{\mathcal{G}}^{*,(1)}]_m &= [\widetilde{\mathcal{J}} \widetilde{\mathcal{V}}^H \underline{G}^*]_m = \sum_{\tau \in [T]} [\widetilde{\mathcal{V}}]_{\tau m}^H [\underline{G}^*]_\tau \\
&= (\bar{p}^{(m)} \otimes \mathbb{1}_{M_m})^\top [\underline{G}^*]_m = (\bar{p}^{(m)} \otimes \mathbb{1}_{M_m})^\top \text{col} \left\{ \text{col} \left\{ \nabla_m J_{\ell_k}^{(m)}(x^*) \right\}_{k \in \mathcal{N}^{(t)}} \right\}_{t \in [T]} \\
&= (p^{(m)} \otimes \mathbb{1}_{M_m})^\top \text{col} \left\{ \nabla_m J_{\ell_k}^{(m)}(x^*) \right\}_{k \in \mathcal{N}^{(m)}} \\
&= \sum_{k \in \mathcal{N}^{(m)}} p_k^{(m)} \nabla_m J_k^{(m)}(x^*) = 0 \cdot \mathbf{1}_{M_m}.
\end{aligned}$$

The last equality is by the definition to the Nash equilibrium. ■

Our goal is to analyze the behavior and upper-bound the growth to $\|\widetilde{X}_i\|^2$. But since

$$\|\widetilde{X}_i\|^2 = \|\widetilde{\mathcal{V}}^{-H} \widetilde{W}_i\|^2 \leq \|\widetilde{\mathcal{V}}^{-1}\|^2 \|\widetilde{W}_i\|^2 = \|\widetilde{\mathcal{V}}^{-1}\|^2 \left(\|\widetilde{W}_i^{(1)}\|^2 + \|\widetilde{W}_i^{(2)}\|^2 \right), \quad (2.76)$$

we can change our goal to upper-bound the growth to $\widetilde{W}_i^{(1)}$ and $\widetilde{W}_i^{(2)}$ instead.

First, let us analyze $\|\widetilde{W}_i^{(1)}\|^2$: taking the expectation conditioned on the filtration $\mathcal{F}_{i-1}$ first, then take the expectation, we have

$$\mathbb{E} \left[\|\widetilde{W}_i^{(1)}\|^2 \right] = \mathbb{E} \left[\left\| \left(\widetilde{\mathcal{J}}^{(11)} - \mu \widetilde{\mathcal{D}}_{i-1}^{(11)} \right) \widetilde{W}_{i-1}^{(1)} - \mu \widetilde{\mathcal{D}}^{(12)} \widetilde{W}_{i-1}^{(2)} \right\|^2 \right] + \mu^2 \mathbb{E} \left[\|\underline{\mathcal{S}}_i^{(1)}\|^2 \right]$$

The first term on the left-hand side can be bounded by using the lemma below:

Lemma 2.15

For $t_k \in (0, 1)$ such that $t := \sum_{k=1}^N t_k < 1$,

$$\left\| \sum_{k=0}^N a_k \right\|^2 = \left\| (1-t) \cdot \frac{a_0}{1-t} + \sum_{k=1}^N t_k \cdot \frac{a_k}{t_k} \right\|^2 \leq \frac{1}{1-t} \|a_0\|^2 + \sum_{k=1}^N \frac{1}{t_k} \|a_k\|^2. \quad (2.77)$$

$$\mathbb{E} \left[\left\| \widetilde{W}_i^{(1)} \right\|^2 \right] \leq \frac{\left\| \widetilde{\mathcal{J}}^{(11)} - \mu \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\|^2}{1-t} \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(1)} \right\|^2 \right] + \frac{\mu^2 \left\| \widetilde{\mathcal{D}}^{(12)} \right\|^2}{t} \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(2)} \right\|^2 \right] + \mu^2 \mathbb{E} \left[\left\| \mathcal{S}_i^{(1)} \right\|^2 \right]$$

Note that

$$\left\| \widetilde{\mathcal{J}}^{(11)} - \mu \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\| = \left\| \mathbb{1}_M - \mu \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\|,$$

we would like to bound this term from above.

Theorem 2.16

Under [Assumption 2.4](#), we have, for all i and μ small enough,

$$\left\| \mathbb{1}_M - \mu \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\| \leq 1 - \mu \varepsilon_J + \mu(T-1)\delta_J. \quad (2.78)$$

Proof. Firstly, let us write down explicitly the matrix $\widetilde{\mathcal{D}}_{i-1}^{(11)}$ and analyze its structure:

$$[\widetilde{\mathcal{D}}_{i-1}^{(11)}]_{mn} = \sum_{k \in \mathcal{N}^{(m)}} p_k^{(m)} H_{k,mn,i-1} = \int_0^1 \sum_{k \in \mathcal{N}^{(m)}} p_k^{(m)} \nabla_{nm}^2 J_k^{(m)}(\cdots - t \widetilde{x}_{k,i-1}^n) dt.$$

By [Assumption 2.4](#), we can see that for $m = n$,

$$[\widetilde{\mathcal{D}}_{i-1}^{(11)}]_{mm} \succeq \varepsilon_J \mathbb{1}_{M_m};$$

for $m \neq n$, via the triangle inequality, we have

$$\left\| [\widetilde{\mathcal{D}}_{i-1}^{(11)}]_{mn} \right\| \leq \delta^{(mn)} \leq \delta_J.$$

By similar procedures as in the proof to [Theorem 2.3](#) and [Theorem 2.4](#), using the block-1-norm and block- ∞ -norm, we can obtain the bound

$$\left\| \mathbb{1} - \mu \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\| \leq \sqrt{\left\| \mathbb{1} - \mu \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\|_{b,1} \left\| \mathbb{1} - \mu \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\|_{b,\infty}} \leq 1 - \mu \varepsilon_J + \mu(T-1)\delta_J$$

for μ small enough. Note that [Equation \(2.16\)](#) is sufficient for $\left\| \mathbb{1} - \mu \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\|$ to be within 1. ■

From the theorem above, it is reasonable to choose $t = \mu(\varepsilon_J - (T-1)\delta_J)$, then

$$\begin{aligned} \mathbb{E} \left[\left\| \widetilde{W}_i^{(1)} \right\|^2 \right] &\leq \left(1 - \mu \varepsilon_J + \mu(T-1)\delta_J + O(\mu^2) \right) \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(1)} \right\|^2 \right] + \dots \\ &\quad \frac{\mu \left\| \widetilde{\mathcal{D}}_{i-1}^{(12)} \right\|^2}{\varepsilon_J - (T-1)\delta_J} \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(2)} \right\|^2 \right] + \mu^2 \mathbb{E} \left[\left\| \mathcal{S}_i^{(1)} \right\|^2 \right] \end{aligned} \quad (2.79)$$

Next, let us analyze $\left\|\widetilde{W}_i^{(2)}\right\|^2$: taking the expectation, we have

$$\begin{aligned}\mathbb{E}\left[\left\|\widetilde{W}_i^{(2)}\right\|^2\right] &= \mathbb{E}\left[\left\|-\mu\widetilde{\mathcal{D}}_{i-1}^{(21)}\widetilde{W}_{i-1}^{(1)} + \left(\widetilde{\mathcal{J}}^{(22)} - \mu\widetilde{\mathcal{D}}_{i-1}^{(22)}\right)\widetilde{W}_{i-1}^{(2)} + \mu\underline{\mathcal{G}}^{*,(2)}\right\|^2\right] + \mu^2\mathbb{E}\left[\left\|\underline{\mathcal{S}}_i^{(2)}\right\|^2\right] \\ &\leq \frac{\left\|\widetilde{\mathcal{J}}^{(22)}\right\|^2}{t}\mathbb{E}\left[\left\|\widetilde{W}_{i-1}^{(2)}\right\|^2\right] + \frac{3\mu^2}{1-t}\left(\left\|\widetilde{\mathcal{D}}_{i-1}^{(21)}\right\|^2\mathbb{E}\left[\left\|\widetilde{W}_{i-1}^{(1)}\right\|^2\right] \dots\right. \\ &\quad \left.+ \left\|\widetilde{\mathcal{D}}_{i-1}^{(22)}\right\|^2\mathbb{E}\left[\left\|\widetilde{W}_{i-1}^{(2)}\right\|^2\right] + \mathbb{E}\left[\left\|\underline{\mathcal{G}}^{*,(2)}\right\|^2\right]\right) + \mu^2\mathbb{E}\left[\left\|\underline{\mathcal{S}}_i^{(2)}\right\|^2\right].\end{aligned}$$

Since $\widetilde{\mathcal{J}}^{(22)}$ is a block diagonal matrix, with each diagonal block being $J_{\varepsilon_\tau} \otimes \mathbb{1}_{M_\tau}$ ($\tau \in [T]$), we have that

$$\left\|\widetilde{\mathcal{J}}^{(22)}\right\|^2 = \max_{\tau \in [T]} \|J_{\varepsilon_\tau}\|^2 = \max_{\tau \in [T]} \lambda_{\max}\left(J_{\varepsilon_\tau}^H J_{\varepsilon_\tau}\right).$$

Note that J_{ε_τ} has ε_τ as its first upper-diagonal, we have

$$J_{\varepsilon_\tau}^H J_{\varepsilon_\tau} = \begin{bmatrix} |\lambda_2|^2 & \varepsilon_\tau \lambda_2^* & \\ \varepsilon_\tau \lambda_2 & |\lambda_2|^2 + \varepsilon_\tau^2 & \ddots \\ & \ddots & \ddots \end{bmatrix}.$$

We can upper-bound its largest eigenvalue (its spectral radius) by any matrix norm, I hence use the 1-norm to upper-bound it and obtained

$$\left\|\widetilde{\mathcal{J}}^{(22)}\right\|^2 \leq \max_{\tau \in [T]} \left\|J_{\varepsilon_\tau}^H J_{\varepsilon_\tau}\right\|_1 \leq \max_{\tau \in [T]} (|\lambda_2| + \varepsilon_\tau)^2 \quad (2.80)$$

Definition 2.3: Spectral Gap

Consider the matrix $P^{(\tau)}$ ($\tau \in [T]$) as defined in Equation (2.54), with J_{ε_τ} being its Jordan block besides the eigenvalue 1.

Let γ_τ denote the spectral gap to $P^{(\tau)}$, which is the difference between the largest and the second largest eigenvalue to $P^{(\tau)}$, i.e., the magnitude of the first element in J_{ε_τ} is

$$|\lambda_2| = 1 - \gamma_\tau. \quad (2.81)$$

Without loss of generality, we set the maximizer to Equation (2.80) to be $\tau = 1$. Choosing $t = 1 - \gamma_1 + \varepsilon_1$, we have

$$\begin{aligned}\mathbb{E}\left[\left\|\widetilde{W}_i^{(2)}\right\|^2\right] &\leq (1 - \gamma_1 + \varepsilon_1)\mathbb{E}\left[\left\|\widetilde{W}_{i-1}^{(2)}\right\|^2\right] + \frac{3\mu^2}{\gamma_1 - \varepsilon_1}\left(\left\|\widetilde{\mathcal{D}}_{i-1}^{(21)}\right\|^2\mathbb{E}\left[\left\|\widetilde{W}_{i-1}^{(1)}\right\|^2\right] \dots\right. \\ &\quad \left.+ \left\|\widetilde{\mathcal{D}}_{i-1}^{(22)}\right\|^2\mathbb{E}\left[\left\|\widetilde{W}_{i-1}^{(2)}\right\|^2\right] + \mathbb{E}\left[\left\|\underline{\mathcal{G}}^{*,(2)}\right\|^2\right]\right) + \mu^2\mathbb{E}\left[\left\|\underline{\mathcal{S}}_i^{(2)}\right\|^2\right].\end{aligned} \quad (2.82)$$

It can finally be seen here that in order to establish stability for the above recursions, it is sufficient to require that the free parameters ε_τ introduced in Equation (2.54) is chosen such that $\varepsilon_\tau < \gamma_\tau$ for all $\tau \in [T]$.

The last term to be analyzed are the gradient noises. Note that by [Assumption 2.5](#) (and the derived [Theorem 2.7](#)), we have

$$\begin{aligned}
 \mathbb{E} \left[\|\underline{\mathcal{S}}_i\|^2 \right] &\leq \|\tilde{\mathcal{J}}\tilde{\mathcal{V}}^H\|^2 \mathbb{E} \left[\|\underline{\mathcal{S}}_i\|^2 \right] \\
 &= \|\tilde{\mathcal{J}}\tilde{\mathcal{V}}^H\|^2 \sum_{\tau, t \in [T]} \sum_{k \in \mathcal{N}^{(t)}} \mathbb{E} \left[\|s_{k,i}^\tau\|^2 \right] \\
 &\leq \|\tilde{\mathcal{J}}\tilde{\mathcal{V}}^H\|^2 \sum_{\tau, t \in [T]} \sum_{k \in \mathcal{N}^{(t)}} \left((\beta_k^t)^2 \|\tilde{x}_{k,i-1}\|^2 + (\sigma_k^t)^2 \right) \\
 &\leq \|\tilde{\mathcal{J}}\tilde{\mathcal{V}}^H\|^2 \left(\beta^2 \|\underline{\tilde{\mathcal{X}}}_{i-1}\|^2 + \sigma^2 \right)
 \end{aligned}$$

for some other constant β and σ . The term $\mu^2 \mathbb{E} \left[\|\underline{\mathcal{S}}_i\|^2 \right]$ thus contributes $O(\mu^2)$.

Finally, let us perform an order analysis on the two recursions [Equation \(2.79\)](#) and [Equation \(2.82\)](#):

$$\begin{bmatrix} \mathbb{E} \left[\|\widetilde{W}_i^{(1)}\|^2 \right] \\ \mathbb{E} \left[\|\widetilde{W}_i^{(2)}\|^2 \right] \end{bmatrix} \leq \underbrace{\begin{bmatrix} 1 - O(\mu) & O(\mu) \\ O(\mu^2) & 1 - \gamma_1 + \varepsilon_1 + O(\mu^2) \end{bmatrix}}_{=: \Gamma} \begin{bmatrix} \mathbb{E} \left[\|\widetilde{W}_{i-1}^{(1)}\|^2 \right] \\ \mathbb{E} \left[\|\widetilde{W}_{i-1}^{(2)}\|^2 \right] \end{bmatrix} + \begin{bmatrix} O(\mu^2) \\ O(\mu^2) \end{bmatrix}. \quad (2.83)$$

The matrix Γ is stable as we can bound its spectral radius by its 1-norm:

$$\rho(\Gamma) \leq \max \left\{ 1 - O(\mu) + O(\mu^2), 1 - \gamma_1 + \varepsilon_1 + O(\mu) \right\}, \quad (2.84)$$

we have $\rho(\Gamma) < 1$ for μ small enough.

The recursion [Equation \(2.83\)](#) is element-wise upper bounded by

$$\limsup_{i \rightarrow \infty} \begin{bmatrix} \mathbb{E} \left[\|\widetilde{W}_i^{(1)}\|^2 \right] \\ \mathbb{E} \left[\|\widetilde{W}_i^{(2)}\|^2 \right] \end{bmatrix} \leq (\mathbb{I} - \Gamma)^{-1} \cdot O(\mu^2) \mathbf{1} = \begin{bmatrix} O(1/\mu) & O(1) \\ O(\mu) & O(1) \end{bmatrix} \begin{bmatrix} O(\mu^2) \\ O(\mu^2) \end{bmatrix} = \begin{bmatrix} O(\mu) \\ O(\mu^2) \end{bmatrix}. \quad (2.85)$$

We thus obtained [Theorem 2.13](#):

$$\begin{aligned}
 \limsup_{i \rightarrow \infty} \mathbb{E} \left[\|\widetilde{W}_i\|^2 \right] &= O(\mu) + O(\mu^2) = O(\mu) \\
 \Rightarrow \limsup_{i \rightarrow \infty} \mathbb{E} \left[\|\widetilde{Z}_i\|^2 \right] &= O(\mu).
 \end{aligned}$$

2.3 Fourth and First Moment Stability

This subsection establishes the following results:

Theorem 2.17: Fourth and First Moment Stability

Under the ATC-ITC recursion as stated in [Theorem 2.11](#), the error vector to Nash equilibrium is stable in the sense of its fourth and first moment for small enough step-sizes:

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \tilde{\underline{X}}_i \right\|^4 \right] = O(\mu^2), \quad (2.86)$$

$$\limsup_{i \rightarrow \infty} \left\| \mathbb{E} \left[\tilde{\underline{X}}_i \right] \right\| = O(\mu). \quad (2.87)$$

The proof for the stability of the fourth moment introduces nothing new, as it follows directly like that in [\[Say14\]](#). One simply goes back to [Theorem 2.14](#) and directly compute the expectation to the fourth moment of $\tilde{\underline{W}}_i^{(1)}$ and $\tilde{\underline{W}}_i^{(2)}$. The bound to each term is mostly exactly the same as that in the previous section.

The proof for the first moment stability, however, requires some new definitions.

2.3.1 Long-Term Dynamic

Definition 2.4

Let us define the Hessians evaluated at the Nash equilibrium to be $\mathcal{H}$, where

$$[\mathcal{H}]_{\tau\tau'tt'} := \begin{cases} \mathcal{H}_{\tau\tau'}^{(t)}, & t = t'; \\ 0 \cdot \mathbf{1}_{K_t \times K_{t'}} \otimes \mathbf{1}_{M_\tau \times M_{\tau'}}, & \text{otherwise,} \end{cases} \quad (2.88)$$

$$\mathcal{H}_{\tau\tau'}^{(t)} := \text{blkdiag}\{H_{k,\tau\tau'}\}_{k \in \mathcal{N}^{(t)}}, \quad (2.89)$$

$$H_{k,\tau\tau'} := \nabla_{\tau'\tau}^2 J_{\ell_k}^{(\tau)}(x^*). \quad (2.90)$$

We can also introduce the Hessian deviation to be

$$\tilde{\mathcal{H}}_{i-1} := \mathcal{H} - \mathcal{H}_{i-1}. \quad (2.91)$$

The error vector recursion of [Equation \(2.50\)](#) thus becomes

$$\tilde{\underline{X}}_i = \mathcal{CA}(\mathbb{1}_{KM} - \mu\mathcal{H})\tilde{\underline{X}}_{i-1} + \mu\mathcal{CA}\tilde{\mathcal{H}}_{i-1}\tilde{\underline{X}}_{i-1} + \mu\mathcal{CAG}^* + \mu\mathcal{CAS}_i. \quad (2.92)$$

Let us denote $\mathcal{B} := \mathcal{CA}(\mathbb{1}_{KM} - \mu\mathcal{H})$ and

$$\underline{H}_{i-1} := \mathbb{E} \left[\tilde{\mathcal{H}}_{i-1} \tilde{\underline{X}}_{i-1} \right], \quad (2.93)$$

we obtain the evolution of the expect value to the error vector:

$$\mathbb{E} \left[\tilde{\underline{X}}_i \right] = \mathcal{B} \cdot \mathbb{E} \left[\tilde{\underline{X}}_{i-1} \right] + \mu\mathcal{CA}\underline{H}_{i-1} + \mu\mathcal{CAG}^*. \quad (2.94)$$

The first key to analysis the above equation is to establish the stability of the coefficient matrix $\mathcal{B}$.

Theorem 2.18: Stability of $\mathcal{B}$

The constant matrix $\mathcal{B}$ is stable with spectral radius given by

$$\rho(\mathcal{B}) = 1 - \mu\nu_R + O(\mu^{\frac{K+1}{K}}), \quad (2.95)$$

where ν_R is as defined in Equation (2.19).

Proof. As with the analysis of the second moment stability, we perform a change of coordinate on $\mathcal{B}$ to obtain

$$\mathcal{B} = \tilde{\mathcal{V}}^{-H} \left(\tilde{\mathcal{J}} - \mu \tilde{\mathcal{D}} \right) \tilde{\mathcal{V}}^H, \quad (2.96)$$

$$[\tilde{\mathcal{D}}]_{mn} := \begin{cases} D_{mn}, & m \in [1 : T]; \\ (J_{\varepsilon_m} \otimes \mathbb{1}_{M_{m-T}}) D_{mn}, & m \in [T+1 : 2T]. \end{cases} \quad (2.97)$$

The only difference between D_{mn} and $D_{mn,i-1}$ is that all the $H_{k,mn,i-1}$ of the latter is replaced by $H_{k,mn}$. The top left block of $m, n \in [1 : T]$, denoted by $\tilde{\mathcal{D}}^{(11)}$, is again of our utmost interest. Note that

$$\tilde{\mathcal{D}}^{(11)} = \begin{bmatrix} \nabla_1^2 J^{(1)}(x^*) & \nabla_{21}^2 J^{(1)}(x^*) & \cdots & \nabla_{T1}^2 J^{(1)}(x^*) \\ \nabla_{12}^2 J^{(2)}(x^*) & \nabla_2^2 J^{(2)}(x^*) & \cdots & \nabla_{T2}^2 J^{(2)}(x^*) \\ \vdots & \vdots & \ddots & \vdots \\ \nabla_{1T}^2 J^{(T)}(x^*) & \nabla_{2T}^2 J^{(T)}(x^*) & \cdots & \nabla_T^2 J^{(T)}(x^*) \end{bmatrix} = \nabla_x \begin{bmatrix} \nabla_1 J^{(1)}(x^*) \\ \nabla_2 J^{(2)}(x^*) \\ \vdots \\ \nabla_T J^{(T)}(x^*) \end{bmatrix} = \nabla_x F(x^*).$$

We can bound the spectrum of $\nabla_x F(x^*)$ by Theorem 2.4:

$$\operatorname{Re}\{\lambda(\nabla_x F(x^*))\} \geq \nu_R \text{ and } |\operatorname{Im}\{\lambda(\nabla_x F(x^*))\}| \leq \nu_I.$$

Next, consider the coordinate transformation below:

$$\mathcal{T} = \operatorname{blkdiag}\{\mu^{1/K} \mathbb{1}_M, \mu^{2/K} \mathbb{1}_M, \dots, \mu^{K/K} \mathbb{1}_M\}.$$

The matrix

$$\mathcal{T}^{-1} \tilde{\mathcal{V}}^H \mathcal{B} \tilde{\mathcal{V}}^{-H} \mathcal{T}$$

has the same spectrum as $\mathcal{B}$ and is of the form

$$\begin{bmatrix} \mathbb{1}_M - \mu \nabla_x F(x^*) & \cdots & O(\mu^{\frac{K+1}{K}}) \\ \vdots & \ddots & \vdots \\ O(\mu^{1/K}) & \cdots & \lambda_L \mathbb{1}_M + O(\mu) \end{bmatrix}.$$

By the **block-Gershgorin theorem** [FV62; ELN18], the eigenvalues to $\mathcal{B}$ are in either

$$\left| \lambda(\mathcal{B}) - \lambda(\mathbb{1}_M - \mu \nabla_x F(x^*)) \right| \leq O(\mu^{\frac{K+1}{K}}) \text{ or } |\lambda(\mathcal{B}) - \lambda_\ell| \leq O(\mu^{1/K}),$$

where λ_ℓ is one of the eigenvalues in J_{ε_τ} for all $\tau \in [T]$. We can thus conclude that

$$\rho(\mathcal{B}) \leq 1 - \mu\nu_R + O(\mu^{\frac{K+1}{K}}) \text{ or } \rho(\mathcal{B}) \leq 1 - \gamma_1 + \varepsilon_1 + O(\mu^{1/K}).$$

If we choose the step-size μ to be small enough, then both the spectral radii above are smaller than 1, thus establishing the stability to $\mathcal{B}$. Furthermore, in this case, the two regions above are actually disjointed. The largest eigenvalue to $\mathcal{B}$ thus reside in the first region. ■

As a further note: the contribution from ν_I is of $O(\mu^2)$, thus omitted.

2.3.2 First Moment Stability

I directly follow the proof in [Say14] from page 547 to 551. The only part that requires further examination is the bounding of the Hessian deviation.

By invoking [Theorem 2.2](#),

$$\begin{aligned}
\left\| \tilde{H}_{k,\tau\tau',i-1} \right\| &= \left\| H_{k,\tau\tau'} - H_{k,\tau\tau',i-1} \right\| \\
&\leq \int_0^1 \left\| \nabla_{\tau'\tau}^2 J_{\ell_k}^{(\tau)}(x^*) - \nabla_{\tau'\tau}^2 J_{\ell_k}^{(\tau)}(\cdots - t\tilde{x}_{k,i-1}^\tau) \right\| dt \\
&\leq \int_0^1 \kappa' \left\| x^* - (\cdots - t\tilde{x}_{k,i-1}^\tau) \right\| dt \\
&\leq \int_0^1 \kappa' \left\| t\tilde{x}_{k,i-1}^\tau \right\| dt.
\end{aligned}$$

One should remember from [Theorem 2.10](#) that the position at which the integrand of $H_{k,\tau\tau',i-1}$ is evaluated is not explicitly mentioned. Henceforth, the placeholder “ $\cdots$ ” is used. Continue on with the bound, we obtain:

$$\left\| \tilde{H}_{k,\tau\tau',i-1} \right\| \leq \frac{\kappa'}{2} \left\| \tilde{x}_{k,i-1}^\tau \right\| \leq \frac{\kappa'}{2} \left\| \tilde{X}_{i-1} \right\|. \quad (2.98)$$

For the whole Hessian deviation, we bound its spectral norm by its block-1-norm and block- ∞ -norm:

$$\left\| \tilde{\mathcal{H}}_{i-1} \right\| \leq \sqrt{\left\| \tilde{\mathcal{H}}_{i-1} \right\|_{b,1} \left\| \tilde{\mathcal{H}}_{i-1} \right\|_{b,\infty}} \leq \frac{\kappa'T}{2} \left\| \tilde{X}_{i-1} \right\|. \quad (2.99)$$

It should be noted here that the constant of proportionality contains T instead of $\sqrt{T}$, this is in contrast to that of the zero local knowledge case, which is further elaborated in [Section 4.1](#).

This helps us establish an upper bound to the perturbation from the Hessian deviation:

$$\left\| \mu \mathcal{CA} \tilde{\mathcal{H}}_{i-1} \right\| \leq \mu \left\| \mathcal{CA} \right\| \mathbb{E} \left[\left\| \tilde{\mathcal{H}}_{i-1} \right\| \left\| \tilde{X}_{i-1} \right\| \right] = \frac{\mu \kappa' T \left\| \mathcal{CA} \right\|}{2} \mathbb{E} \left[\left\| \tilde{X}_{i-1} \right\|^2 \right] \xrightarrow{\limsup_{i \rightarrow \infty}} O(\mu^2), \quad (2.100)$$

where the last line utilizes the second moment stability established in [Theorem 2.13](#).

Combining with the result of the fourth moment stability, we can achieve the first moment stability.

As a remark: what if we define the term $\mathcal{H}$ as that containing only the Hessian from own team, and that the local knowledge is zero? The norm to the Hessian deviation will be proportional to $\sqrt{T}$ instead! But its contribution to the stability and MSD performance is only of higher order in μ .

2.4 MSD Performance of ATC-ITC

This subsection establishes the following result:

Theorem 2.19: MSD Performance

Under the ATC-ITC recursion as stated in [Theorem 2.11](#), the error vector to Nash equilibrium is stable in the sense of its second moment for small enough step-sizes:

$$\limsup_{i \rightarrow \infty} \frac{1}{K} \mathbb{E} \left[\left\| \tilde{\underline{X}}_i \right\|^2 \right] = \text{MSD}_1 + O(\mu^{\min\{1+\gamma', 3/2\}}), \quad (2.101)$$

where $\gamma' := \min\{1, \gamma/2\} \in (0, 1]$. The MSD performance is characterized by

$$\text{MSD}_1 = \frac{1}{K} \mu^2 \text{Tr} \{ \mathcal{X} \} = O(\mu), \quad (2.102)$$

where $\mathcal{X}$ is the matrix that satisfies the following discrete-time Lyapunov equation:

$$\mathcal{Y} = \mathcal{X} - \mathcal{B}\mathcal{X}\mathcal{B}^\top, \text{ where} \quad (2.103)$$

$$\mathcal{Y} = \mathcal{C}\mathcal{A}\mathcal{R}\mathcal{A}^\top\mathcal{C}^\top, \quad (2.104)$$

$$\mathcal{R} := \text{blkdiag} \left\{ \text{blkdiag} \left\{ \text{blkdiag} \{ R_k^\top \}_{k \in \mathcal{N}^{(t)}} \right\}_{t \in [T]} \right\}_{\tau \in [T]} \quad (2.105)$$

with $R_k^\top$'s as defined in [Definition 2.1](#). And for large enough i , the convergence rate of the error variance $\mathbb{E} \left[\left\| \tilde{\underline{X}}_i \right\|^2 \right]$ towards the steady-state region is **upper-bounded by**

$$1 - 2\mu\nu_S - O\left(\mu^{\frac{K+1}{K}}\right) \leq \alpha \leq 1 - 2\mu\nu_R + O\left(\mu^{\frac{K+1}{K}}\right). \quad (2.106)$$

2.4.1 Long-Term Error Model

Consider again the error recursion of [Equation \(2.92\)](#):

$$\tilde{\underline{X}}_i = \mathcal{B}\tilde{\underline{X}}_{i-1} + \mu\mathcal{C}\mathcal{A}\tilde{\underline{H}}_{i-1}\tilde{\underline{X}}_{i-1} + \mu\mathcal{C}\mathcal{A}\underline{G}^* + \mu\mathcal{C}\mathcal{A}\underline{S}_i. \quad (2.107)$$

We can establish

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \mathcal{C}\mathcal{A}\tilde{\underline{H}}_{i-1}\tilde{\underline{X}}_{i-1} \right\|^2 \right] = O(\mu) \quad (2.108)$$

through similar steps as in [Equation \(2.100\)](#). This suggests that the perturbation from the Hessian deviation can be omitted as it does not affect the MSD performance of the recursion. We thus have the following theorem.

Theorem 2.20

Consider the following recursion:

$$\tilde{\underline{Y}}_i = \mathcal{B}\tilde{\underline{Y}}_{i-1} + \mu\mathcal{C}\mathcal{A}\underline{G}^* + \mu\mathcal{C}\mathcal{A}\underline{S}_i(\underline{X}_{i-1}). \quad (2.109)$$

Note that the dependence of $\underline{S}_i$ on $\underline{X}_{i-1}$ is explicitly written out. Then we have that $\tilde{\underline{X}}_i$ is well-

approximated by $\underline{Y}_i$ in that

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \underline{\tilde{X}}_i - \underline{\tilde{Y}}_i \right\|^2 \right] = O(\mu^2), \quad (2.110)$$

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \underline{\tilde{X}}_i \right\|^2 \right] = \limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \underline{\tilde{Y}}_i \right\|^2 \right] + O(\mu^{3/2}). \quad (2.111)$$

Proof. To simplify the notation, let the difference between the two recursion be

$$\underline{Z}_i := \underline{\tilde{X}}_i - \underline{\tilde{Y}}_i = \mathcal{B}\underline{Z}_{i-1} + \mu\mathcal{CA}\tilde{\mathcal{H}}_{i-1}\underline{\tilde{X}}_{i-1}.$$

Performing a change of coordinate, we have

$$\tilde{\mathcal{V}}^H \underline{Z}_i = \left(\tilde{\mathcal{J}} - \mu\tilde{\mathcal{D}} \right) \tilde{\mathcal{V}}^H \underline{Z}_{i-1} + \mu\tilde{\mathcal{J}}\tilde{\mathcal{V}}^H \tilde{\mathcal{H}}_{i-1} \underline{\tilde{X}}_{i-1}, \quad (2.112)$$

$$\begin{bmatrix} \underline{W}_i^{(1)} \\ \underline{W}_i^{(2)} \end{bmatrix} = \begin{bmatrix} \mathbb{I}_M - \mu\tilde{\mathcal{D}}^{(11)} & -\mu\tilde{\mathcal{D}}^{(12)} \\ -\mu\tilde{\mathcal{D}}^{(21)} & \tilde{\mathcal{J}}^{(22)} - \mu\tilde{\mathcal{D}}^{(22)} \end{bmatrix} \begin{bmatrix} \underline{W}_{i-1}^{(1)} \\ \underline{W}_{i-1}^{(2)} \end{bmatrix} + \mu \begin{bmatrix} \tilde{\underline{H}}_{i-1}^{(1)} \\ \tilde{\underline{H}}_{i-1}^{(2)} \end{bmatrix}. \quad (2.113)$$

Similar to the proofs to the stability of different moments of $\underline{\tilde{X}}_i$ we have established, we will finally obtain

$$\begin{bmatrix} \mathbb{E} \left[\left\| \underline{W}_i^{(1)} \right\|^2 \right] \\ \mathbb{E} \left[\left\| \underline{W}_i^{(2)} \right\|^2 \right] \end{bmatrix} \leq \begin{bmatrix} 1 - O(\mu) & O(\mu) \\ O(\mu^2) & 1 - \gamma_1 + \varepsilon_1 + O(\mu^2) \end{bmatrix} \begin{bmatrix} \mathbb{E} \left[\left\| \underline{W}_{i-1}^{(1)} \right\|^2 \right] \\ \mathbb{E} \left[\left\| \underline{W}_{i-1}^{(2)} \right\|^2 \right] \end{bmatrix} + \begin{bmatrix} O(\mu) \\ O(\mu^2) \end{bmatrix} \mathbb{E} \left[\left\| \underline{\tilde{X}} \right\|^4 \right].$$

In the limit, we have

$$\limsup_{i \rightarrow \infty} \begin{bmatrix} \mathbb{E} \left[\left\| \underline{W}_i^{(1)} \right\|^2 \right] \\ \mathbb{E} \left[\left\| \underline{W}_i^{(2)} \right\|^2 \right] \end{bmatrix} = \begin{bmatrix} O(1/\mu) & O(1) \\ O(\mu) & O(1) \end{bmatrix} \begin{bmatrix} O(\mu^3) \\ O(\mu^4) \end{bmatrix} = \begin{bmatrix} O(\mu^2) \\ O(\mu^4) \end{bmatrix}.$$

Hence,

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \underline{Z}_i \right\|^2 \right] = O(\mu^2) + O(\mu^4) = O(\mu^2).$$

It then follows that

$$\begin{aligned} \limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \underline{\tilde{Y}}_i \right\|^2 \right] &= \limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \underline{\tilde{Y}}_i - \underline{\tilde{X}}_i + \underline{\tilde{X}}_i \right\|^2 \right] \\ &\leq \limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \underline{Z}_i \right\|^2 \right] + \mathbb{E} \left[\left\| \underline{\tilde{X}}_i \right\|^2 \right] + 2 \left| \mathbb{E} \left[\underline{Z}_i^T \underline{\tilde{X}}_i \right] \right| \\ &\leq \limsup_{i \rightarrow \infty} \underbrace{\mathbb{E} \left[\left\| \underline{Z}_i \right\|^2 \right]}_{O(\mu^2)} + \mathbb{E} \left[\left\| \underline{\tilde{X}}_i \right\|^2 \right] + 2 \underbrace{\sqrt{\mathbb{E} \left[\left\| \underline{Z}_i \right\|^2 \right] \mathbb{E} \left[\left\| \underline{\tilde{X}}_i \right\|^2 \right]}}_{O(\mu^{3/2})}. \end{aligned}$$

Thus it is shown. ■

Theorem 2.21

The variable $\underline{\tilde{Y}}_i$ is stable in that we have

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \underline{\tilde{Y}}_i \right\|^2 \right] = O(\mu), \quad (2.114)$$

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \tilde{Y}_i \right\|^4 \right] = O(\mu^2), \quad (2.115)$$

$$\limsup_{i \rightarrow \infty} \left\| \mathbb{E} \left[\tilde{Y}_i \right] \right\| = O(\mu). \quad (2.116)$$

Proof. The proofs for the three different results on the stability of $\tilde{Y}_i$ is nearly the same as that of $\tilde{X}_i$. The only difference is that $\mathcal{B}_i$ is now replaced by $\mathcal{B}$, and that the noise term actually depends on $\tilde{X}_i$ instead of $\tilde{Y}_i$. However, the derivation is mostly identical and we will not spend extra time here on the details. ■

2.4.2 Noise Covariance

From [Assumption 2.6](#) and the derived [Theorem 2.8](#), we know that

$$\left\| \lim_{i \rightarrow \infty} \mathbb{E} \left[s_{k,i}^t(x) s_{k,i}^t(x)^\top \mid \mathcal{F}_{i-1} \right] - G_k^t \right\| \leq \max \left\{ L \|x - x^*\|^\gamma, L' \|x - x^*\|^2 \right\}$$

for all t and k , where

$$G_k^t := \lim_{i \rightarrow \infty} \mathbb{E} \left[s_{k,i}^t(x^*) s_{k,i}^t(x^*)^\top \mid \mathcal{F}_{i-1} \right].$$

As a consequence, following the notations introduced in [Definition 2.1](#), we have

Theorem 2.22

Under [Assumption 2.6](#),

$$\lim_{i \rightarrow \infty} \mathbb{E} \left[s_{k,i}^t(x_{k,i-1}) s_{k,i}^t(x_{k,i-1})^\top \right] = R_k^t + O(\mu^{\gamma'}), \quad (2.117)$$

where $\gamma' := \min\{1, \gamma/2\}$. Consequently, the trace satisfies:

$$\text{Tr} \left\{ R_k^t \right\} - O(\mu^{\gamma'}) \leq \lim_{i \rightarrow \infty} \mathbb{E} \left[\left\| s_{k,i}^t(x_{k,i-1}) \right\|^2 \right] \leq \text{Tr} \left\{ R_k^t \right\} + O(\mu^{\gamma'}). \quad (2.118)$$

Proof.

$$\begin{aligned} & \left\| \lim_{i \rightarrow \infty} \mathbb{E} \left[s_{k,i}^t(x_{k,i-1}) s_{k,i}^t(x_{k,i-1})^\top \right] - R_k^t \right\| \\ &= \left\| \mathbb{E} \left[\lim_{i \rightarrow \infty} \mathbb{E} \left[s_{k,i}^t(x_{k,i-1}) s_{k,i}^t(x_{k,i-1})^\top \mid \mathcal{F}_{i-1} \right] - G_k^t \right] \right\| \\ &\leq \mathbb{E} \left[\left\| \lim_{i \rightarrow \infty} \mathbb{E} \left[s_{k,i}^t(x_{k,i-1}) s_{k,i}^t(x_{k,i-1})^\top \mid \mathcal{F}_{i-1} \right] - G_k^t \right\|^2 \right] \\ &\leq \limsup_{i \rightarrow \infty} \mathbb{E} \left[\max \left\{ L \|x_{k,i-1} - x^*\|^\gamma, L' \|x_{k,i-1} - x^*\|^2 \right\} \right] \end{aligned}$$

Then, from previous sections, we have that

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\|x_{k,i-1} - x^*\|^2 \right] \leq \limsup_{i \rightarrow \infty} \mathbb{E} \left[\|\underline{X}_i - \underline{X}^*\|^2 \right] = O(\mu)$$

and

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\|x_{k,i-1} - x^*\|^\gamma \right] = \limsup_{i \rightarrow \infty} \mathbb{E} \left[\left(\|x_{k,i-1} - x^*\|^4 \right)^{\gamma/4} \right]$$

$$\begin{aligned}
 &\leq \limsup_{i \rightarrow \infty} \mathbb{E} \left[\left(\|\underline{X}_i - \underline{X}^*\|^4 \right)^{\gamma/4} \right] \\
 &\leq \limsup_{i \rightarrow \infty} \left(\mathbb{E} \left[\|\underline{X}_i - \underline{X}^*\|^4 \right] \right)^{\gamma/4} = O(\mu^{\gamma/2}).
 \end{aligned}$$

Combining the two equations, we have

$$\left\| \lim_{i \rightarrow \infty} \mathbb{E} \left[s_{k,i}^t(x_{k,i-1}) s_{k,i}^t(x_{k,i-1})^\top \right] - R_k^t \right\| = \max \left\{ L \cdot O(\mu^{\gamma/2}), L' \cdot O(\mu) \right\} = O(\mu^{\gamma'}).$$

Thus it is proven. ■

2.4.3 Mean-Square-Error Performance

Let us prove [Theorem 2.19](#) in this subsection. Starting from the long-term error model of [Equation \(2.109\)](#), we can consider the centered variable:

$$\underline{Z}_i := \tilde{Y}_i - \mathbb{E} \left[\tilde{Y}_i \right], \tag{2.119}$$

which follows the recursion

$$\underline{Z}_i = \mathcal{B} \underline{Z}_{i-1} + \mu \mathcal{C} \mathcal{A} \underline{S}_i(\underline{X}_{i-1}). \tag{2.120}$$

To analyze the MSD performance of $\tilde{X}_i$, we can just consider that of $\underline{Z}_i$ since

$$\begin{aligned}
 \limsup_{i \rightarrow \infty} \mathbb{E} \left[\|\underline{Z}_i\|^2 \right] &= \limsup_{i \rightarrow \infty} \mathbb{E} \left[\|\tilde{Y}_i\|^2 \right] - \left(\mathbb{E} \left[\|\tilde{Y}_i\| \right] \right)^2 \\
 &= \limsup_{i \rightarrow \infty} \mathbb{E} \left[\|\tilde{Y}_i\|^2 \right] - O(\mu^2) \\
 &= \limsup_{i \rightarrow \infty} \mathbb{E} \left[\|\tilde{X}_i\|^2 \right] + O(\mu^{3/2}).
 \end{aligned} \tag{2.121}$$

The performance of the mean-square value to $\underline{Z}_i$ is analyzed by relying on energy conservation arguments: Let Σ denote an arbitrary positive semi-definite matrix that will be determined later. Computing the squared error of $\underline{Z}_i$ with respect to Σ , we obtain

$$\mathbb{E} \left[\|\underline{Z}_i\|_\Sigma^2 \right] = \mathbb{E} \left[\|\underline{Z}_{i-1}\|_{\mathcal{B}^\top \Sigma \mathcal{B}}^2 \right] + \mu^2 \mathbb{E} \left[\|\underline{S}_i\|_{\mathcal{A}^\top \mathcal{C}^\top \Sigma \mathcal{C} \mathcal{A}}^2 \right]. \tag{2.122}$$

The second term on the right-hand side is

$$\mathbb{E} \left[\|\underline{S}_i\|_{\mathcal{A}^\top \mathcal{C}^\top \Sigma \mathcal{C} \mathcal{A}}^2 \right] = \text{Tr} \left\{ \mathcal{A}^\top \mathcal{C}^\top \Sigma \mathcal{C} \mathcal{A} \cdot \mathbb{E} \left[\underline{S}_i \underline{S}_i^\top \right] \right\}. \tag{2.123}$$

From [Assumption 2.5](#), we know that

$$\mathbb{E} \left[\underline{S}_i \underline{S}_i^\top \right] = \text{blkdiag} \left\{ \text{blkdiag} \left\{ \text{blkdiag} \left\{ \mathbb{E} \left[s_{k,i-1}^\tau(x_{k,i-1}) s_{k,i-1}^\tau(x_{k,i-1})^\top \right] \right\}_{k \in \mathcal{N}^{(t)}} \right\}_{t \in [T]} \right\}_{\tau \in [T]}. \tag{2.124}$$

Hence by applying the result of [Theorem 2.22](#) and , we have that

$$\limsup_{i \rightarrow \infty} \left\| \mathbb{E} \left[\underline{S}_i \underline{S}_i^\top \right] - \mathcal{R} \right\| = O(\mu^{\gamma'}),$$

where

$$\mathcal{R} := \text{blkdiag} \left\{ \text{blkdiag} \left\{ \text{blkdiag} \{ R_k^\tau \}_{k \in \mathcal{N}^{(t)}} \right\}_{t \in [T]} \right\}_{\tau \in [T]}. \quad (2.125)$$

Thus,

$$\begin{aligned} \limsup_{i \rightarrow \infty} \left\| \mathcal{A}^\top \mathcal{C}^\top \Sigma \mathcal{C} \mathcal{A} \left(\mathbb{E} \left[\underline{S}_i \underline{S}_i^\top \right] - \mathcal{R} \right) \right\| &= \left\| \mathcal{A}^\top \mathcal{C}^\top \Sigma \mathcal{C} \mathcal{A} \right\| \cdot O(\mu^{\gamma'}) \\ &= \text{Tr} \{ \Sigma \} \cdot O(\mu^{\gamma'}), \end{aligned}$$

where the equalities are by the sub-multiplicative property of matrix norms and the fact that $\text{Tr} \{ \Sigma \} \geq \|\Sigma\|$. Taking the trace, and set $\mathcal{Y} := \mathcal{C} \mathcal{A} \mathcal{R} \mathcal{A}^\top \mathcal{C}^\top$,

$$\limsup_{i \rightarrow \infty} \left| \mathbb{E} \left[\|\underline{S}_i\|_{\mathcal{A}^\top \mathcal{C}^\top \Sigma \mathcal{C} \mathcal{A}}^2 \right] - \text{Tr} \{ \Sigma \mathcal{Y} \} \right| = \text{Tr} \{ \Sigma \} \cdot O(\mu^{\gamma'}) =: b_1.$$

Henceforth, if we let $b_0 = \text{Tr} \{ \Sigma \} \cdot O(\mu^{\gamma'})$ be some positive constant larger than b_1 , then

$$\begin{aligned} \limsup_{i \rightarrow \infty} \mathbb{E} \left[\|\underline{Z}_i\|_{\Sigma - \mathcal{B}^\top \Sigma \mathcal{B}}^2 \right] &\leq \mu^2 (\text{Tr} \{ \Sigma \mathcal{Y} \} + b_0), \\ \liminf_{i \rightarrow \infty} \mathbb{E} \left[\|\underline{Z}_i\|_{\Sigma - \mathcal{B}^\top \Sigma \mathcal{B}}^2 \right] &\geq \mu^2 (\text{Tr} \{ \Sigma \mathcal{Y} \} - b_0). \end{aligned}$$

Lastly, based on the above results, let us determine what Σ should be. Obviously, if

$$\Sigma - \mathcal{B}^\top \Sigma \mathcal{B} = \mathbb{1}_{KM}, \quad (2.126)$$

then we can obtain our desired result of MSD.

Theorem 2.23: Solving Discrete-Time Lyapunov Equation

For Σ that satisfies the discrete-time Lyapunov equation of the form

$$\Sigma - \mathcal{B}^\top \Sigma \mathcal{B} = \mathbb{1} \quad (2.127)$$

with $\mathcal{B}$ being stable. The scalar

$$\text{Tr} \{ \Sigma \mathcal{Y} \} = \text{Tr} \{ \mathcal{X} \}, \quad (2.128)$$

where $\mathcal{X}$ satisfies another discrete-time Lyapunov equation of the form

$$\mathcal{X} - \mathcal{B} \mathcal{X} \mathcal{B}^\top = \mathcal{Y}. \quad (2.129)$$

Proof. The Lyapunov equation of Equation (2.127) can be solved efficiently by using matrix vectorization:

$$\begin{aligned} \underline{\Sigma} - (\mathcal{B}^\top \otimes \mathcal{B}^\top) \underline{\Sigma} &= \underline{\mathbb{1}}, \\ \Rightarrow \underline{\Sigma} &= (\mathbb{1} - \mathcal{B}^\top \otimes \mathcal{B}^\top)^{-1} \underline{\mathbb{1}} \\ &= \sum_{k=0}^{\infty} (\mathcal{B}^\top \otimes \mathcal{B}^\top)^k \underline{\mathbb{1}}, \\ \Rightarrow \Sigma &= \sum_{k=0}^{\infty} (\mathcal{B}^\top)^k \cdot \mathbb{1} \cdot (\mathcal{B})^k. \end{aligned}$$

Note that the inverse to $(\mathbb{1} - \mathcal{B}^\top \otimes \mathcal{B}^\top)$ and the corresponding series expansion exist because $\mathcal{B}$ is stable. Now, the trace is computed to be

$$\begin{aligned} \text{Tr} \{ \Sigma \mathcal{Y} \} &= \sum_{k=0}^{\infty} \text{Tr} \left\{ \left(\mathcal{B}^\top \right)^k (\mathcal{B})^k \mathcal{Y} \right\} \\ &= \text{Tr} \left\{ \sum_{k=0}^{\infty} (\mathcal{B})^k \mathcal{Y} \left(\mathcal{B}^\top \right)^k \right\} = \text{Tr} \{ \mathcal{X} \}. \end{aligned}$$

The fact that $\mathcal{X}$ satisfies Equation (2.129) is easily observed by comparing it with the series form of Σ . Thus it is proven. $\blacksquare$

We have finally established the MSD performance of the ATC-ITC algorithm:

$$\limsup_{i \rightarrow \infty} \frac{1}{K} \mathbb{E} \left[\|Z_i\|^2 \right] = \underbrace{\frac{\mu^2}{K} \text{Tr} \{ \mathcal{X} \}}_{\text{MSD}_1} + \mu^2 \text{Tr} \{ \Sigma \} \cdot O(\mu^{\gamma'}), \quad (2.130)$$

$$\Rightarrow \limsup_{i \rightarrow \infty} \frac{1}{K} \mathbb{E} \left[\|\tilde{X}_i\|^2 \right] = \text{MSD}_1 + \mu^2 \text{Tr} \{ \Sigma \} \cdot O(\mu^{\gamma'}) + O(\mu^{3/2}). \quad (2.131)$$

A few more questions still concerns us, though. Firstly, we need to show that the dominating term of the MSD performance is $O(\mu)$, and the extra terms are of higher orders. Secondly, we need to show the convergence rate to steady-state of the ATC-ITC algorithm is α as stated in Theorem 2.19.

First, the order of magnitude of the terms constituting the MSD:

Theorem 2.24

For sufficiently small step-sizes, it holds that

$$\left(\mathbb{1} - \mathcal{B}^\top \otimes \mathcal{B}^\top \right)^{-1} = O(1/\mu). \quad (2.132)$$

Proof. Firstly, let us perform the Jordan decomposition on $\mathcal{B}^\top$:

$$\begin{aligned} \mathcal{B}^\top &= \tilde{\mathcal{V}} \left(\tilde{\mathcal{J}} - \mu \tilde{\mathcal{D}} \right)^\text{H} \tilde{\mathcal{V}}^{-1} \\ &= \tilde{\mathcal{V}} \left[\begin{array}{c|c} \mathbb{1}_M - \mu \tilde{\mathcal{D}}^{(11)} & -\mu \tilde{\mathcal{D}}^{(12)} \\ \hline -\mu \tilde{\mathcal{D}}^{(21)} & \tilde{\mathcal{J}}^{(22)} - \mu \tilde{\mathcal{D}}^{(22)} \end{array} \right]^\text{H} \tilde{\mathcal{V}}^{-1}. \end{aligned}$$

Then, associated to the block-partitioning above, define a column permutation $\mathcal{T}$ such that it satisfies $\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} = \left(\tilde{\mathcal{V}} \otimes \tilde{\mathcal{V}} \right) \mathcal{T}$. The exact form to this block-Kronecker product is discussed in the proof of Theorem 2.27. Since both $\tilde{\mathcal{V}}$ and $\tilde{\mathcal{V}}^{-\text{H}}$ are column-partitioned, while $\tilde{\mathcal{J}}$ and $\tilde{\mathcal{D}}$ are both row- and column-partitioned, we have

$$\begin{aligned} \mathcal{B}^\top \otimes \mathcal{B}^\top &= \left(\tilde{\mathcal{V}} \otimes \tilde{\mathcal{V}} \right) \left(\left(\tilde{\mathcal{J}} - \mu \tilde{\mathcal{D}} \right) \otimes \left(\tilde{\mathcal{J}} - \mu \tilde{\mathcal{D}} \right) \right)^\text{H} \left(\tilde{\mathcal{V}} \otimes \tilde{\mathcal{V}} \right)^{-1} \\ &= \left(\tilde{\mathcal{V}} \otimes \tilde{\mathcal{V}} \right) \mathcal{T} \left(\mathcal{T}^{-1} \left(\tilde{\mathcal{J}} - \mu \tilde{\mathcal{D}} \right) \otimes \left(\tilde{\mathcal{J}} - \mu \tilde{\mathcal{D}} \right) \mathcal{T} \right)^\text{H} \mathcal{T}^{-1} \left(\tilde{\mathcal{V}} \otimes \tilde{\mathcal{V}} \right)^{-1} \\ &= \left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right) \left(\left(\tilde{\mathcal{J}} - \mu \tilde{\mathcal{D}} \right) \otimes_b \left(\tilde{\mathcal{J}} - \mu \tilde{\mathcal{D}} \right) \right)^\text{H} \left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right)^{-1}. \end{aligned}$$

Therefore,

$$\begin{aligned} & (\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}})^{-1} (\mathcal{B}^\top \otimes \mathcal{B}^\top) (\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}}) \dots \\ &= \left(\begin{bmatrix} \mathbb{1}_M - \mu \tilde{\mathcal{D}}^{(11)} & -\mu \tilde{\mathcal{D}}^{(12)} \\ -\mu \tilde{\mathcal{D}}^{(21)} & \tilde{\mathcal{J}}^{(22)} - \mu \tilde{\mathcal{D}}^{(22)} \end{bmatrix} \otimes_b \begin{bmatrix} \mathbb{1}_M - \mu \tilde{\mathcal{D}}^{(11)} & -\mu \tilde{\mathcal{D}}^{(12)} \\ -\mu \tilde{\mathcal{D}}^{(21)} & \tilde{\mathcal{J}}^{(22)} - \mu \tilde{\mathcal{D}}^{(22)} \end{bmatrix} \right)^H. \end{aligned}$$

where $\otimes_b$ is the block-Kronecker product with respect to the block partition of a matrix as introduced in [Theorem 2.14](#). Partition the resulting matrix into the form of

$$= \begin{bmatrix} \mathbb{1} - \mathcal{P}_{11} & \mathcal{P}_{12} \\ \mathcal{P}_{21} & \mathbb{1} - \bar{\mathcal{P}}_{22} \end{bmatrix}^H = \begin{bmatrix} (\mathbb{1}_M - \mu \tilde{\mathcal{D}}^{(11)}) \otimes (\mathbb{1}_M - \mu \tilde{\mathcal{D}}^{(11)}) & \mathcal{P}_{12} \\ \bar{\mathcal{P}}_{21} & \bar{\mathcal{P}}_{22} \end{bmatrix}^H.$$

We thus have that $\mathcal{P}_{11} = O(\mu) \in \mathbb{R}^{M^2 \times M^2}$, $\mathcal{P}_{12} = O(\mu) \in \mathbb{C}^{M^2 \times (K^2-1)M^2}$, $\mathcal{P}_{21} = O(\mu) \in \mathbb{C}^{(K^2-1)M^2 \times M^2}$, and $\mathcal{P}_{22} = O(1) \in \mathbb{C}^{(K^2-1)M^2 \times (K^2-1)M^2}$. This is same as how we analyzed the big-O representation of $(\mathbb{1} - \Gamma)^{-1}$ in [Equation \(2.85\)](#).

Next, since $\rho(\mathcal{B}^\top \otimes \mathcal{B}^\top) = \rho(\mathcal{B})^2 < 1$, $(\mathbb{1} - \mathcal{B}^\top \otimes \mathcal{B}^\top)$ is invertible. We apply the matrix inversion lemma to obtain

$$\begin{aligned} & (\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}})^{-1} (\mathbb{1} - \mathcal{B}^\top \otimes \mathcal{B}^\top)^{-1} (\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}}) = \begin{bmatrix} \mathcal{P}_{11} & \mathcal{P}_{12} \\ \bar{\mathcal{P}}_{21} & \bar{\mathcal{P}}_{22} \end{bmatrix}^{-\top} \\ &= \begin{bmatrix} \mathcal{P}_{11}^{-1} & 0 \\ 0 & 0 \end{bmatrix}^\top + \begin{bmatrix} -\mathcal{P}_{11}^{-1} \mathcal{P}_{12} \Delta^{-1} \mathcal{P}_{21} \mathcal{P}_{11}^{-1} & -\mathcal{P}_{11}^{-1} \mathcal{P}_{12} \Delta^{-1} \\ -\Delta^{-1} \bar{\mathcal{P}}_{21} \mathcal{P}_{11}^{-1} & \Delta^{-1} \end{bmatrix}^\top \\ &= \begin{bmatrix} O(1/\mu) & O(1) \\ O(1) & O(1) \end{bmatrix}, \end{aligned}$$

with $\Delta = \mathcal{P}_{22} - \mathcal{P}_{21} \mathcal{P}_{11}^{-1} \mathcal{P}_{12} = O(1)$ for μ small enough. The result above is because $\mathcal{P}_{11}^{-1} = O(1/\mu)$, $\mathcal{P}_{11}^{-1} \mathcal{P}_{12} \Delta^{-1} \mathcal{P}_{21} \mathcal{P}_{11}^{-1} = O(1)$, $\mathcal{P}_{11}^{-1} \mathcal{P}_{12} \Delta^{-1} = O(1)$, $\Delta^{-1} \mathcal{P}_{21} \mathcal{P}_{11}^{-1} = O(1)$, and $\Delta^{-1} = O(1)$. The whole matrix is $O(1/\mu)$. Henceforth, $\mathbb{1} - \mathcal{B}^\top \otimes \mathcal{B}^\top = O(1/\mu)$ for μ small enough. ■

Directly from the above theorem, we can establish the size of the terms forming the MSD performance:

Theorem 2.25: Order of Σ and $\mathcal{X}$

With Σ and $\mathcal{X}$ as defined in [Theorem 2.23](#), they satisfy

$$\text{Tr} \{\Sigma\} = O(1/\mu), \quad (2.133)$$

$$\text{Tr} \{\mathcal{X}\} = O(1/\mu). \quad (2.134)$$

Proof. Note that Σ and $\mathcal{X}$ are both $KM \times KM$ matrices. In the vectorize form:

$$\|\underline{\Sigma}\| \leq \left\| \left(\mathbb{1} - \mathcal{B}^\top \otimes \mathcal{B}^\top \right)^{-1} \right\| \|\underline{\mathbb{1}}\| \leq KM \left\| \left(\mathbb{1} - \mathcal{B}^\top \otimes \mathcal{B}^\top \right)^{-1} \right\|_1 \|\underline{\mathbb{1}}\| = O(1/\mu).$$

Next, by Jensen's inequality:

$$(\text{Tr} \{\Sigma\})^2 \leq KM \|\underline{\Sigma}\|^2 = O(1/\mu^2).$$

Finally, since $\mathcal{Y} = O(1)$, $\|\underline{\mathcal{X}}\| = \|(\mathbb{1} - \mathcal{B} \otimes \mathcal{B})^{-1} \underline{\mathcal{Y}}\| = O(1/\mu)$, the same derivation as above applies here as well. Thus it is shown. ■

Returning to Equation (2.131):

$$\begin{aligned} \limsup_{i \rightarrow \infty} \frac{1}{K} \mathbb{E} \left[\left\| \tilde{X}_i \right\|^2 \right] &= \text{MSD}_1 + \mu^2 \text{Tr} \{ \Sigma \} \cdot O(\mu^{\gamma'}) + O(\mu^{3/2}) \\ &= \text{MSD}_1 + O(\mu^{1+\gamma'}) + O(\mu^{3/2}) = \text{MSD}_1 + O(\mu^{\min\{1+\gamma', 3/2\}}). \end{aligned}$$

Thus far, we have shown another result of Theorem 2.19.

Lastly, let us discuss the speed of convergence to stability. Invoking Equation (2.122) and plugging in the results we obtained: since $\text{Tr} \{ (\mathcal{B}^\top \mathcal{B})^k \} = O(1)$,

$$\begin{aligned} \mathbb{E} \left[\left\| \underline{Z}_i \right\|_\Sigma^2 \right] &= \mathbb{E} \left[\left\| \underline{Z}_{i-1} \right\|_{\mathcal{B}^\top \Sigma \mathcal{B}}^2 \right] + \mu^2 \text{Tr} \{ \Sigma \mathcal{Y} \} + \mu^2 \text{Tr} \{ \Sigma \} \cdot o(1), \\ \Rightarrow \mathbb{E} \left[\left\| \underline{Z}_i \right\|^2 \right] &= \mathbb{E} \left[\left\| \underline{Z}_{i-1} \right\|_{\mathcal{B}^\top \mathcal{B}}^2 \right] + \mu^2 \text{Tr} \{ \mathcal{Y} \} + o(\mu^2) \\ &= \mathbb{E} \left[\left\| \underline{Z}_{i-2} \right\|_{(\mathcal{B}^\top)^2 \mathcal{B}^2}^2 \right] + \mu^2 \left(\text{Tr} \{ \mathcal{B}^\top \mathcal{B} \mathcal{Y} \} + \text{Tr} \{ \mathcal{Y} \} \right) + o(\mu^2) \\ &= \dots \\ &= \mathbb{E} \left[\left\| \underline{Z}_0 \right\|_{(\mathcal{B}^\top)^i \mathcal{B}^i}^2 \right] + \mu^2 \sum_{k=0}^{i-1} \text{Tr} \left\{ \left(\mathcal{B}^\top \mathcal{B} \right)^k \mathcal{Y} \right\} + o(\mu^2). \end{aligned}$$

The first term on the right-hand side is the transient term that decays to zero as time goes on, and the second term converges to $\text{Tr} \{ \mathcal{X} \}$, which is the first-order term in the MSD performance. The speed in which the first term decays is characterized below:

Theorem 2.26

The term

$$\mathbb{E} \left[\left\| \underline{Z}_0 \right\|_{(\mathcal{B}^\top)^i \mathcal{B}^i}^2 \right]$$

decays exponentially in the form of α^i , where the rate of convergence α is upper-bounded by

$$\alpha \leq 1 - 2\mu\nu_R + O(\mu^{\frac{K+1}{K}}). \quad (2.135)$$

Proof.

$$\begin{aligned} \mathbb{E} \left[\left\| \underline{Z}_0 \right\|_{(\mathcal{B}^\top)^i \mathcal{B}^i}^2 \right] &= \mathbb{E} \left[\underline{Z}_0^\top (\mathcal{B}^\top)^i \mathcal{B}^i \underline{Z}_0 \right] \\ &= \text{Tr} \left\{ (\mathcal{B}^\top)^i \mathcal{B}^i \cdot \underline{Z}_0 \underline{Z}_0^\top \right\} \\ &= \text{vec} \left(\underline{Z}_0 \underline{Z}_0^\top \right)^\top \text{vec} \left((\mathcal{B}^\top)^i \mathcal{B}^i \right) \\ &= \text{vec} \left(\underline{Z}_0 \underline{Z}_0^\top \right)^\top \left((\mathcal{B}^\top)^i \otimes (\mathcal{B}^\top)^i \right) \text{vec}(\mathbb{1}) \\ &= \text{vec} \left(\underline{Z}_0 \underline{Z}_0^\top \right)^\top \left(\mathcal{B}^\top \otimes \mathcal{B}^\top \right)^i \text{vec}(\mathbb{1}). \end{aligned}$$

The rate of convergence α is equal to the spectral radius of $\mathcal{B}^\top \otimes \mathcal{B}^\top$, which is

$$\alpha = \rho(\mathcal{B}^\top \otimes \mathcal{B}^\top) = \rho(\mathcal{B})^2 \leq \left(1 - \mu\nu_R + O(\mu^{\frac{K+1}{K}}) \right)^2 = 1 - 2\mu\nu_R + O(\mu^{\frac{K+1}{K}}).$$

Thus it is proven. ■

All statements in Theorem 2.19 is now proven. Note that a more general version with team-dependent step-sizes is given in Section 3.1.

2.4.4 Explicit Computation of MSD

From the last subsection, we have finally determined that

$$\text{MSD}_1 = \frac{\mu^2}{K} \text{Tr} \{ \mathcal{X} \}.$$

However, as the number of players K and strategy size M grows, the computation required for obtaining $\mathcal{X}$ grows as well, requiring inversion of a matrix of size $(KM)^2 \times (KM)^2$. In terms of computation speed, this is unacceptable, and we can do much better.

For us to obtain a simpler expression for the MSD, a deeper analysis into the discrete-time Lyapunov equation [Equation \(2.129\)](#) is required:

Theorem 2.27: Simpler Expression for MSD Performance

For $\mathcal{X}$ defined as the solution to [Equation \(2.129\)](#), the first-order term component to the MSD can be expressed as

$$\overline{\text{MSD}}_1 := \mu \cdot \left(\lim_{\mu \rightarrow 0} \frac{1}{\mu} \text{MSD}_1 \right) = \mu \sum_{\tau \in [T]} \text{Tr} \left\{ \mathcal{W}_\tau \sum_{k \in \mathcal{N}^{(\tau)}} (p_k^{(\tau)})^2 R_k^\tau \right\}, \quad (2.136)$$

where $\mathcal{W}_\tau \in \mathbb{R}^{M_\tau \times M_\tau}$ are the diagonal blocks to $\mathcal{W} \in \mathbb{R}^{M \times M}$ that satisfies the continuous-time Lyapunov equation

$$\mathcal{W} \tilde{\mathcal{D}}^{(11)} + (\tilde{\mathcal{D}}^{(11)})^\top \mathcal{W} = \mathbb{1}_M, \quad (2.137)$$

with $\tilde{\mathcal{D}}^{(11)} = \nabla_x F(x^*)$.

Note that in the previous subsection, the term MSD_1 contains the first-order term, but it is not independent of $o(\mu)$. Henceforth, the newly introduced $\overline{\text{MSD}}_1$ is defined so such that it contains solely the first-order component of the MSD. Further, from this result, we can also see that $\overline{\text{MSD}}_1$ is only dependent on $A^{(\tau)}$'s and not on any of the $C^{(t\tau)}$'s.

Proof. From its definition, and following parts of the proof in [Theorem 2.24](#):

$$\begin{aligned} \overline{\text{MSD}}_1 &= \mu \cdot \left(\lim_{\mu \rightarrow 0} \frac{\mu}{K} \text{Tr} \{ \mathcal{X} \} \right) \\ &= \mu \cdot \left(\lim_{\mu \rightarrow 0} \frac{\mu}{K} \text{vec}(\mathbb{1}_{KM})^\top \text{vec}(\mathcal{X}) \right) \\ &= \mu \cdot \left(\lim_{\mu \rightarrow 0} \frac{\mu}{K} \text{vec}(\mathbb{1})^\top (\mathbb{1}_{M^2} - \mathcal{B} \otimes \mathcal{B})^{-1} \text{vec}(\mathcal{Y}) \right) \\ &= \mu \cdot \lim_{\mu \rightarrow 0} \frac{\mu}{K} \cdot \text{vec}(\mathbb{1})^\top (\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}})^{-\top} \left[\begin{array}{c|c} \mathcal{P}_{11} & \mathcal{P}_{12} \\ \hline \mathcal{P}_{21} & \mathcal{P}_{22} \end{array} \right]^{-*} (\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}})^\top \text{vec}(\mathcal{Y}) \\ &= \mu \cdot \lim_{\mu \rightarrow 0} \frac{\mu}{K} \left(\text{vec}(\mathbb{1})^\top (\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}})^{-\top} \left[\begin{array}{c|c} \mathcal{P}_{11}^{-1} & 0 \\ \hline 0 & 0 \end{array} \right] (\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}})^\top \text{vec}(\mathcal{Y}) + O(1) \right). \end{aligned}$$

The $O(1)$ terms disappears in the limit, hence we only need to consider the $O(1/\mu)$ component of $\mathcal{P}_{11}^{-1}$, which is denoted by $\mathcal{Q}_{11}^{-1}$:

$$\mathcal{Q}_{11}^{-1} := \left(\mu \tilde{\mathcal{D}}^{(11)} \otimes \mathbb{1}_M + \mu \mathbb{1}_M \otimes \tilde{\mathcal{D}}^{(11)} \right)^{-1}$$

$$= \left(\mu \tilde{\mathcal{D}}^{(11)} \otimes \mathbb{1}_M + \mu \mathbb{1}_M \otimes \tilde{\mathcal{D}}^{(11)} + O(\mu^2) \right)^{-1} + O(1) = \mathcal{P}_{11}^{-1} + O(1).$$

Viz., we have

$$\overline{\text{MSD}}_1 = \mu \cdot \lim_{\mu \rightarrow 0} \frac{\mu}{K} \cdot \text{vec}(\mathbb{1})^\top \left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right)^{-\top} \begin{bmatrix} \mathcal{Q}_{11}^{-1} & \vdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \vdots & 0 \end{bmatrix} \left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right)^\top \text{vec}(\mathcal{Y}).$$

The terms between $\text{vec}(\mathbb{1})$ and $\text{vec}(\mathcal{Y})$ can be further simplified. Firstly, note that

$$\left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right)^{-\top} \begin{bmatrix} \mathcal{Q}_{11}^{-1} & \vdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \vdots & 0 \end{bmatrix} \left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right)^\top = \left(\tilde{\mathcal{V}}^{-\text{H}} \otimes_b \tilde{\mathcal{V}}^{-\text{H}} \right)^* \begin{bmatrix} \mathcal{Q}_{11}^{-1} & \vdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \vdots & 0 \end{bmatrix} \left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right)^\top,$$

where, from Equation (2.63) and Equation (2.64), $\tilde{\mathcal{V}}$ and $\tilde{\mathcal{V}}^{-\text{H}}$ admit the column partition:

$$\begin{aligned} \tilde{\mathcal{V}} &= \begin{bmatrix} \tilde{\mathcal{V}}^{(1)} & \vdots & \tilde{\mathcal{V}}^{(2)} \end{bmatrix}, \\ \tilde{\mathcal{V}}^{(1)} &= [\tilde{\mathcal{V}}]_{\tau \in [T], m \in [1:T]}, \\ \tilde{\mathcal{V}}^{(2)} &= [\tilde{\mathcal{V}}]_{\tau \in [T], m \in [T+1:2T]}, \\ \tilde{\mathcal{V}}^{-\text{H}} &= \begin{bmatrix} \tilde{\mathcal{U}}^{(1)} & \vdots & \tilde{\mathcal{U}}^{(2)} \end{bmatrix}, \\ \tilde{\mathcal{U}}^{(1)} &= [\tilde{\mathcal{V}}^{-\text{H}}]_{\tau \in [T], m \in [1:T]}, \\ \tilde{\mathcal{U}}^{(2)} &= [\tilde{\mathcal{V}}^{-\text{H}}]_{\tau \in [T], m \in [T+1:2T]}. \end{aligned}$$

By careful examination of the size of each block, we arrived at

$$\left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right)^{-\top} \begin{bmatrix} \mathcal{Q}_{11}^{-1} & \vdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \vdots & 0 \end{bmatrix} \left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right)^\top = \left(\tilde{\mathcal{U}}^{(1)} \otimes \tilde{\mathcal{U}}^{(1)} \right) \mathcal{Q}_{11}^{-1} \left(\tilde{\mathcal{V}}^{(1)} \otimes \tilde{\mathcal{V}}^{(1)} \right)^\top.$$

One should note that both $\tilde{\mathcal{V}}^{(1)}$ and $\tilde{\mathcal{U}}^{(1)}$ are real.

Continue on,

$$\begin{aligned} \left(\tilde{\mathcal{U}}^{(1)} \otimes \tilde{\mathcal{U}}^{(1)} \right)^\top \text{vec}(\mathbb{1}_{KM}) &= \text{vec} \left((\tilde{\mathcal{U}}^{(1)})^\top \cdot \mathbb{1}_{KM} \cdot \tilde{\mathcal{U}}^{(1)} \right), \text{ with} \\ (\tilde{\mathcal{U}}^{(1)})^\top \tilde{\mathcal{U}}^{(1)} &= \text{blkdiag} \{ \mathbf{1}_K \otimes \mathbb{1}_{M_m} \}_{m \in [T]}^\top \text{blkdiag} \{ \mathbf{1}_K \otimes \mathbb{1}_{M_m} \}_{m \in [T]} \\ &= \text{blkdiag} \{ K \mathbb{1}_{M_m} \}_{m \in [T]} = K \mathbb{1}_M. \end{aligned}$$

Similarly,

$$\left(\tilde{\mathcal{V}}^{(1)} \otimes \tilde{\mathcal{V}}^{(1)} \right)^\top \text{vec}(\mathcal{Y}) = \text{vec} \left((\tilde{\mathcal{V}}^{(1)})^\top \mathcal{Y} \tilde{\mathcal{V}}^{(1)} \right).$$

Next, a nice trick is applied to help us analyze the whole term: we want to find a vector $\text{vec}(\mathcal{W})$, with $\mathcal{W} \in \mathbb{R}^{M \times M}$, that satisfies

$$\begin{aligned} \frac{K}{\mu} \text{vec}(\mathcal{W}) &:= \mathcal{Q}_{11}^{-\top} \left(\tilde{\mathcal{U}}^{(1)} \otimes \tilde{\mathcal{U}}^{(1)} \right)^\top \text{vec}(\mathbb{1}_{KM}) \\ \Leftrightarrow \frac{K}{\mu} \mathcal{Q}_{11}^\top \text{vec}(\mathcal{W}) &= \left(\tilde{\mathcal{U}}^{(1)} \otimes \tilde{\mathcal{U}}^{(1)} \right)^\top \text{vec}(\mathbb{1}_{KM}) = K \mathbb{1}_M \\ \Leftrightarrow \mathcal{W} \tilde{\mathcal{D}}^{(11)} + (\tilde{\mathcal{D}}^{(11)})^\top \mathcal{W} &= \mathbb{1}_M, \end{aligned} \tag{2.138}$$

i.e., $\mathcal{W}$ is a solution to Equation (2.137), a continuous-time Lyapunov equation. The solution is unique and one immediately sees that it has to be symmetric.

To wrap things up, let us plug $\mathcal{W}$ back into the expression for $\overline{\text{MSD}}_1$:

$$\begin{aligned}\overline{\text{MSD}}_1 &= \mu \cdot \lim_{\mu \rightarrow 0} \frac{\mu}{K} \cdot \frac{K}{\mu} \text{vec}(\mathcal{W})^\top \text{vec} \left((\tilde{\mathcal{V}}^{(1)})^\top \mathcal{Y} \tilde{\mathcal{V}}^{(1)} \right) \\ &= \mu \text{Tr} \left\{ \mathcal{W} (\tilde{\mathcal{V}}^{(1)})^\top \mathcal{Y} \tilde{\mathcal{V}}^{(1)} \right\} \\ &= \mu \text{Tr} \left\{ \mathcal{W} \cdot (\tilde{\mathcal{V}}^{(1)})^\top \mathcal{CA} \cdot \mathcal{R} \cdot \mathcal{A}^\top \mathcal{C}^\top \tilde{\mathcal{V}}^{(1)} \right\}.\end{aligned}$$

Observe that $\tilde{\mathcal{V}}$ have columns that coincides with the left-eigenvectors to $\mathcal{CA}$ (see [Theorem 2.9](#)), hence,

$$(\tilde{\mathcal{V}}^{(1)})^\top \mathcal{CA} = (\tilde{\mathcal{V}}^{(1)})^\top.$$

Moreover,

$$\begin{aligned}(\tilde{\mathcal{V}}^{(1)})^\top \mathcal{R} \tilde{\mathcal{V}}^{(1)} &= \text{blkdiag} \left\{ \bar{p}^{(m)} \otimes \mathbb{1}_{M_m} \right\}_{m \in [T]}^\top \cdots \\ &\quad \cdot \text{blkdiag} \left\{ \text{blkdiag} \{ R_k^\tau \}_{k \in \mathcal{N}} \right\}_{\tau \in [T]} \cdot \text{blkdiag} \left\{ \bar{p}^{(m)} \otimes \mathbb{1}_{M_m} \right\}_{m \in [T]} \\ &= \text{blkdiag} \left\{ \sum_{k \in \mathcal{N}^{(\tau)}} (p_k^{(\tau)})^2 R_k^\tau \right\}_{\tau \in [T]}.\end{aligned}$$

where one should remind themselves that $p^{(m)}$ is the Perron vector, and $\bar{p}^{(m)}$ is the extended Perron vector of length K . Thus,

$$\overline{\text{MSD}}_1 = \mu \text{Tr} \left\{ \mathcal{W} \cdot (\tilde{\mathcal{V}}^{(1)})^\top \mathcal{R} \tilde{\mathcal{V}}^{(1)} \right\} = \mu \sum_{\tau \in [T]} \text{Tr} \left\{ \mathcal{W}_\tau \sum_{k \in \mathcal{N}^{(\tau)}} (p_k^{(\tau)})^2 R_k^\tau \right\},$$

where $\mathcal{W}_\tau$ are the $M_\tau \times M_\tau$ blocks on the diagonal of $\mathcal{W}$. Finally, the statement is proven. ■

It seems a bit unsatisfactory that we did not have an analytic formula for $\mathcal{W}_\tau$ in the last theorem. Luckily, an integral formula does exist for $\mathcal{W}$ if $-\tilde{\mathcal{D}}^{(11)}$ is stable (in the sense of *Hurwitz stability*, having its eigenvalues in the left-hand side of the complex plane):

Lemma 2.28: Integral Form of $\mathcal{W}$

For $\mathcal{W}$ satisfying the following continuous-time Lyapunov equation

$$\mathcal{W} \tilde{\mathcal{D}}^{(11)} + (\tilde{\mathcal{D}}^{(11)})^\top \mathcal{W} = \mathbb{1},$$

since $-\tilde{\mathcal{D}}^{(11)}$ is Hurwitz stable, we have

$$\mathcal{W} = \int_0^\infty e^{-(\tilde{\mathcal{D}}^{(11)})^\top t} e^{-\tilde{\mathcal{D}}^{(11)} t} dt. \quad (2.139)$$

Using the lemma above, we can manipulate the expressions for $\overline{\text{MSD}}_1$ into something more interesting. To begin with,

$$\begin{aligned}\text{Tr} \left\{ \mathcal{W} \cdot (\tilde{\mathcal{V}}^{(1)})^\top \mathcal{R} \tilde{\mathcal{V}}^{(1)} \right\} &= \int_0^\infty \text{Tr} \left\{ e^{-\tilde{\mathcal{D}}^{(11)} t} (\tilde{\mathcal{V}}^{(1)})^\top \mathcal{R} \tilde{\mathcal{V}}^{(1)} e^{-(\tilde{\mathcal{D}}^{(11)})^\top t} \right\} dt \\ &= \int_0^\infty \left\| \exp \left(-(\tilde{\mathcal{D}}^{(11)})^\top t \right) \right\|_{(\tilde{\mathcal{V}}^{(1)})^\top \mathcal{R} \tilde{\mathcal{V}}^{(1)}}^2 dt.\end{aligned}$$

Theorem 2.29: Integral Form of $\overline{\text{MSD}}_1$

The first-order component to the MSD of the ATC-ITC algorithm has the integral form of

$$\overline{\text{MSD}}_1 = \mu \int_0^\infty \left\| \exp \left(- (\nabla_x F(x^\star))^\top t \right) \right\|_{\tilde{\mathcal{R}}}^2 dt, \quad (2.140)$$

where the weight to the matrix norm is

$$\tilde{\mathcal{R}} := (\tilde{\mathcal{V}}^{(1)})^\top \mathcal{R} \tilde{\mathcal{V}}^{(1)} = \text{blkdiag} \left\{ \sum_{k \in \mathcal{N}^{(\tau)}} (p_k^{(\tau)})^2 R_k^\tau \right\}_{\tau \in [T]} \succeq 0. \quad (2.141)$$

2.5 Step-Size and Iteration Analysis

Throughout the derivation above, we required that the step-sizes are small enough and often consider behaviors that are "in the limit." However, how exactly small / large should these terms be? This section gives a more detailed analysis on the acceptable step-sizes and the number of iterations required for the squared error to reach steady state.

First, let us relate the step-size to the number of iterations required:

2.6 Results

2.6.1 Summary of Chapter

A summary of all the assumptions and theorems with how they are interconnected is given below.

2.6.2 Quadratic Game

Let us consider a simple *quadratic game* where the local loss function of each agent $k \in \mathcal{N}^{(t)}$ is

$$J_k^{(t)}(x) = \frac{1}{2}x^t{}^\top A_k x^t + \sum_{\tau \in [T] \setminus \{t\}} x^t{}^\top C_{k,\tau} x^\tau + b_k^\top x^t,$$

where $A_k \in \mathbb{R}^{M_t \times M_t}$ is positive definite, $C_{k,\tau} \in \mathbb{R}^{M_t \times M_\tau}$, and $b_k \in \mathbb{R}^{M_t \times 1}$. The gradients and Hessians are calculated to be

$$\nabla_t J_k^{(t)}(x) = A_k x^t + \sum_{\tau \in [T] \setminus \{t\}} C_{k,\tau} x^\tau + b_k,$$

$$\nabla_t^2 J_k^{(t)}(x) = A_k,$$

$$\nabla_{\tau t}^2 J_k^{(t)}(x) = C_{k,\tau}.$$

The aggregated loss function is of the form

$$J^{(t)}(x) = \frac{1}{2}x^t{}^\top A_{(t)} x^t + \sum_{\tau \in [T] \setminus \{t\}} x^t{}^\top C_{(t),\tau} x^\tau + b_{(t)}^\top x^t, \text{ with}$$

$$A_{(t)} = \sum_{k \in \mathcal{N}^{(t)}} p_k^{(t)} A_k,$$

$$C_{(t),\tau} = \sum_{k \in \mathcal{N}^{(t)}} p_k^{(t)} C_{k,\tau},$$

$$b_{(t)} = \sum_{k \in \mathcal{N}^{(t)}} p_k^{(t)} b_k.$$

Under [Assumption 2.4](#) and [Theorem 2.3](#), it is required that the matrix

$$\nabla_x F(x) = \begin{bmatrix} A_{(1)} & C_{(1),2} & C_{(1),3} & \cdots \\ C_{(2),1} & A_{(2)} & C_{(2),3} & \cdots \\ C_{(3),1} & C_{(3),2} & A_{(3)} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

satisfies $\lambda_{\min}(\nabla_x F(x)) > 0$ to ensure stability of the ATC-ITC algorithm. A sufficient condition is to enforce $A_k \succeq \varepsilon_J \mathbb{I}_{M_t}$ and $\|C_{k,\tau}\| \leq \delta_J$ for all k , with $\varepsilon_J > (T-1)\delta_J$.

From the proof of the second moment stability, the step-size and iteration should be set to

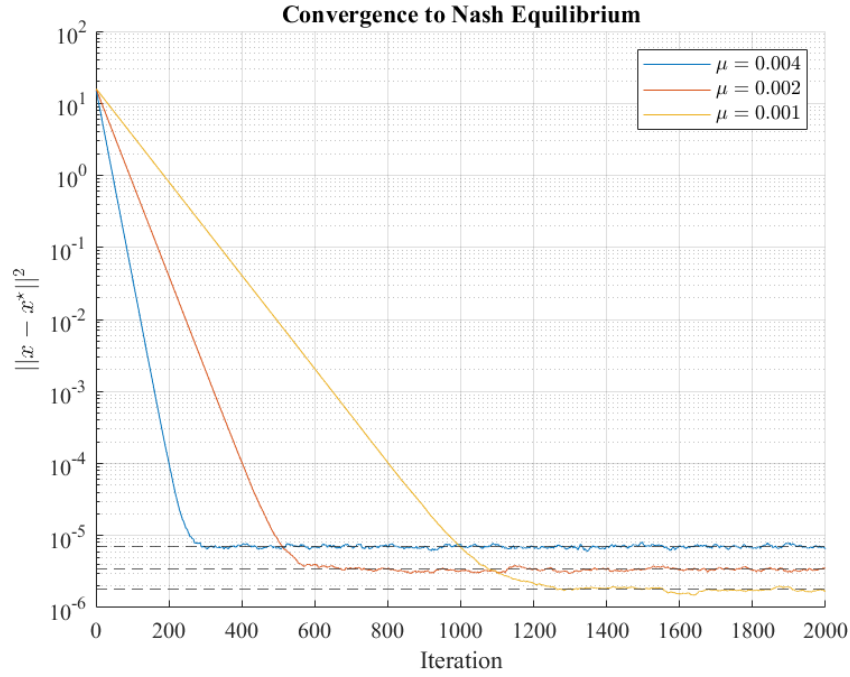


Figure 2.1 The time evolution of squared error the ATC-ITC algorithm over a network of three teams with three different step-sizes. The local loss functions within a team are identical. The dashed lines are the theoretical bound $K \cdot \text{MSD}_1 = \mu^2 \text{Tr}\{\mathcal{X}\}$ for the squared error at steady state.

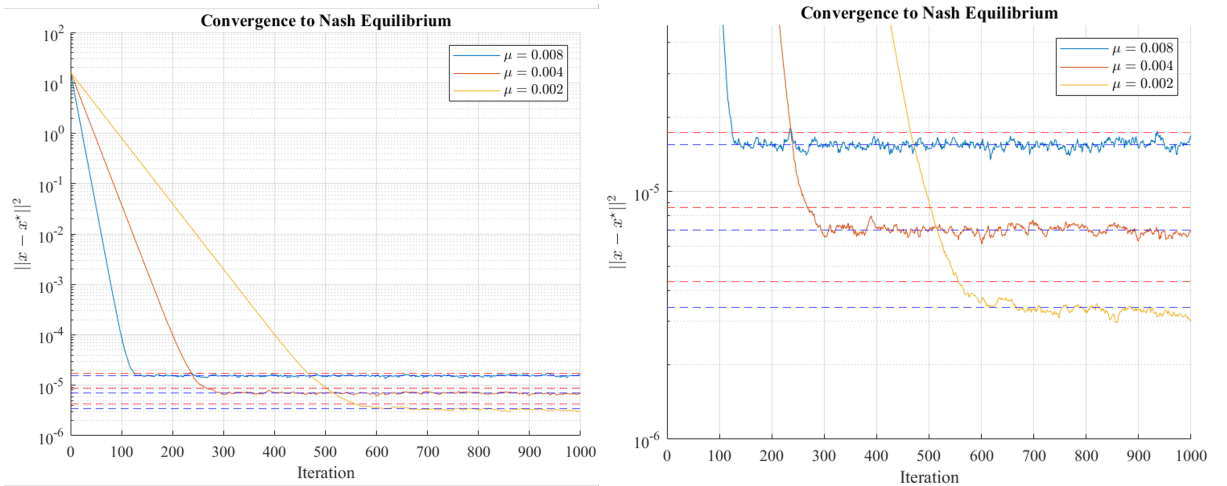


Figure 2.2 Comparison between MSD_1 and $\overline{\text{MSD}}_1$, the former is represented by the dashed lines of color blue, while the latter is of color red. The local loss functions within a team are identical.

3 Variable Step-Sizes

It is suggested that the step-sizes for the ATC-ITC algorithm can vary with respect to team, or even with respect to the iteration as well. A reformulation of the results and a few assumptions are required.

The change to the formulation of the ATC-ITC algorithm is described below:

(a) Strategy update (ATC):

$$x_{k,i}^t = \sum_{\ell \in \mathcal{N}^{(t)}} a_{\ell k}^{(t)} \left(x_{\ell,i-1}^t - \mu_i^t \nabla_t J_\ell^{(t)}(x_{\ell,i-1}^t) \right) \quad (3.1)$$

for some step-size μ_i^t . The gradient ∇_t is applied with respect to the strategy x^t , the set $x_\ell = \{x_\ell^\tau\}_{\tau \in [T]}$ includes the strategy and estimates of agent $\ell \in \mathcal{N}^{(t)}$.

(b) Estimates update (ITC): for all $\tau \neq t$,

$$x_{k,i}^\tau = \sum_{\ell \in \mathcal{N}^{(t)}} c_{\ell k}^{(tt)} \left(x_{\ell,i-1}^\tau - \mu_i^\tau g_{\ell,i-1}^\tau \right) + \sum_{\ell \in \mathcal{N}^{(\tau)}} c_{\ell k}^{(\tau t)} x_{\ell,i}^\tau. \quad (3.2)$$

The vector $g_{\ell,i-1}^\tau$ is some local knowledge of agent $\ell \in \mathcal{N}^{(t)}$ has on team τ . The estimate is updated by aggregating only the information from neighbors of the same team and agents in team τ .

For the sake of simplicity and facilitating further comparisons between ATC-ITC with other algorithms, let us denote

$$\mathcal{P} := \mathcal{CA} \quad (3.3)$$

as the matrix that describes the propagation of information across the agents within the network.

Following [Definition 2.2](#), by introducing the matrix

$$\mathcal{M}_i := \text{blkdiag} \{ \mu_i^\tau \mathbb{1}_{KM_\tau} \}_{\tau \in [T]}, \quad (3.4)$$

we have the state recursion relation

$$\underline{X}_i = \mathcal{P} \left(\underline{X}_i - \mathcal{M}_i \hat{\underline{G}}_i \right) = \mathcal{P} \left(\underline{X}_{i-1} - \mathcal{M}_i \underline{G}_i \right) - \mathcal{P} \mathcal{M}_i \underline{S}_i. \quad (3.5)$$

Similarly, we can define the error recursion:

$$\tilde{\underline{X}}_i = \mathcal{P} \left(\tilde{\underline{X}}_{i-1} + \mathcal{M}_i \underline{G}_i \right) + \mathcal{P} \mathcal{M}_i \underline{S}_i \quad (3.6)$$

$$= \underbrace{\mathcal{P} (\mathbb{I} - \mathcal{M}_i \mathcal{H}_{i-1})}_{\tilde{\mathcal{B}}_i} \tilde{\underline{X}}_{i-1} + \mathcal{P} \mathcal{M}_i \underline{G}^* + \mathcal{P} \mathcal{M}_i \underline{S}_i \quad (3.7)$$

$$= \underbrace{\mathcal{P} (\mathbb{I} - \mathcal{M}_i \mathcal{H})}_{\mathcal{B}_i} \tilde{\underline{X}}_{i-1} + \mathcal{P} \mathcal{M}_i \tilde{\mathcal{H}}_{i-1} \tilde{\underline{X}}_{i-1} + \mathcal{P} \mathcal{M}_i \underline{G}^* + \mathcal{P} \mathcal{M}_i \underline{S}_i. \quad (3.8)$$

Before any further discussions, let us consider a change of coordinates to make the derivations cleaner.

First of all, let us apply a change of coordinate so that those information residing in the space spanned by the Perron vectors are grouped together:

Definition 3.1: Error Recursion in a New Coordinate

From Equation (3.7), by defining $\widetilde{W}_i := \widetilde{\mathcal{V}}^H \widetilde{X}_i$, we have

$$\widetilde{W}_i = \left(\widetilde{\mathcal{J}} - \widetilde{\mathcal{M}}_i \widetilde{\mathcal{D}}_{i-1} \right) \widetilde{W}_{i-1} + \widetilde{\mathcal{M}}_i \underline{\mathcal{G}}^* + \widetilde{\mathcal{M}}_i \underline{\mathcal{S}}_i, \quad (3.9)$$

with

$$\begin{aligned} \widetilde{\mathcal{M}}_i &:= \widetilde{\mathcal{V}}^H \mathcal{M}_i \widetilde{\mathcal{V}}^{-H} \\ &= \text{blkdiag} \left\{ \text{blkdiag} \left\{ \mu_i^\tau \mathbb{1}_{M_\tau} \right\}_{\tau \in [T]}, \text{blkdiag} \left\{ \mu_i^\tau \mathbb{1}_{(K-1)} \otimes \mathbb{1}_{M_\tau} \right\}_{\tau \in [T]} \right\}. \end{aligned} \quad (3.10)$$

The recursion relation above can immediately be established by noticing that the two matrices $\widetilde{\mathcal{J}}$ and $\widetilde{\mathcal{M}}_i$ commutes. Similar to Theorem 2.14, the matrices above can each be partitioned into four blocks, with the top-left block being of size $M \times M$.

Next, let us consider the spectrum of $\widetilde{\mathcal{J}} - \widetilde{\mathcal{M}}_i \widetilde{\mathcal{D}}_{i-1}$. To reiterate, let us restate Assumption 2.3 and Assumption 2.4 as below:

Assumption 3.1: Bounds on Loss Functions

For all $t, \tau \in [T]$ and $k \in \mathcal{N}^{(t)}$,

- (a) The local risk functions $J_k^{(t)}$ are twice differentiable with respect to all arguments.
- (b) Smoothness: for all $x_1, x_2 \in \mathbb{R}^{M \times 1}$,

$$\left\| \nabla_t J_k^{(t)}(x_1) - \nabla_t J_k^{(t)}(x_2) \right\| \leq \delta_k^{(t)} \|x_1 - x_2\| \quad (3.11)$$

holds for some $\delta_k^{(t)} > 0$.

- (c) Hessians and mixed Hessians are globally Lipschitz continuous about the Nash equilibrium:

$$\left\| \nabla_{\tau t}^2 J_k^{(t)}(x^* + \Delta x) - \nabla_{\tau t}^2 J_k^{(t)}(x^*) \right\| \leq \kappa' \|\Delta x\| \quad (3.12)$$

holds for some $\kappa' > 0$.

- (d) There exist $\varepsilon^{(t)} > 0$ such that

$$\sum_{k \in \mathcal{N}^{(t)}} p_k^{(t)} \nabla_t^2 J_k^{(t)}(x_k) \succeq \varepsilon^{(t)} \mathbb{1}_{M_t}. \quad (3.13)$$

- (e) There exist constants $\delta^{(t\tau)} > 0$ such that

$$\delta^{(t\tau)} := \sup_{\{x_k\}_{k \in \mathcal{N}^{(t)}}} \left\| \sum_{k \in \mathcal{N}^{(t)}} p_k^{(t)} \nabla_{\tau t}^2 J_k^{(t)}(x_k) \right\|. \quad (3.14)$$

Assumption 3.2: Game Parameters Condition

The constants defined in and the step-sizes are assumed to satisfy

$$\varepsilon^{(t)} - \sum_{t' \neq t} \delta^{(tt')} > 0 \quad (3.15)$$

and

$$\mu_i^t \varepsilon^{(t)} - \sum_{t' \neq t} \mu_i^{t'} \delta^{(t't)} > 0 \quad (3.16)$$

for all $t \in [T]$.

As a consequence, we have:

Theorem 3.1: Spectrum of $\widetilde{\mathcal{M}}_i^{(11)} \nabla_x F(x)$

Under [Assumption 3.1](#), the spectrum of $\widetilde{\mathcal{M}}_i^{(11)} \nabla_x F(x)$ is within the region defined by:

$$\min_{t \in [T]} \mu_i^t \left(\varepsilon^{(t)} - \sum_{t' \neq t} \delta^{(tt')} \right) \leq \operatorname{Re}\{\lambda\} \leq \max_{t \in [T]} \mu_i^t \sum_{t' \in [T]} \delta^{(tt')}, \quad (3.17)$$

$$|\operatorname{Im}\{\lambda\}| \leq \max_{t \in [T]} \mu_i^t \sum_{t' \neq t} \delta^{(tt')}. \quad (3.18)$$

Under [Assumption 3.2](#), this shows that the real and imaginary parts to the eigenvalues have orders of $\operatorname{Re}\{\lambda\} = \Theta(\mathcal{M}_i)$ and $\operatorname{Im}\{\lambda\} = O(\mathcal{M}_i)$.

Since each $\mu_i^\tau > 0$ is independent of those with different τ , and the ratio between any two is finite, the abuse of notation $O(\mathcal{M}_i) = O(\mu_i^\tau)$ for any τ . The same abuse of notation applies to $\Theta(\cdot)$, $\Omega(\cdot)$ and their lower case variants.

Proof. Note that the $M \times M$ matrix in question can be partitioned into blocks, with the (t, t') th block being of size $M_t \times M_{t'}$, and is of the form

$$\left[\widetilde{\mathcal{M}}_i^{(11)} \nabla_x F(x) \right]_{tt'} = \mu_i^t \cdot \nabla_{t't}^2 J^{(t)}(x).$$

From [Assumption 3.1](#), the diagonal blocks are symmetric and satisfy

$$\left[\widetilde{\mathcal{M}}_i^{(11)} \nabla_x F(x) \right]_{tt} \succeq \mu_i^t \varepsilon^{(t)} \mathbb{I}, \quad \left\| \left[\widetilde{\mathcal{M}}_i^{(11)} \nabla_x F(x) \right]_{tt} \right\| \leq \mu_i^t \delta^{(tt)},$$

while the off-diagonal blocks satisfy

$$\left\| \left[\widetilde{\mathcal{M}}_i^{(11)} \nabla_x F(x) \right]_{tt'} \right\| \leq \mu_i^t \delta^{(tt')}.$$

These conditions can be used to bound the spectrum of $\widetilde{\mathcal{M}}_i^{(11)} \nabla_x F(x)$.

We can establish a perturbation bound on the eigenvalues. Let us define the whole matrix as the diagonal blocks D perturbed by E from the off-diagonal blocks. Using the block-Gershgorin's theorem, the eigenvalues to $D + E$ lies in the union of the regions

$$G_t := \left\{ z \in \mathbb{C} \mid \min |z - \lambda(D_{tt})| \leq \sum_{t' \neq t} \|E_{tt'}\| \right\}$$

$$\subseteq \left\{ z \in \mathbb{C} \mid \min_t |z - \lambda(D_{tt})| \leq \mu_i^t \sum_{t' \neq t} \delta^{(tt')} \right\}.$$

Since the diagonal blocks D_{tt} are symmetric, it has real eigenvalues within the region $[\mu_i^t \varepsilon^{(t)}, \mu_i^t \delta^{(tt)}]$. We thus obtain a bound on the region of eigenvalues as

$$\lambda(\widetilde{\mathcal{M}}_i^{(11)} \nabla_x F(x)) \subseteq \bigcup_{t \in [T]} \left\{ z \in \mathbb{C} \mid \exists s \in [\mu_i^t \varepsilon^{(t)}, \mu_i^t \delta^{(tt)}], |z - s| \leq \mu_i^t \sum_{t' \neq t} \delta^{(tt')} \right\}.$$

Thus the statement is shown. $\blacksquare$

Theorem 3.2: Spectral Norm of $\widetilde{\mathcal{J}}^{(11)} - \widetilde{\mathcal{M}}_i^{(11)} \widetilde{\mathcal{D}}_{i-1}^{(11)}$

Under [Assumption 3.1](#) and [Assumption 3.2](#), the spectral norm of the upper-left block of $\widetilde{\mathcal{J}} - \widetilde{\mathcal{M}}_i \widetilde{\mathcal{D}}_{i-1}$ can be bounded by

$$\left\| \mathbb{1}_M - \widetilde{\mathcal{M}}_i^{(11)} \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\| = 1 - \Omega(\mathcal{M}_i). \quad (3.19)$$

Proof. Again, let us bound its spectral norm via its block-1-norm and block- ∞ -norm:

$$\begin{aligned} \left\| \mathbb{1} - \widetilde{\mathcal{M}}_i^{(11)} \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\|^2 &\leq \left\| \mathbb{1} - \widetilde{\mathcal{M}}_i^{(11)} \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\|_{b,1} \left\| \mathbb{1} - \widetilde{\mathcal{M}}_i^{(11)} \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\|_{b,\infty} \\ &\leq \max_{t'} \left(1 - \mu_i^{t'} \varepsilon^{(t')} + \sum_{t \neq t'} \mu_i^t \delta^{(tt')} \right) \max_t \left(1 - \mu_i^t \varepsilon^{(t)} + \mu_i^t \sum_{t' \neq t} \delta^{(tt')} \right) \\ &= 1 - \Theta(\mathcal{M}_i). \end{aligned}$$

Thus it is shown. $\blacksquare$

Theorem 3.3: Spectral Radius of $\mathcal{B}_i = \mathcal{P}(\mathbb{1} - \mathcal{M}_i \mathcal{H})$

The spectral radius of $\mathcal{B}_i$ is of order

$$\rho(\mathcal{B}_i) = 1 - \Theta(\mathcal{M}_i). \quad (3.20)$$

Proof. The proof of this theorem utilizes the block-Gershgorin's theorem and [Theorem 3.1](#).

Consider the same change of coordinates as in the proof to [Theorem 2.18](#): we obtain

$$\mathcal{T}^{-1} \widetilde{\mathcal{V}}^H \mathcal{B}_i \widetilde{\mathcal{V}}^{-H} \mathcal{T} = \begin{bmatrix} \mathbb{1}_M - \widetilde{\mathcal{M}}_i^{(11)} \nabla_x F(x^*) & \cdots & O(\mathcal{M}_i^{\frac{K+1}{K}}) & \cdots \\ \vdots & \lambda_2 \mathbb{1}_M + O(\mathcal{M}_i) & \cdots & O(\mathcal{M}_i^{1/K}) \\ O(\mathcal{M}_i^{1/K}) & \vdots & \ddots & \vdots \\ \vdots & O(\mathcal{M}_i^{1/K}) & \cdots & \lambda_L \mathbb{1}_M + O(\mathcal{M}_i) \end{bmatrix},$$

where the λ_k 's in the bottom-right block are the eigenvalues to $\widetilde{\mathcal{J}}^{(22)}$. The eigenvalues to $\lambda_k \mathbb{1}_M + O(\mathcal{M}_i)$ is perturbed from λ_k by a perturbation of $O(\mathcal{M}_i)$.

Next, by the block-Gershgorin's theorem, we have that the eigenvalues to $\mathcal{B}_i$ lie in the region below

$$\left\{ z \in \mathbb{C} \left| \min_k \left| z - \lambda_k \left(\mathbb{1} - \widetilde{\mathcal{M}}_i^{(11)} \nabla_x F(x^*) \right) \right| \leq O(\mathcal{M}_i^{\frac{K+1}{K}}) \right\} \dots \\ \cup \left\{ z \in \mathbb{C} \left| \min_k |z - \lambda_k| \leq O(\mathcal{M}_i^{1/K}) \right. \right\}$$

If each step-sizes are made small enough, the two sets above are in fact *disjointed*. Hence, the eigenvalue with the largest norm must lie in the first set. Thus, we obtained that the spectral radius to $\mathcal{B}_i$ is

$$1 - \max_{t \in [T]} \mu_i^t \sum_{t' \in [T]} \delta^{(tt')} - O(\mathcal{M}_i^{\frac{K+1}{K}}) \leq \rho(\mathcal{B}_i) \leq 1 - \min_{t \in [T]} \mu_i^t \left(\varepsilon^{(t)} - \sum_{t' \neq t} \delta^{(tt')} \right) + O(\mathcal{M}_i^{\frac{K+1}{K}}).$$

I.e., it is of order $1 - \Theta(\mathcal{M}_i)$. ■

Note that [Theorem 3.2](#) is used to establish the stability of the algorithm, while [Theorem 3.3](#) is required to establish the rate of convergence to the ATC-ITC algorithm.

3.1 Constant Step-Sizes

Assume that the step-sizes are constant with respect to the iteration i , i.e. $\mu_i^\tau = \mu^\tau$ for all i , i.e., the matrix $\mathcal{M}_i = \mathcal{M}$. In comparison to [Chapter 2](#), the change of the step-sizes to be team-dependent results in modifications of some constants. To be specific, the following should be re-evaluated:

- The relations between the constants defined in [Assumption 2.3](#) is to be modified as a function depending on the ratios between the step-sizes.
- The establishment of [Theorem 2.3](#) on the strong-monotonicity of the game.
- Consequently from the change of the above item, the criteria for which $\mathbb{1} - \mu\tilde{\mathcal{D}}_{i-1}$ is stable, the spectrum to $\mathbb{1} - \mu\tilde{\mathcal{D}}$ should be accordingly adjusted.
- Some minor changes to the constants in front of the bounding of $\underline{S}_i$ and $\underline{G}^*$.
- The rate of convergence to steady-state is going to be different.
- The MSD performance may change as well. Needs further inspection.

3.1.1 Stability

Theorem 3.4: Second, Fourth and First Moment Stability

Under the ATC-ITC recursion as stated in [Equation \(3.7\)](#), the error vector to Nash equilibrium is stable in the sense of:

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \tilde{\underline{X}}_i \right\|^2 \right] = O(\mathcal{M}), \quad (3.21)$$

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\left\| \tilde{\underline{X}}_i \right\|^4 \right] = O(\mathcal{M}^2), \quad (3.22)$$

$$\limsup_{i \rightarrow \infty} \left\| \mathbb{E} \left[\tilde{\underline{X}}_i \right] \right\| = O(\mathcal{M}) \quad (3.23)$$

for step-sizes small enough.

Proof. The proof is almost written verbatim as that in [Section 2.2](#) and [Section 2.3](#), with the main difference being replacing $\mu\tilde{\mathcal{D}}_{i-1}$ with $\tilde{\mathcal{M}}\tilde{\mathcal{D}}_{i-1}$.

In establishing the second-moment stability, the bounding to $\left\| \tilde{\mathcal{J}} \right\|^{(11)} - \tilde{\mathcal{M}}\tilde{\mathcal{D}}_{i-1}^{(11)}$ is again achieved by applying the block-1-norm and block- ∞ -norm.

For the fourth- and first-moment stability, we utilize [Theorem 3.3](#) for the upper bound of the spectral radius of $\mathcal{B} = \mathcal{P}(\mathbb{1} - \mathcal{M}\mathcal{H})$. The bound for the term $\tilde{\mathcal{H}}_{i-1}$ remains proportional to T as we did not change form of the local knowledge here. ■

3.1.2 First-Order MSD Performance

Theorem 3.5: First-Order MSD Performance

For $\mathcal{X}$ defined as the solution to [Equation \(2.129\)](#), the first-order term component to the MSD can

be expressed as, for any t ,

$$\overline{\text{MSD}}_1 := \mu^t \cdot \left(\lim_{\mu^t \rightarrow 0} \frac{1}{\mu^t} \text{MSD}_1 \right) = \sum_{\tau \in [T]} \mu^\tau \cdot \text{Tr} \left\{ \mathcal{W}_\tau \sum_{k \in \mathcal{N}(\tau)} (p_k^{(\tau)})^2 R_k^\tau \right\}, \quad (3.24)$$

where $\mathcal{W}_\tau \in \mathbb{R}^{M_\tau \times M_\tau}$ are the diagonal blocks to $\mathcal{W} \in \mathbb{R}^{M \times M}$ that satisfies the continuous-time Lyapunov equation

$$\mathcal{W} \tilde{\mathcal{D}}^{(11)} + (\tilde{\mathcal{D}}^{(11)})^\top \mathcal{W} = \mathbb{1}_M, \quad (3.25)$$

with $\tilde{\mathcal{D}}^{(11)} = \nabla_x F(x^*)$.

Proof. Again starting from the long-term error model with the centered recursion:

$$\underline{Z}_i = \mathcal{B} \underline{Z}_{i-1} + \mathcal{P} \mathcal{M} \underline{S}_i(\underline{X}_{i-1}),$$

we also have

$$\limsup_{i \rightarrow \infty} \mathbb{E} \left[\|\underline{Z}_i\|^2 \right] = \limsup_{i \rightarrow \infty} \mathbb{E} \left[\|\tilde{\underline{X}}_i\|^2 \right] + O(\mathcal{M}^{3/2}).$$

Taking the weighted norm of the centered recursion, we obtain

$$\mathbb{E} \left[\|\underline{Z}_i\|_\Sigma^2 \right] = \mathbb{E} \left[\|\underline{Z}_{i-1}\|_{\mathcal{B}^\top \Sigma \mathcal{B}}^2 \right] + \mathbb{E} \left[\|\underline{S}_i\|_{\mathcal{M}^\top \mathcal{P}^\top \Sigma \mathcal{P} \mathcal{M}}^2 \right],$$

with

$$\limsup_{i \rightarrow \infty} \left| \mathbb{E} \left[\|\underline{S}_i\|_{\mathcal{M}^\top \mathcal{P}^\top \Sigma \mathcal{P} \mathcal{M}}^2 \right] - \text{Tr} \{ \Sigma \mathcal{Y} \} \right| = \text{Tr} \{ \Sigma \} \cdot O(\mathcal{M}^{2+\gamma'})$$

and $\mathcal{Y} := \mathcal{P} \mathcal{M} \mathcal{R} \mathcal{M}^\top \mathcal{P}^\top$.

Again, let Σ satisfy the discrete-time Lyapunov equation $\Sigma - \mathcal{B}^\top \Sigma \mathcal{B} = \mathbb{1}$. We have $\text{Tr} \{ \Sigma \mathcal{Y} \} = \text{Tr} \{ \mathcal{X} \}$, where $\mathcal{X} - \mathcal{B} \mathcal{X} \mathcal{B}^\top = \mathcal{Y}$. Therefore,

$$\limsup_{i \rightarrow \infty} \frac{1}{K} \mathbb{E} \left[\|\tilde{\underline{X}}_i\|^2 \right] = \underbrace{\frac{1}{K} \text{Tr} \{ \mathcal{X} \}}_{\text{MSD}_1} + \text{Tr} \{ \Sigma \} \cdot O(\mathcal{M}^{2+\gamma'}) + O(\mathcal{M}^{3/2}).$$

Note that, once again, $\text{Tr} \{ \Sigma \} = O(\mathcal{M}^{-1})$. But the newly defined $\mathcal{X}$ satisfies $\text{Tr} \{ \mathcal{X} \} = O(\mathcal{M})$.

The last step is to find an explicit form to

$$\overline{\text{MSD}}_1 := \mu^t \cdot \left(\lim_{\mu^t \rightarrow 0} \right) \frac{1}{\mu^t} \text{MSD}_1.$$

This can be obtained by finding the $O(\mathcal{M}^{-1})$ component to $(\mathbb{1} - \mathcal{B} \otimes \mathcal{B})^{-1}$. By defining

$$\mathcal{Q}_{11}^{-1} := \left(\tilde{\mathcal{M}}^{(11)} \tilde{\mathcal{D}}^{(11)} \otimes \mathbb{1}_M + \mathbb{1}_M \otimes \tilde{\mathcal{M}}^{(11)} \tilde{\mathcal{D}}^{(11)} \right)^{-1},$$

we have

$$\begin{aligned} \overline{\text{MSD}}_1 &= \mu^t \cdot \lim_{\mu^t \rightarrow 0} \frac{1}{K \mu^t} \cdot \text{vec}(\mathbb{1})^\top \left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right)^{-\top} \begin{bmatrix} \mathcal{Q}_{11}^{-1} & \vdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \vdots & 0 \end{bmatrix} \left(\tilde{\mathcal{V}} \otimes_b \tilde{\mathcal{V}} \right)^\top \text{vec}(\mathcal{Y}) \\ &= \mu^t \cdot \lim_{\mu^t \rightarrow 0} \frac{1}{K \mu^t} \cdot K \mathbb{1}_M^\top \mathcal{Q}_{11}^{-1} \text{vec} \left((\tilde{\mathcal{V}}^{(1)})^\top \mathcal{Y} \tilde{\mathcal{V}}^{(1)} \right). \end{aligned}$$

By finding a vector $\widetilde{\mathcal{W}}$ that satisfies the continuous-time Lyapunov equation

$$\begin{aligned} K \text{vec}(\widetilde{\mathcal{W}}) &= \mathcal{Q}_{11}^{-\top} \cdot K \mathbb{1}_M \\ \Leftrightarrow \widetilde{\mathcal{W}} \widetilde{\mathcal{M}}^{(11)} \widetilde{\mathcal{D}}^{(11)} + (\widetilde{\mathcal{M}}^{(11)} \widetilde{\mathcal{D}}^{(11)})^\top \widetilde{\mathcal{W}} &= \mathbb{1}_M. \end{aligned}$$

Note that since $\widetilde{\mathcal{W}}$ is symmetric, $\widetilde{\mathcal{W}} \widetilde{\mathcal{M}}^{(11)} = \widetilde{\mathcal{M}}^{(11)} \widetilde{\mathcal{W}} =: \mathcal{W}$. This new matrix satisfies

$$\mathcal{W} \widetilde{\mathcal{D}}^{(11)} + (\widetilde{\mathcal{D}}^{(11)})^\top \mathcal{W} = \mathbb{1}_M.$$

Since for any μ^t , $\mu^t \cdot \lim_{\mu^t \rightarrow 0} \frac{1}{\mu^t} \mathcal{M} = \mathcal{M}$, we obtain

$$\begin{aligned} \overline{\text{MSD}}_1 &= \mu^t \cdot \lim_{\mu^t \rightarrow 0} \frac{1}{\mu^t} \text{vec}(\widetilde{\mathcal{W}})^\top \text{vec}((\widetilde{\mathcal{V}}^{(1)})^\top \mathcal{Y} \widetilde{\mathcal{V}}^{(1)}) \\ &= \mu^t \cdot \lim_{\mu^t \rightarrow 0} \frac{1}{\mu^t} \text{Tr} \left\{ (\widetilde{\mathcal{M}}^{(11)})^{-1} \mathcal{W} (\widetilde{\mathcal{V}}^{(1)})^\top \mathcal{Y} \widetilde{\mathcal{V}}^{(1)} \right\} \\ &= \mu^t \cdot \lim_{\mu^t \rightarrow 0} \frac{1}{\mu^t} \text{Tr} \left\{ (\widetilde{\mathcal{M}}^{(11)})^{-1} \mathcal{W} \cdot (\widetilde{\mathcal{V}}^{(1)})^\top \mathcal{P} \mathcal{M} \cdot \mathcal{R} \cdot \mathcal{M}^\top \mathcal{P}^\top \widetilde{\mathcal{V}}^{(1)} \right\} \\ &= \mu^t \cdot \lim_{\mu^t \rightarrow 0} \frac{1}{\mu^t} \text{Tr} \left\{ \widetilde{\mathcal{M}}^{(11)} \mathcal{W} \cdot (\widetilde{\mathcal{V}}^{(1)})^\top \mathcal{R} \widetilde{\mathcal{V}}^{(1)} \right\} \\ &= \text{Tr} \left\{ \widetilde{\mathcal{M}}^{(11)} \mathcal{W} \cdot (\widetilde{\mathcal{V}}^{(1)})^\top \mathcal{R} \widetilde{\mathcal{V}}^{(1)} \right\} \\ &= \sum_{\tau \in [T]} \mu^\tau \text{Tr} \left\{ \mathcal{W}_\tau \sum_{k \in \mathcal{N}(\tau)} (p_k^{(\tau)})^2 R_k^\tau \right\}. \end{aligned}$$

It should be noted that each $M_\tau \times M_\tau$ diagonal block $\mathcal{W}_\tau$ of $\mathcal{W}$ is independent of $\mathcal{M}$.

Lastly, we can see that the result is independent of the $t \in [T]$ chosen as it is assumed that any two step-sizes have a finite ratio. The statement is proven. $\blacksquare$

3.1.3 Rate of Convergence

Theorem 3.6: Rate of Convergence

Under the ATC-ITC recursion as stated in Equation (3.7), the convergence rate of the squared deviation $\mathbb{E} \left[\left\| \widetilde{\underline{X}}_i \right\|^2 \right]$ towards the steady-state region is α^i , where

$$\alpha = 1 - \Theta(\mathcal{M}). \quad (3.26)$$

Proof. The rate of convergence of $\widetilde{\underline{X}}_i$ to the steady state is equal to the spectral radius of $\mathcal{B}^\top \otimes \mathcal{B}^\top$, which is, by Theorem 3.3

$$1 - 2 \max_{t \in [T]} \mu_i^t \sum_{t' \in [T]} \delta^{(tt')} - O(\mathcal{M}_i^{\frac{K+1}{K}}) \leq \rho(\mathcal{B}_i) \leq 1 - 2 \min_{t \in [T]} \mu_i^t \left(\varepsilon^{(t)} - \sum_{t' \neq t} \delta^{(tt')} \right) + O(\mathcal{M}_i^{\frac{K+1}{K}}).$$

I.e., $\alpha = 1 - \Theta(\mathcal{M})$. $\blacksquare$

3.1.4 Analytic Formula for MSD

We wish to obtain a clearer expression for MSD such that the effects from each parameters can be read off explicitly.

3.2 Decreasing Step-Sizes

Starting from [Definition 3.1](#), we will establish that under a decreasing step-size scheme, the ATC-ITC algorithm converges to the Nash equilibrium with probability one. Some assumptions on show the step-sizes decrease are needed:

Assumption 3.3: Decreasing Step-Sizes

It is assumed that the ratios between the step-sizes of each team are finite. Furthermore, in order to establish convergence, assume

$$\sum_{i=1}^{\infty} \mu_i^{\tau} = \infty \text{ and } \sum_{i=1}^{\infty} (\mu_i^{\tau})^2 < \infty \quad (3.27)$$

for any τ .

Since

$$\widetilde{W}_i^{(1)} = \left(\mathbb{1}_M - \widetilde{\mathcal{M}}_i^{(11)} \widetilde{\mathcal{D}}_{i-1}^{(11)} \right) \widetilde{W}_{i-1}^{(1)} - \widetilde{\mathcal{M}}_i^{(11)} \widetilde{\mathcal{D}}_{i-1}^{(12)} \widetilde{W}_{i-1}^{(2)} + \widetilde{\mathcal{M}}_i^{(11)} \underline{\mathcal{S}}_i^{(1)},$$

by first taking the expectation with respect to the filtration $\mathcal{F}_{i-1}$, then taking the remaining expectation, we have:

$$\begin{aligned} \mathbb{E} \left[\left\| \widetilde{W}_i^{(1)} \right\|^2 \right] &\leq \frac{\left\| \mathbb{1}_M - \widetilde{\mathcal{M}}_i^{(11)} \widetilde{\mathcal{D}}_{i-1}^{(11)} \right\|^2}{1-t} \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(1)} \right\|^2 \right] + \frac{\left\| \widetilde{\mathcal{M}}_i^{(11)} \right\|^2 \left\| \widetilde{\mathcal{D}}_{i-1}^{(12)} \right\|^2}{t} \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(2)} \right\|^2 \right] \dots \\ &\quad + \left\| \widetilde{\mathcal{M}}_i^{(11)} \right\|^2 \mathbb{E} \left[\left\| \underline{\mathcal{S}}_i^{(1)} \right\|^2 \right]. \end{aligned}$$

From [Theorem 3.2](#), we know to set $t = 1 - \Theta(\mathcal{M}_i)$.

Similarly, since

$$\widetilde{W}_i^{(2)} = \left(\widetilde{\mathcal{J}}^{(22)} - \widetilde{\mathcal{M}}_i^{(22)} \widetilde{\mathcal{D}}_{i-1}^{(22)} \right) \widetilde{W}_{i-1}^{(2)} - \widetilde{\mathcal{M}}_i^{(22)} \widetilde{\mathcal{D}}_{i-1}^{(21)} \widetilde{W}_{i-1}^{(1)} + \widetilde{\mathcal{M}}_i^{(22)} \underline{\mathcal{G}}^{*,(2)} + \widetilde{\mathcal{M}}_i^{(22)} \underline{\mathcal{S}}_i^{(2)},$$

we have

$$\begin{aligned} \mathbb{E} \left[\left\| \widetilde{W}_i^{(2)} \right\|^2 \right] &\leq \frac{\left\| \widetilde{\mathcal{J}}^{(22)} \right\|^2}{t} \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(2)} \right\|^2 \right] + \frac{3 \left\| \widetilde{\mathcal{M}}_i^{(22)} \right\|^2}{1-t} \left(\left\| \widetilde{\mathcal{D}}_{i-1}^{(21)} \right\|^2 \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(1)} \right\|^2 \right] \dots \right. \\ &\quad \left. + \left\| \widetilde{\mathcal{D}}_{i-1}^{(22)} \right\|^2 \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(2)} \right\|^2 \right] + \mathbb{E} \left[\left\| \underline{\mathcal{G}}^{*,(2)} \right\|^2 \right] \right) + \left\| \widetilde{\mathcal{M}}_i^{(22)} \right\|^2 \mathbb{E} \left[\left\| \underline{\mathcal{S}}_i^{(2)} \right\|^2 \right]. \end{aligned}$$

Same as before, we choose $t = 1 - \gamma_1 + \varepsilon_1$.

Combining the inequalities above to obtain the recursion:

$$\begin{bmatrix} \mathbb{E} \left[\left\| \widetilde{W}_i^{(1)} \right\|^2 \right] \\ \mathbb{E} \left[\left\| \widetilde{W}_i^{(2)} \right\|^2 \right] \end{bmatrix} \leq \begin{bmatrix} 1 - \Theta(\mathcal{M}_i) & \Theta(\mathcal{M}_i) \\ \Theta(\mathcal{M}_i^2) & 1 - \gamma_1 + \varepsilon_1 + \Theta(\mathcal{M}_i^2) \end{bmatrix} \begin{bmatrix} \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(1)} \right\|^2 \right] \\ \mathbb{E} \left[\left\| \widetilde{W}_{i-1}^{(2)} \right\|^2 \right] \end{bmatrix} + \begin{bmatrix} \Theta(\mathcal{M}_i^2) \\ \Theta(\mathcal{M}_i^2) \end{bmatrix}.$$

By the theorem below, the supremum to this sequence converges to identically zero under [Assumption 3.3](#). Since the sequence is strictly positive, we have shown its convergence to the Nash equilibrium.

Theorem 3.7: Convergence of a Numerical Sequence

Consider the positive sequence

$$\begin{bmatrix} z_{1,t} \\ z_{2,t} \end{bmatrix} \leq \begin{bmatrix} 1 - \Theta(1/t) & \Theta(1/t) \\ \Theta(1/t^2) & \varepsilon + \Theta(1/t^2) \end{bmatrix} \begin{bmatrix} z_{1,t-1} \\ z_{2,t-1} \end{bmatrix} + \begin{bmatrix} \Theta(1/t^2) \\ \Theta(1/t^2) \end{bmatrix}, \quad (3.28)$$

where $\varepsilon \in [0, 1)$. The supremum of the sequence converges identically to zero. I.e., the sequence converges to zero.

Proof. By defining $z_t = z_{1,t} + z_{2,t}$, we have, for large enough t ,

$$\begin{aligned} z_t &\leq \left(1 - \Theta(1/t) + \Theta(1/t^2)\right) z_{1,t-1} + \left(\varepsilon + \Theta(1/t) + \Theta(1/t^2)\right) z_{2,t-1} + \Theta(1/t^2) \\ &\leq \left(1 - \Theta(1/t) + \Theta(1/t^2)\right) z_{t-1} + \Theta(1/t^2), \end{aligned}$$

where the second inequality is due to $\varepsilon < 1$. Then by invoking the result in [Pol87, pp. 45–50], we can show that $z_t \rightarrow 0$. ■

4 Comparison between Algorithms

To facilitate the comparison, let us define

$$\mathcal{P} := \mathcal{CA} \tag{4.1}$$

as the matrix that determines the propagation of information throughout the network.

4.1 Local Knowledge

Instead of having

$$g_{k,i-1}^{\tau} = \nabla_{\tau} J_{\ell_k}^{(\tau)}(x_{k,i-1}^{\tau}),$$

what happens if we set the local knowledge to be identically zero?

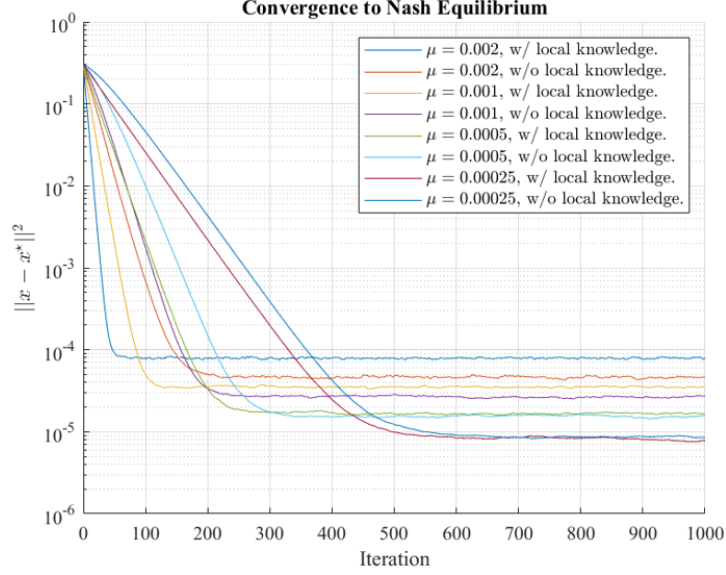


Figure 4.1 Comparison of MSD performance between with or without local knowledge. The result is obtained by taking the average of 500 samples.

The above Figure 4.1 is a comparison of the MSD performance between those with local knowledge and those without. It can be seen that the former ones converges faster than the latter ones. However, their steady state performances are also worse than the latter ones. But these are for the cases where step-sizes are large. As the step-sizes decrease, the performance of the two becomes indistinguishable.

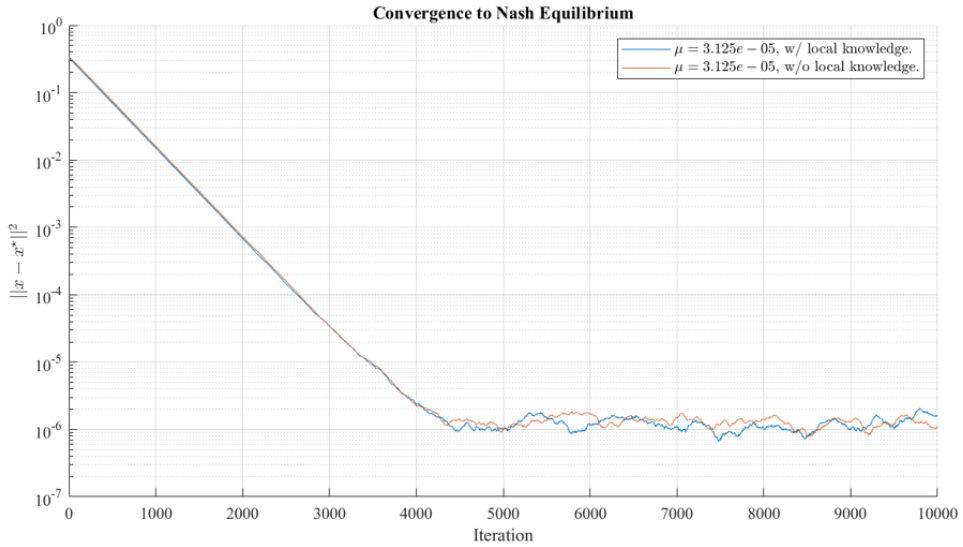


Figure 4.2 Comparison of MSD performance between with or without local knowledge. The result is obtained by taking the average of 10 samples. The performances are also indistinguishable for really small step-sizes.

The changes lies in the form of the stochastic gradient: using the subscript $\circ$ to represent zero local knowledge, we have

$$[\widehat{G}_{\circ,i}]_{\tau t} := \begin{cases} [\widehat{G}_i]_{\tau t}, & \tau = t; \\ 0, & \text{otherwise.} \end{cases} \quad (4.2)$$

Henceforth, the state recursion relation becomes

$$\underline{X}_i = \mathcal{P} \left(\underline{X}_{i-1} - \mathcal{M}_i \widehat{G}_{\circ,i} \right) = \mathcal{P} \left(\underline{X}_{i-1} - \mathcal{M}_i \underline{G}_{\circ,i} \right) - \mathcal{P} \mathcal{M}_i \underline{S}_{\circ,i}, \quad (4.3)$$

and the error recursion becomes:

$$\widetilde{X}_i = \mathcal{P} \left(\widetilde{X}_{i-1} + \mathcal{M}_i \underline{G}_{\circ,i} \right) + \mathcal{P} \mathcal{M}_i \underline{S}_{\circ,i} \quad (4.4)$$

$$= \underbrace{\mathcal{P} (\mathbb{1} - \mathcal{M}_i \mathcal{H}_{\circ,i-1})}_{\widetilde{B}_{\circ,i}} \widetilde{X}_{i-1} + \mathcal{P} \mathcal{M}_i \underline{G}_{\circ}^* + \mathcal{P} \mathcal{M}_i \underline{S}_{\circ,i} \quad (4.5)$$

$$= \underbrace{\mathcal{P} (\mathbb{1} - \mathcal{M}_i \mathcal{H}_{\circ})}_{\mathcal{B}_{\circ,i}} \widetilde{X}_{i-1} + \mathcal{P} \mathcal{M}_i \widetilde{\mathcal{H}}_{\circ,i-1} \widetilde{X}_{i-1} + \mathcal{P} \mathcal{M}_i \underline{G}_{\circ}^* + \mathcal{P} \mathcal{M}_i \underline{S}_{\circ,i}. \quad (4.6)$$

Many of the terms appearing above are defined later.

We will establish the three main performances within this section: the stability of the algorithm, its first-order MSD performance, and its rate of convergence.

Theorem 4.1: Performance of ATC-ITC with Zero Local Knowledge

The ATC-ITC algorithm remains stable under the scenario of zero local knowledge. Furthermore, its MSD performance and rate of convergence are *equivalent* to that of the non-zero local knowledge *up to first order* of the step-sizes.

The effects of the different local knowledge thus can be determined to be only of higher nonlinear orders in the step-sizes. This result is immediately acceptable as the equations of all the terms introduced above exhibit no explicit references to the local knowledge.

4.1.1 Stability

The proof to the stability lies in bounding the spectral norm to $\widetilde{B}_{\circ,i} := \mathcal{P}(\mathbb{1} - \mathcal{M}_i \mathcal{H}_{\circ,i-1})$.

$$[\mathcal{H}]_{\tau\tau'tt'} := \begin{cases} \mathcal{H}_{\tau\tau'}^{(t)}, & t = t'; \\ 0 \cdot \mathbf{1}_{K_t \times K_{t'}} \otimes \mathbf{1}_{M_\tau \times M_{\tau'}}, & \text{otherwise,} \end{cases} \quad (4.7)$$

$$\mathcal{H}_{\tau\tau'}^{(t)} := \text{blkdiag}\{H_{k,\tau\tau'}\}_{k \in \mathcal{N}^{(t)}}, \quad (4.8)$$

$$H_{k,\tau\tau'} := \nabla_{\tau'\tau}^2 J_{\ell_k}^{(\tau)}(x^*). \quad (4.9)$$

4.1.2 First-Order MSD Performance

4.1.3 Rate of Convergence

4.2 Consensus Diffusion

4.3 **ATC-ITC Versus CD**

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